EXERCISE · WEEK 02

From your laptop to GitHub

Monday 9 November · 10:30–12:20

START

10:38 Student package
GITHUB

10:46 Meet GitHub
GITHUB

10:51 Meet VS Code
VS CODE

10:55 wsl → code .
WSL

11:02 Sign in to GitHub
VS CODE

11:07 Fork
GITHUB

11:12 Clone
VS CODE

11:18 Switch to develop
VS CODE

11:24 Your details
VS CODE

----- 11:34 BREAK ----------

11:44 Push develop
VS CODE → GITHUB

11:50 Merge: a conflict
VS CODE → GITHUB

12:07 Peer review
GITHUB

Standup

Three people, two minutes each: where you are, what broke.

THE ROUTE · 12 STAGES

Today, one route

I click, you click, we check. Mac: Cmd for Ctrl, except in Ctrl+` and Ctrl+Shift+G.

STARTS	STAGE	TOOL	BRANCH	STEPS
10:38	**Student package**	GITHUB		○○○○○○○○○○○○
10:46	**Meet GitHub**	GITHUB		○○○○○○
10:51	**Meet VS Code**	VS CODE		○○○○○○
10:55	**wsl → code .**	WSL		○○○○○○○○○○
11:02	**Sign in to GitHub**	VS CODE		○○○○○○
11:07	**Fork**	GITHUB		○○○○◌
11:12	**Clone**	VS CODE	main	○○○○○○
11:18	**Switch to develop**	VS CODE	main → develop	○○○○○○
11:24	**Your details**	VS CODE	develop	○○○○○○○○
11:34	BREAK, BACK AT 11:44			
11:44	**Push develop**	VS CODE → GITHUB	develop	○○◌○○○
11:50	**Merge: a conflict**	VS CODE → GITHUB	develop → main	○○○○○○○○○○○○○○
12:07	**Peer review**	GITHUB		○○○○○○

Before you apply: four things

GitHub checks them first · set them up at home, then apply

1 TWO-FACTOR AUTHENTICATION
On: Settings → Password and authentication

2 BILLING INFORMATION
Your full legal name and address · no card needed

3 PROFILE NAME
Your full name, as on your Technion document

4 TECHNION E-MAIL
Added and verified: Settings → Emails

Turn on two-factor authentication

A code from your phone, as well as your password

1 AN AUTHENTICATOR APP

Install one on your phone (a TOTP app)

2 SETTINGS → PASSWORD AND AUTHENTICATION

Avatar, top right → Settings → Access

3 ENABLE TWO-FACTOR AUTHENTICATION

Scan the QR code, type the 6-digit code

4 RECOVERY CODES

Download them, keep them safe → I have saved my recovery codes

WATCH STUDENT PACKAGE · GITHUB · 3 OF 13

2FA: scan, type, save the codes

Settings → Password and authentication → Enable two-factor authentication

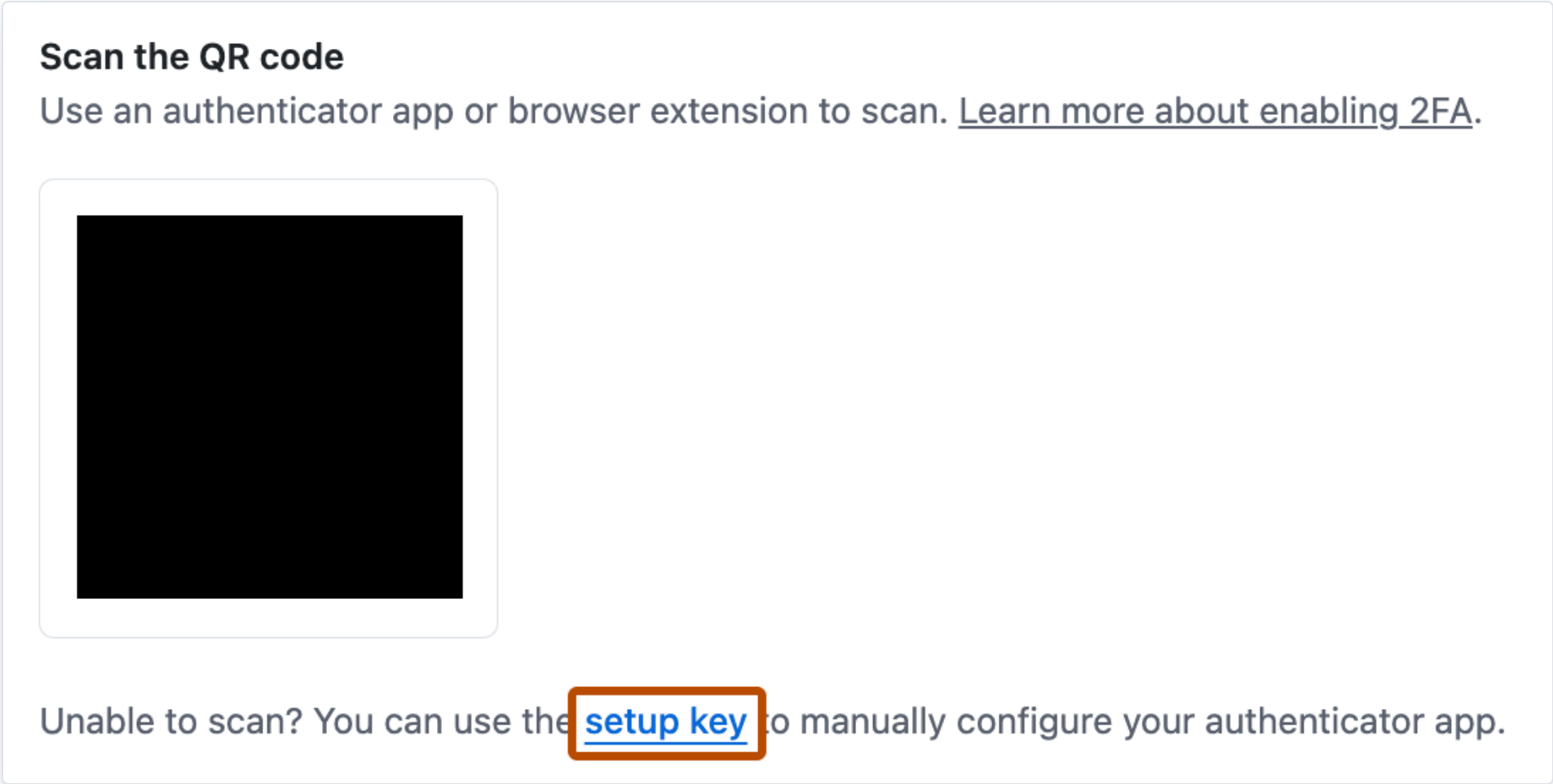


Image: GitHub Docs, CC BY 4.0

Billing information: your legal name

GitHub wants it on file · no credit card needed

1 SETTINGS → BILLING AND LICENSING
Then Payment information

2 BILLING INFORMATION → EDIT
The section under Payment information

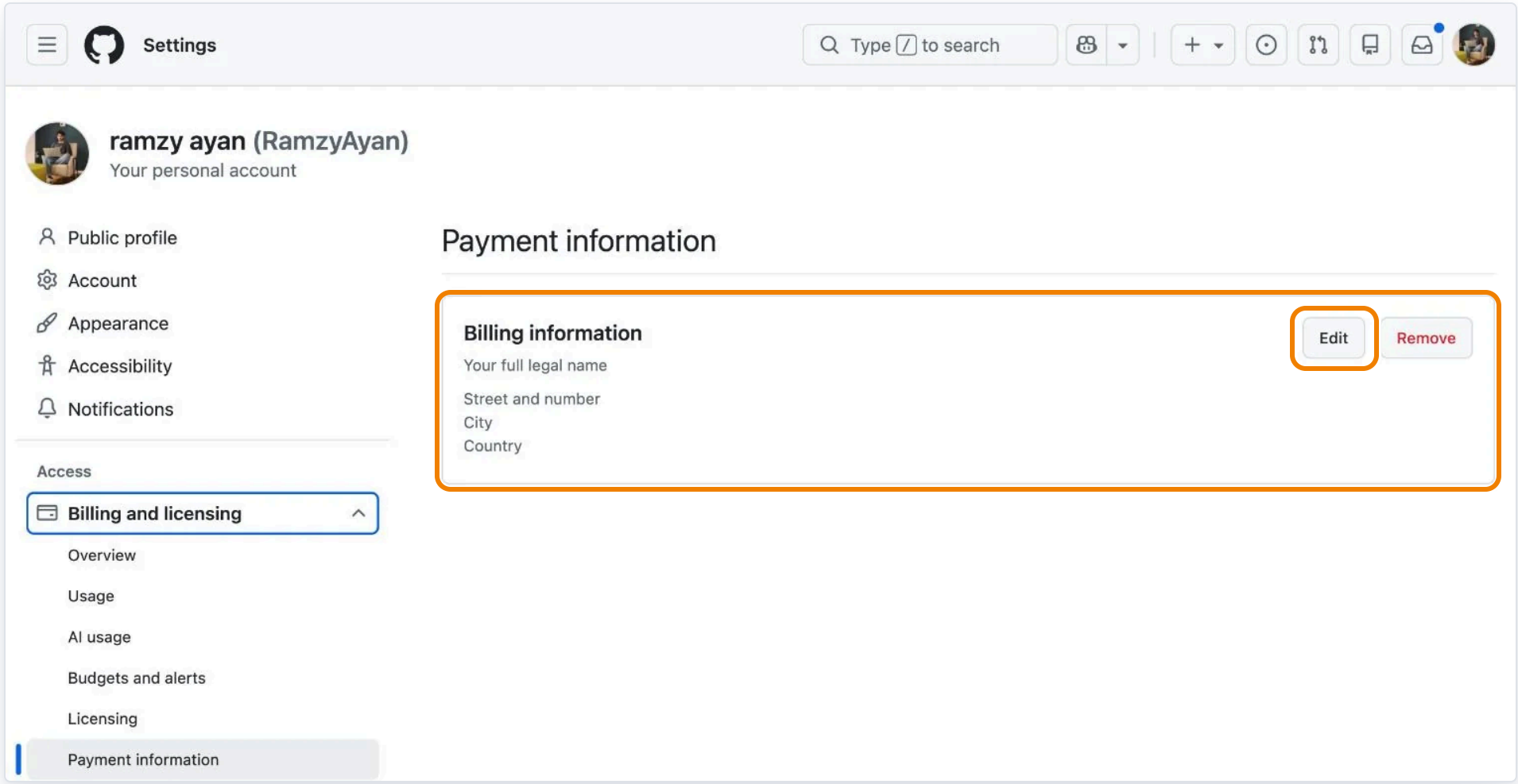
3 FULL LEGAL NAME AND ADDRESS
Exactly as on your Technion document

4 SAVE BILLING INFORMATION
Leave the payment method empty

WATCH STUDENT PACKAGE · GITHUB · 5 OF 13

Billing information → Edit

Settings → Billing and licensing → Payment information



The TA's settings page; the name and address are placeholders

Profile name: the same name

Settings → Public profile → Name

1 SETTINGS → PUBLIC PROFILE
The first page of Settings

2 NAME
Your full name, the same as on your Technion document

3 UPDATE PROFILE
At the bottom of the page

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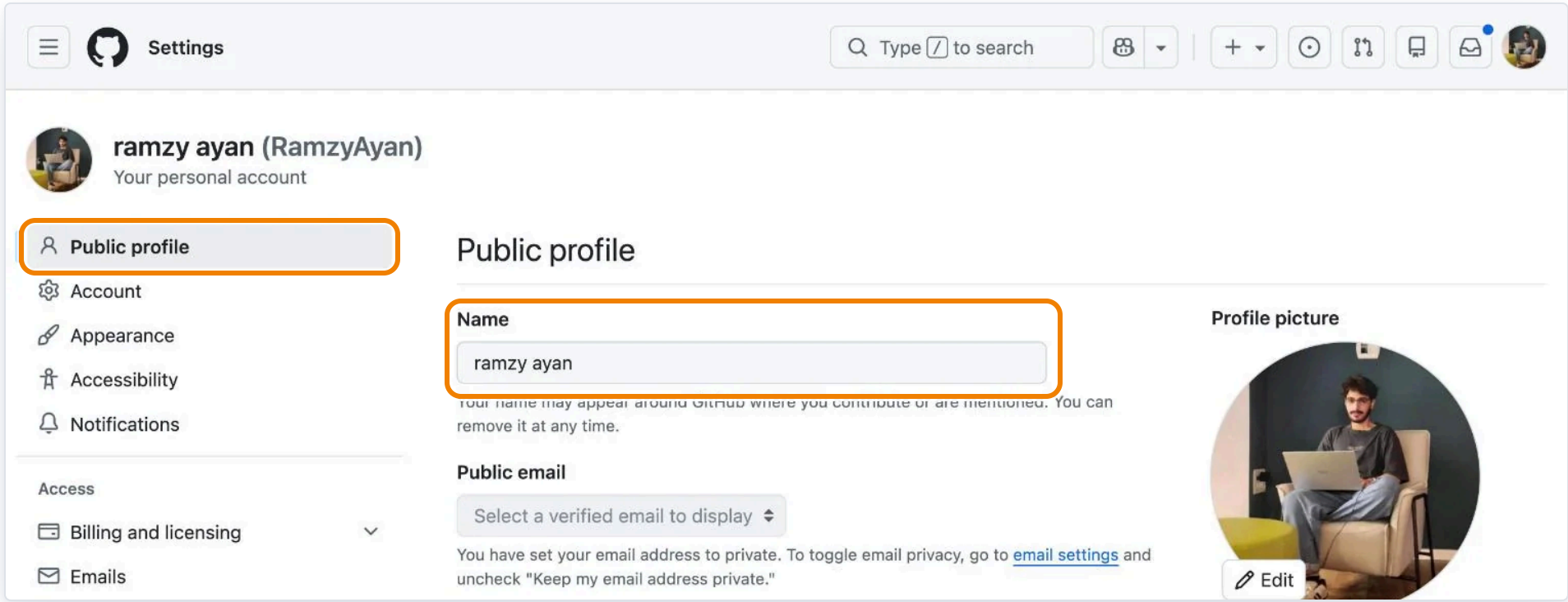
Public profile → Name

The same name as on your Technion document · Update profile



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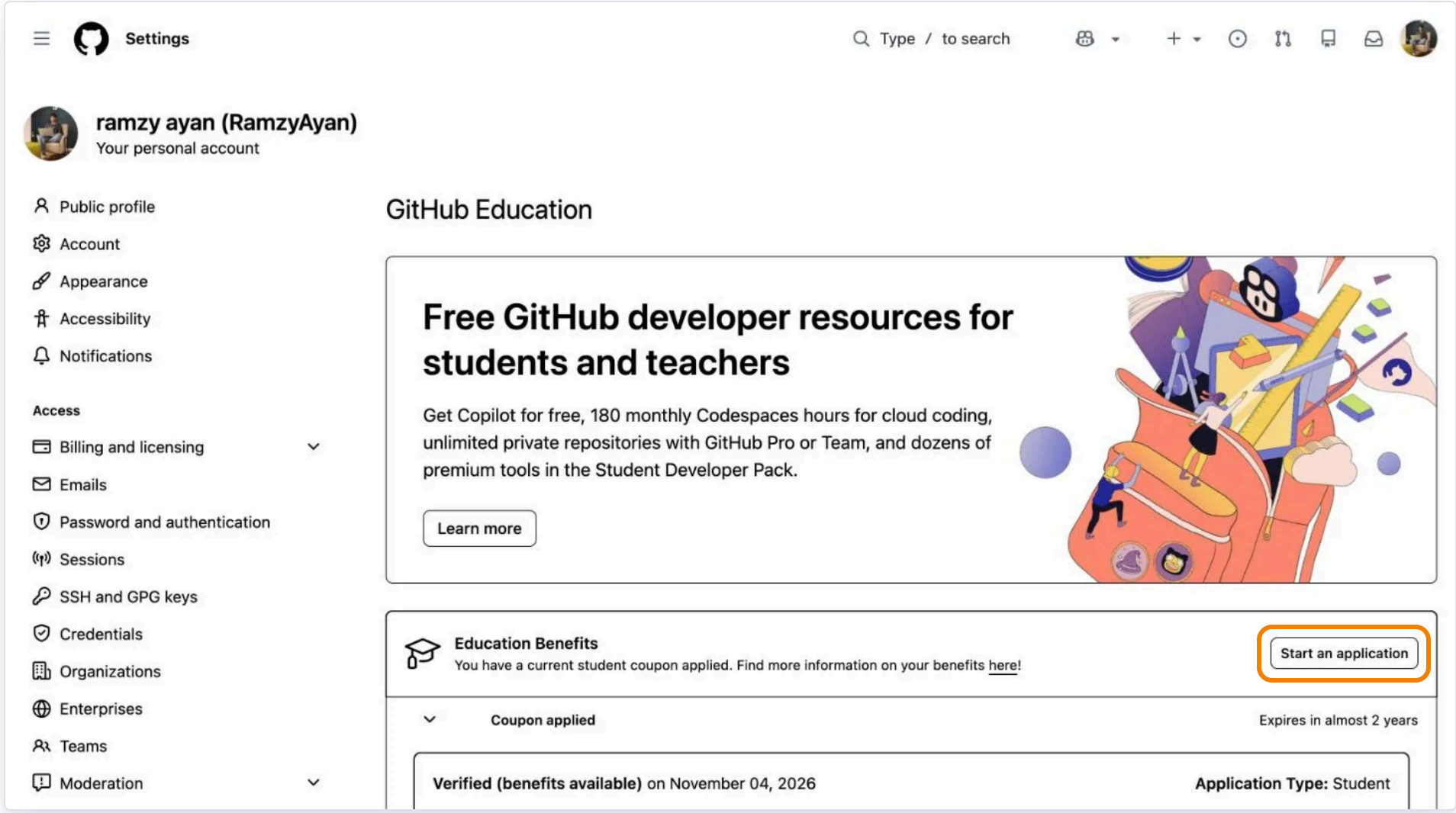
Start an application

github.com/settings/education/benefits · yours is green



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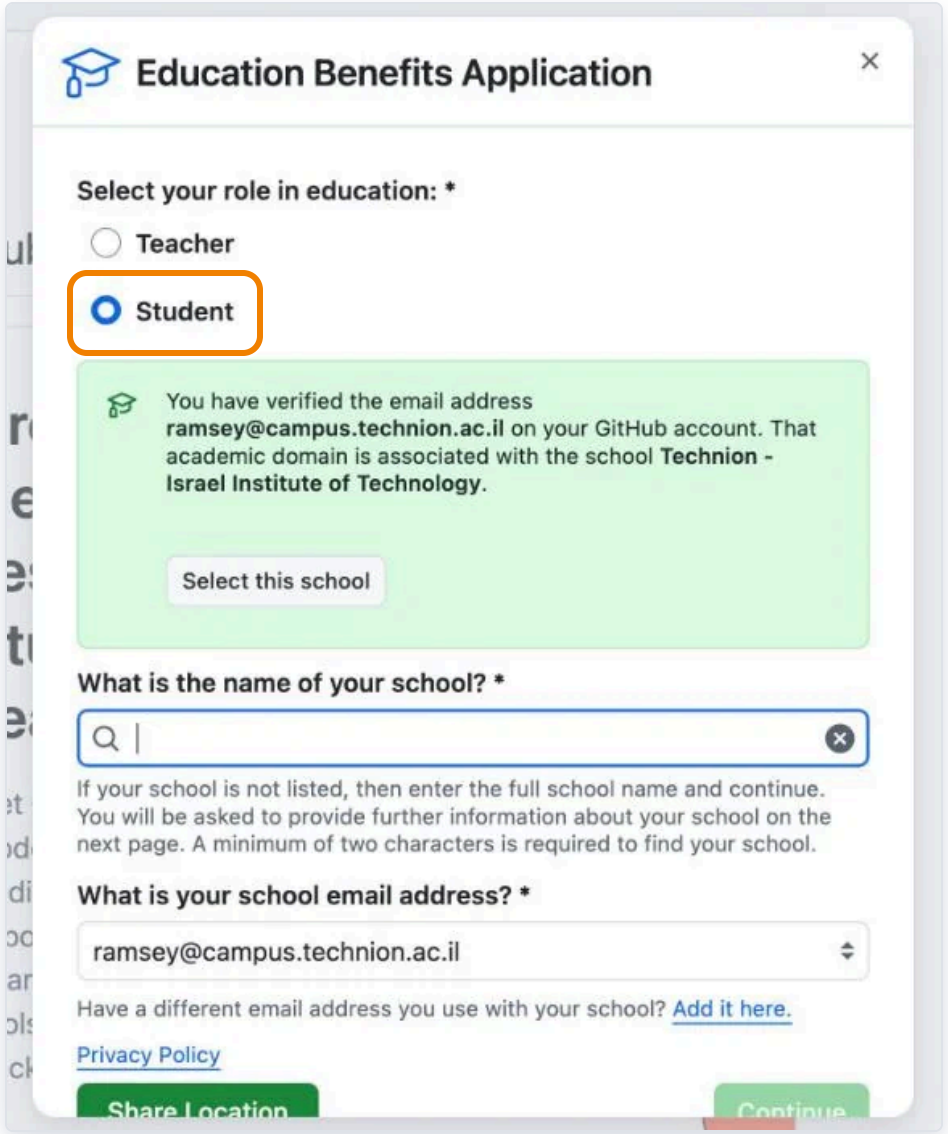
Student



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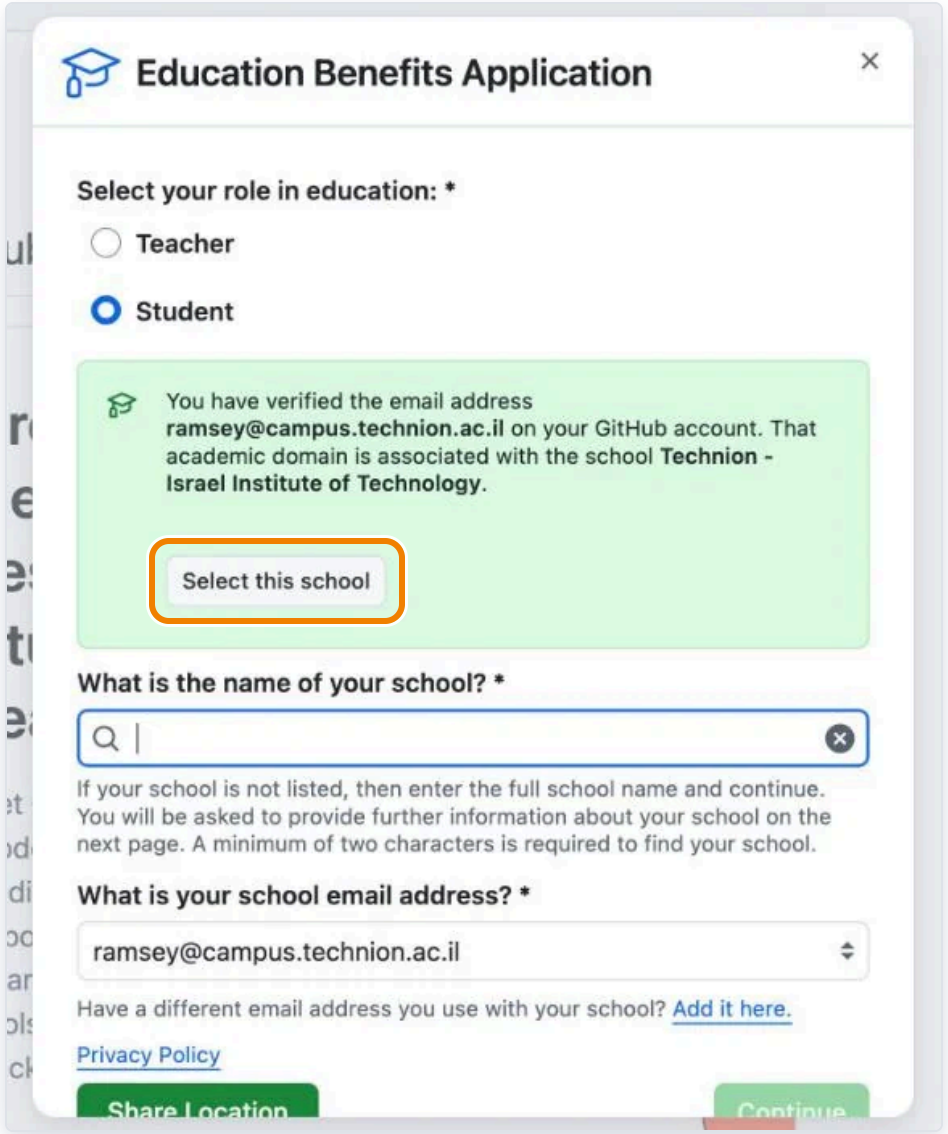
Select this school



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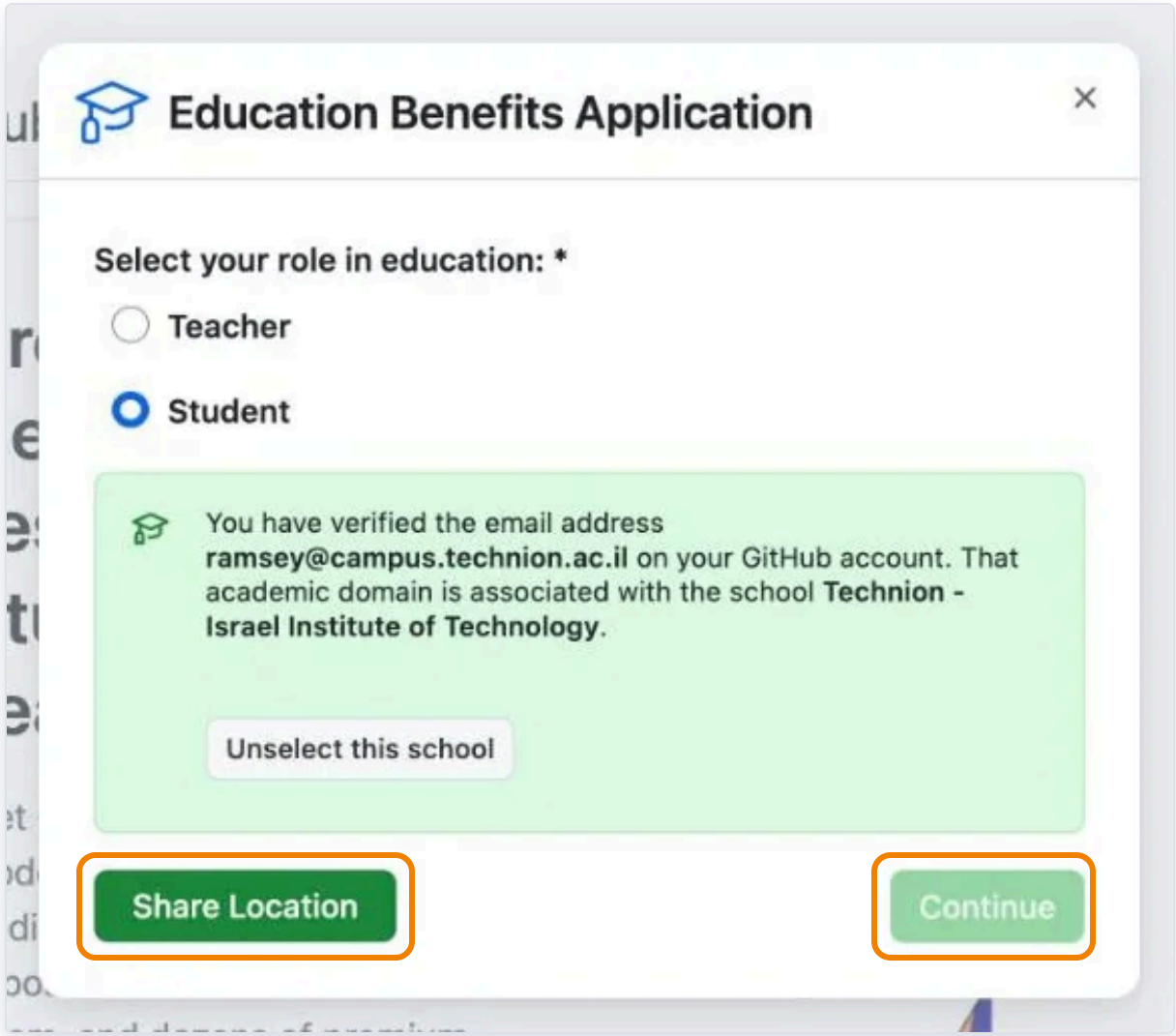
Share Location → Continue

The browser asks for your location: Allow. VPN off.



Take a picture · Proof Type

Student ID, transcript or enrolment letter · allow the camera

Please upload proof of your academic status.

Snap a picture of your qualifying proof of current academic status using your HD webcam or smartphone camera.

Use your computer's camera to take a picture now.

Take a picture

Proof Type *

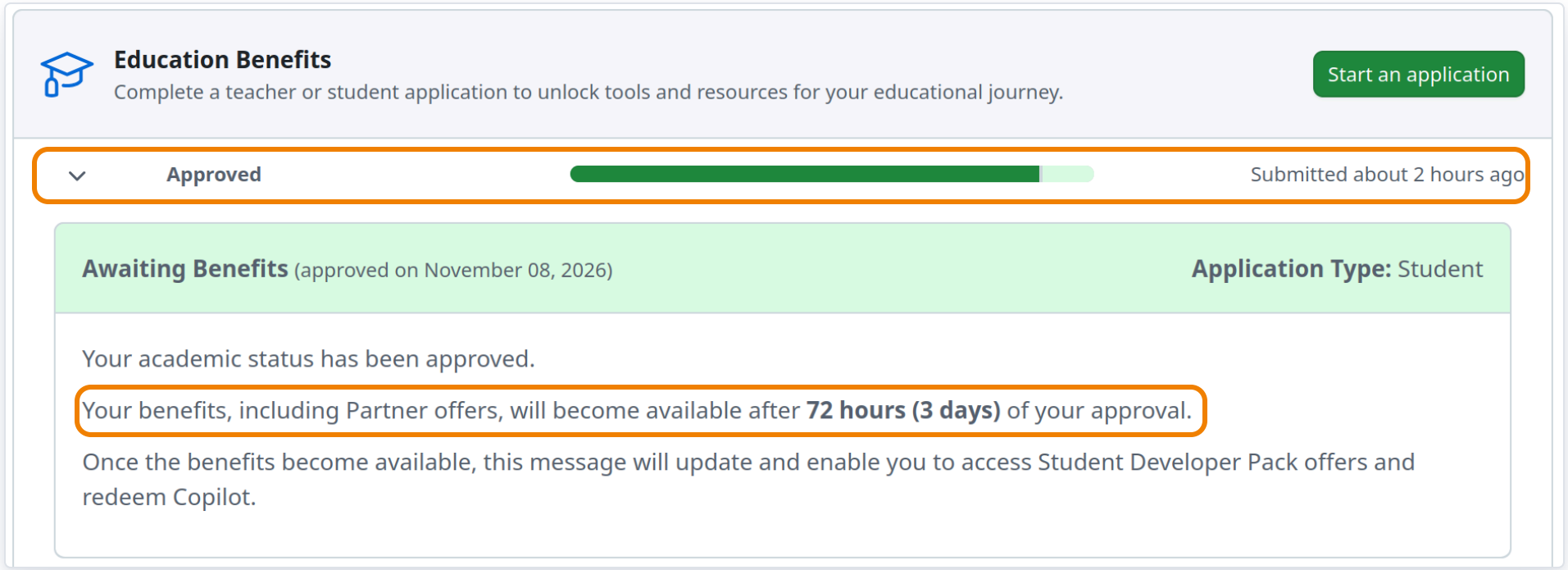
--Select proof type--

Image: GitHub Docs, CC BY 4.0

DONE WHEN STUDENT PACKAGE · GITHUB · 13 OF 13

Then: approved, benefits in 72 hours

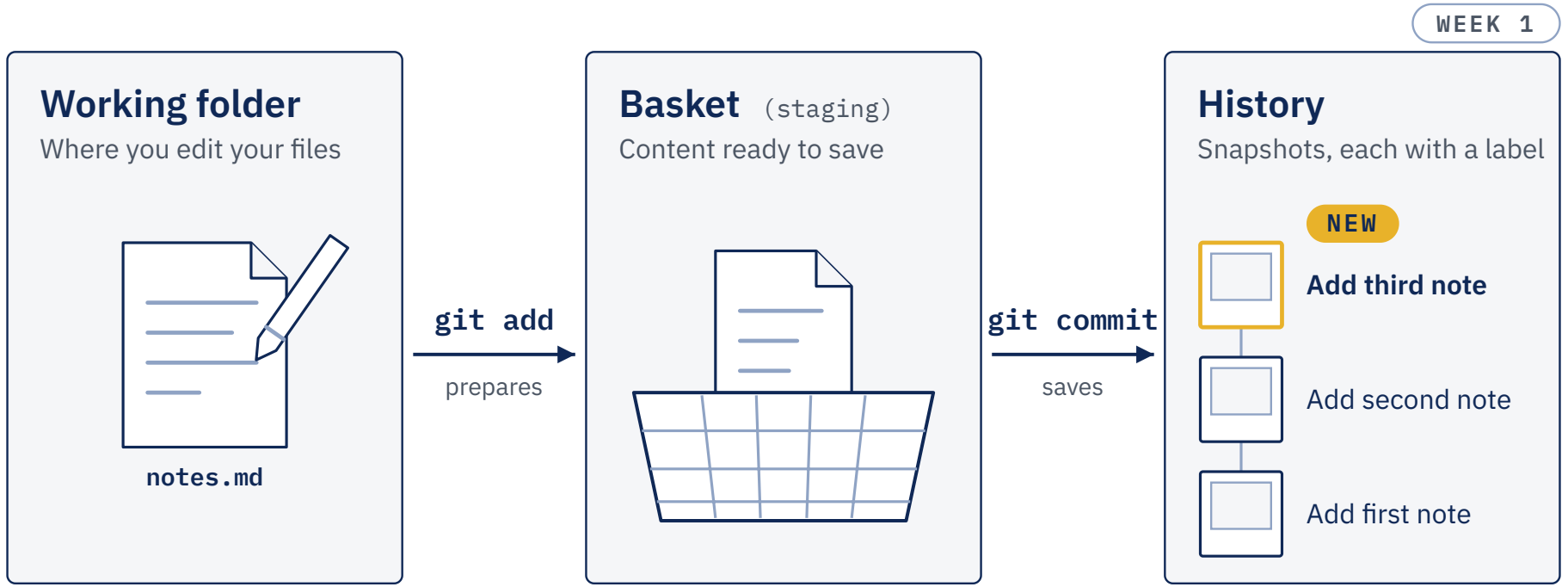
Here: approved 2 hours after submitting · benefits and Copilot 3 days later



WATCH MEET GITHUB · GITHUB · 1 OF 6

Last week: stage, then commit

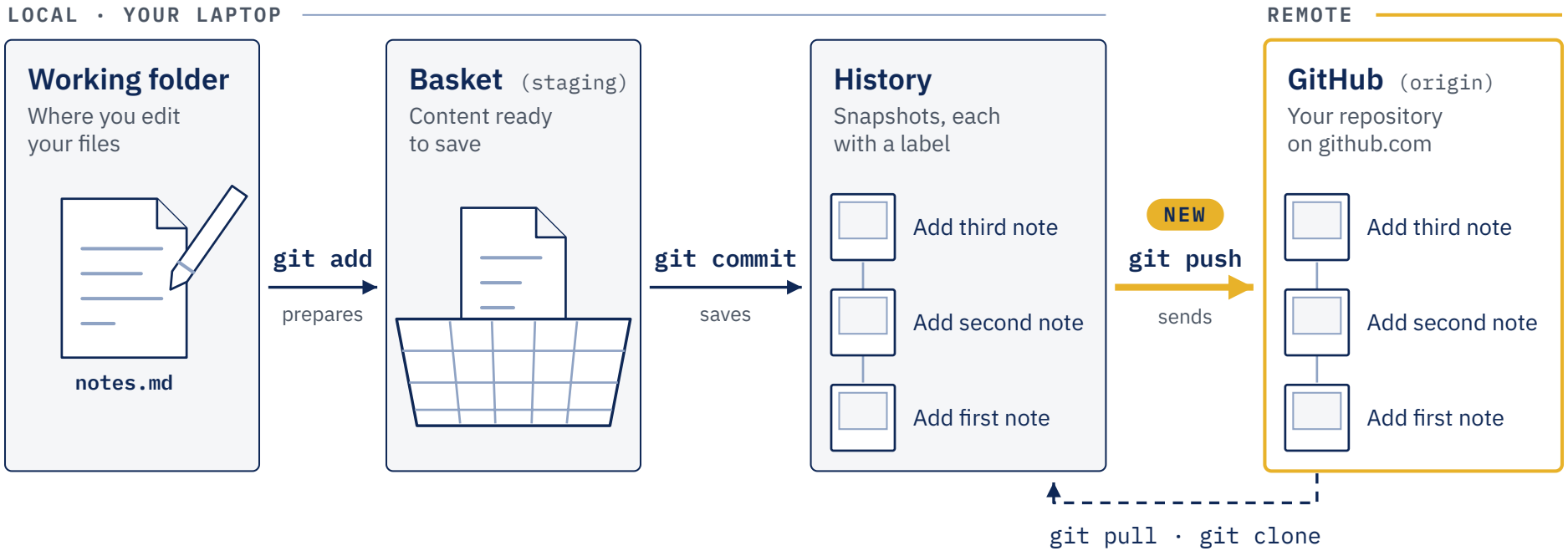
git add prepares. git commit saves. So far, all of it stays on your laptop.



WATCH MEET GITHUB · GITHUB · 2 OF 6

This week: stage, commit, and now push

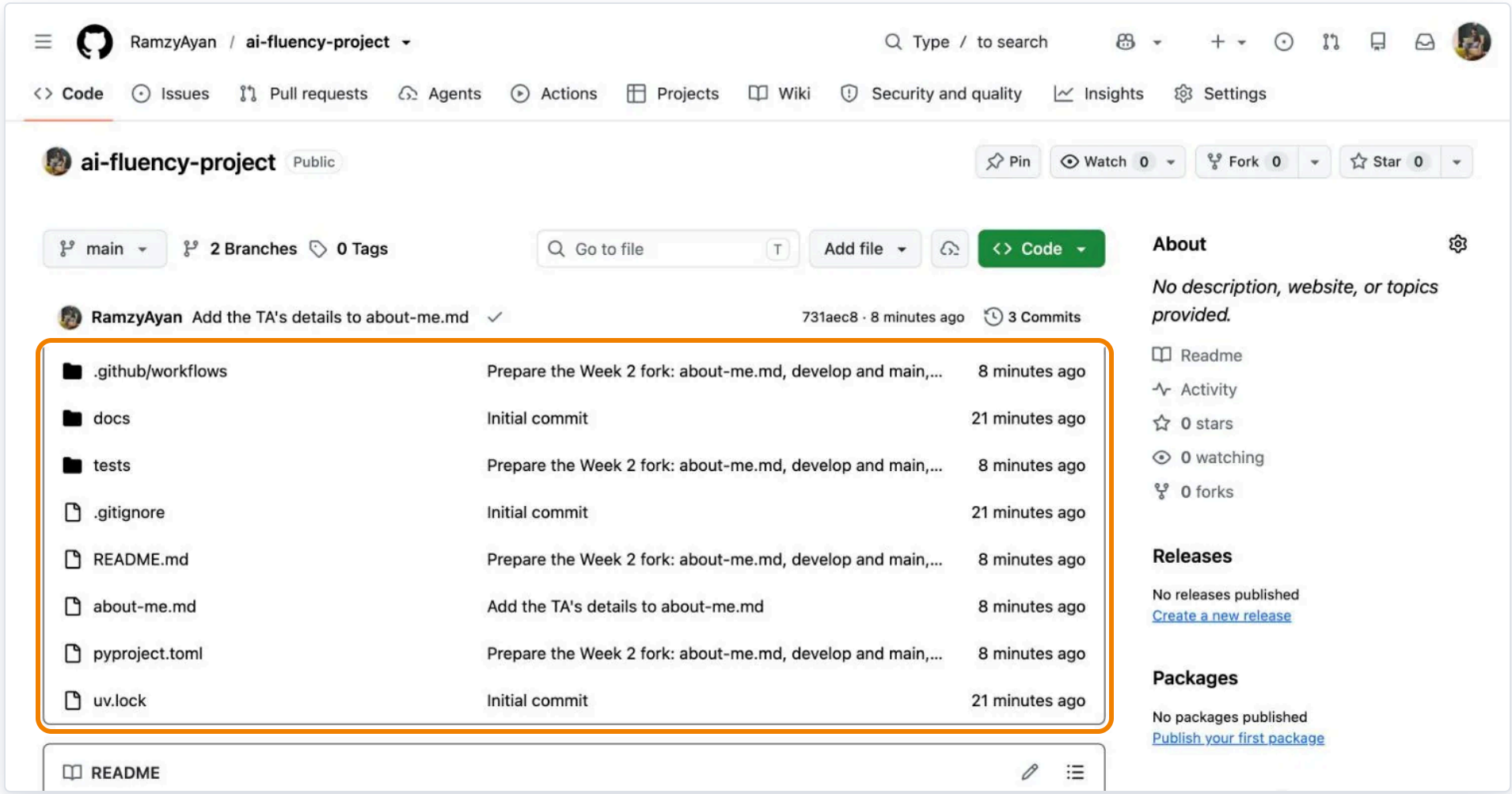
Push sends commits only: uncommitted work stays on your laptop.



WATCH MEET GITHUB · GITHUB · 3 OF 6

A repository

A project's files and their history, on GitHub



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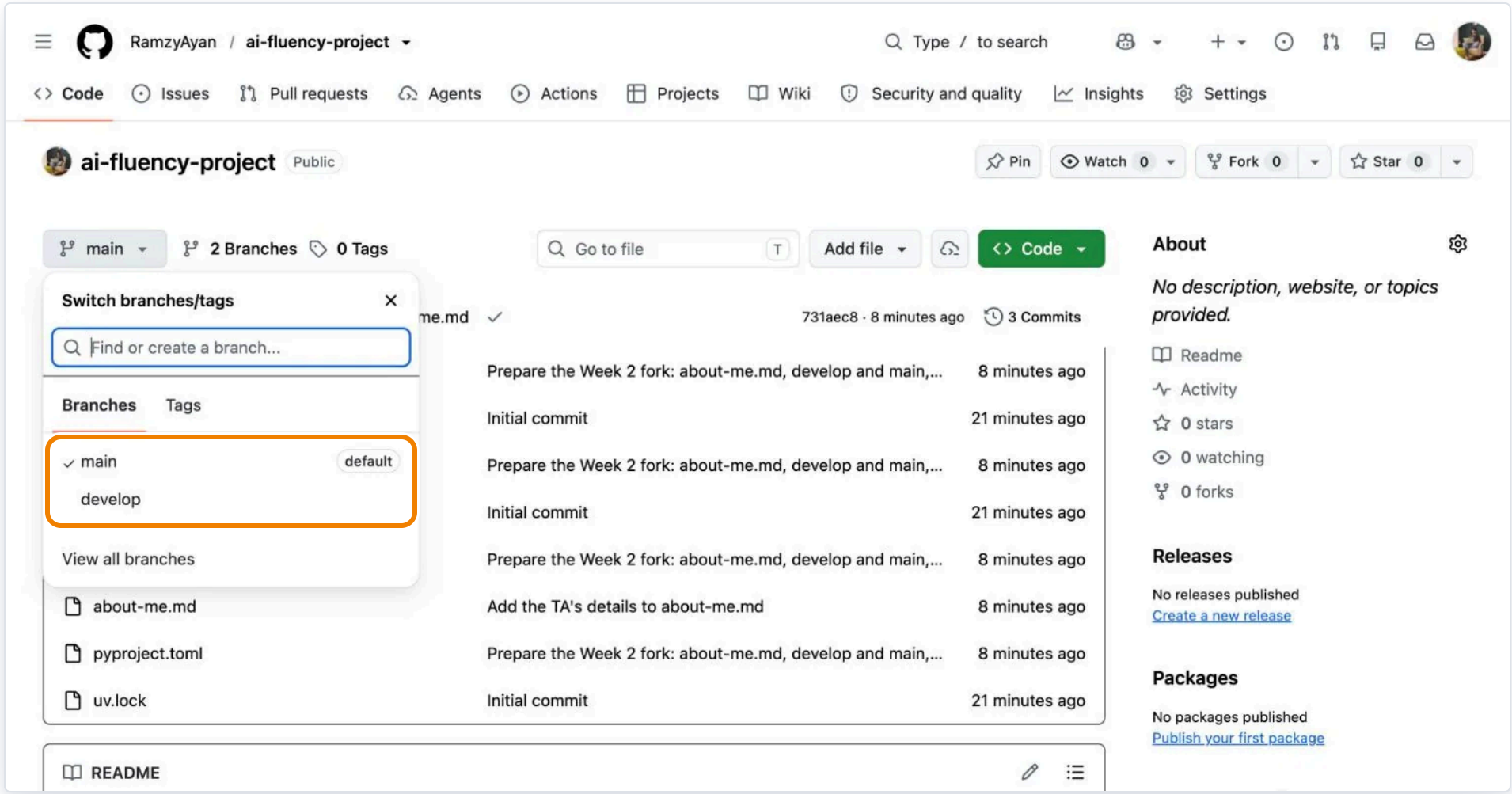
Branches: lines of work

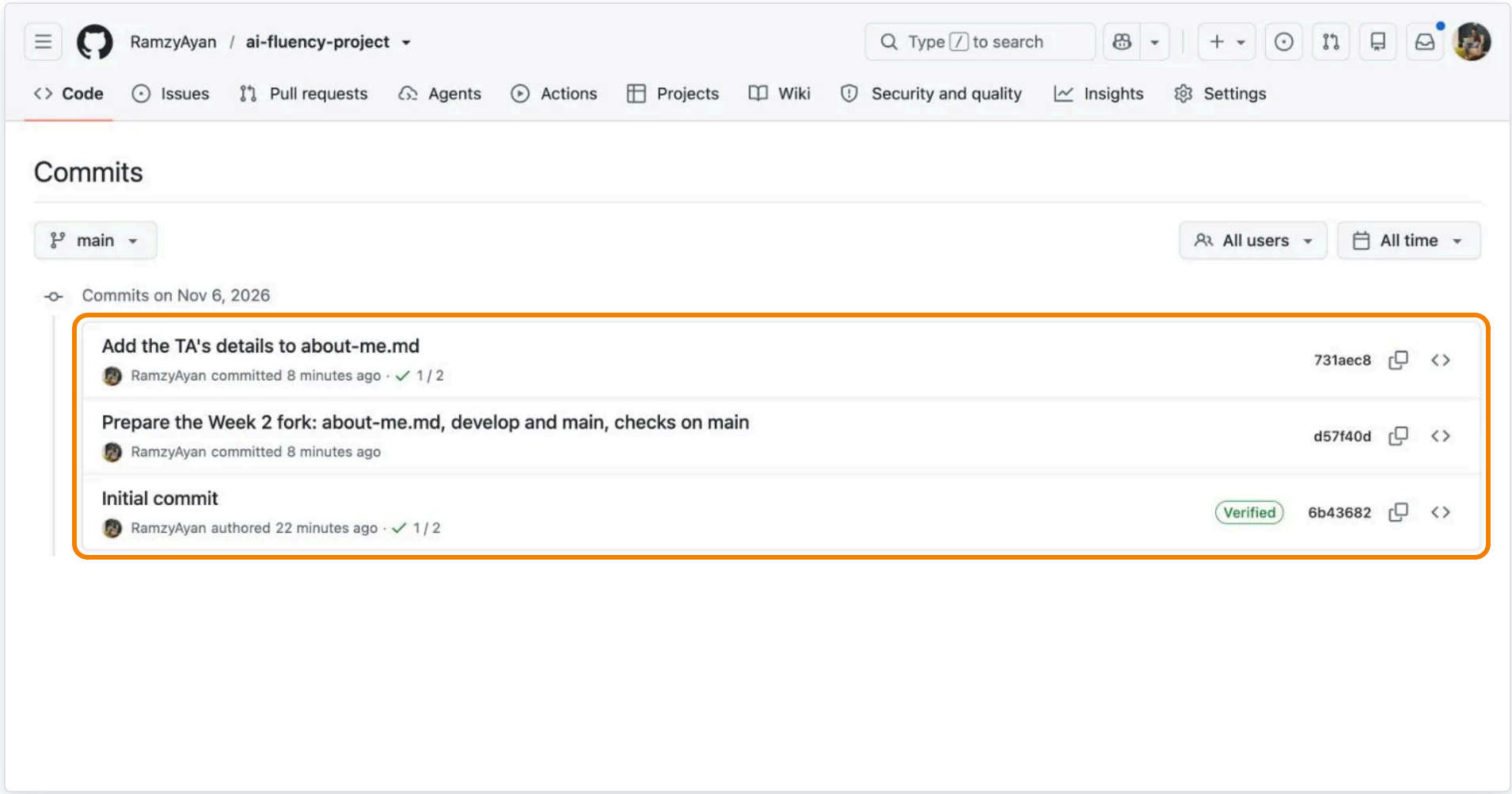
main and develop



WATCH MEET GITHUB · GITHUB · 5 OF 6

Commits: the history

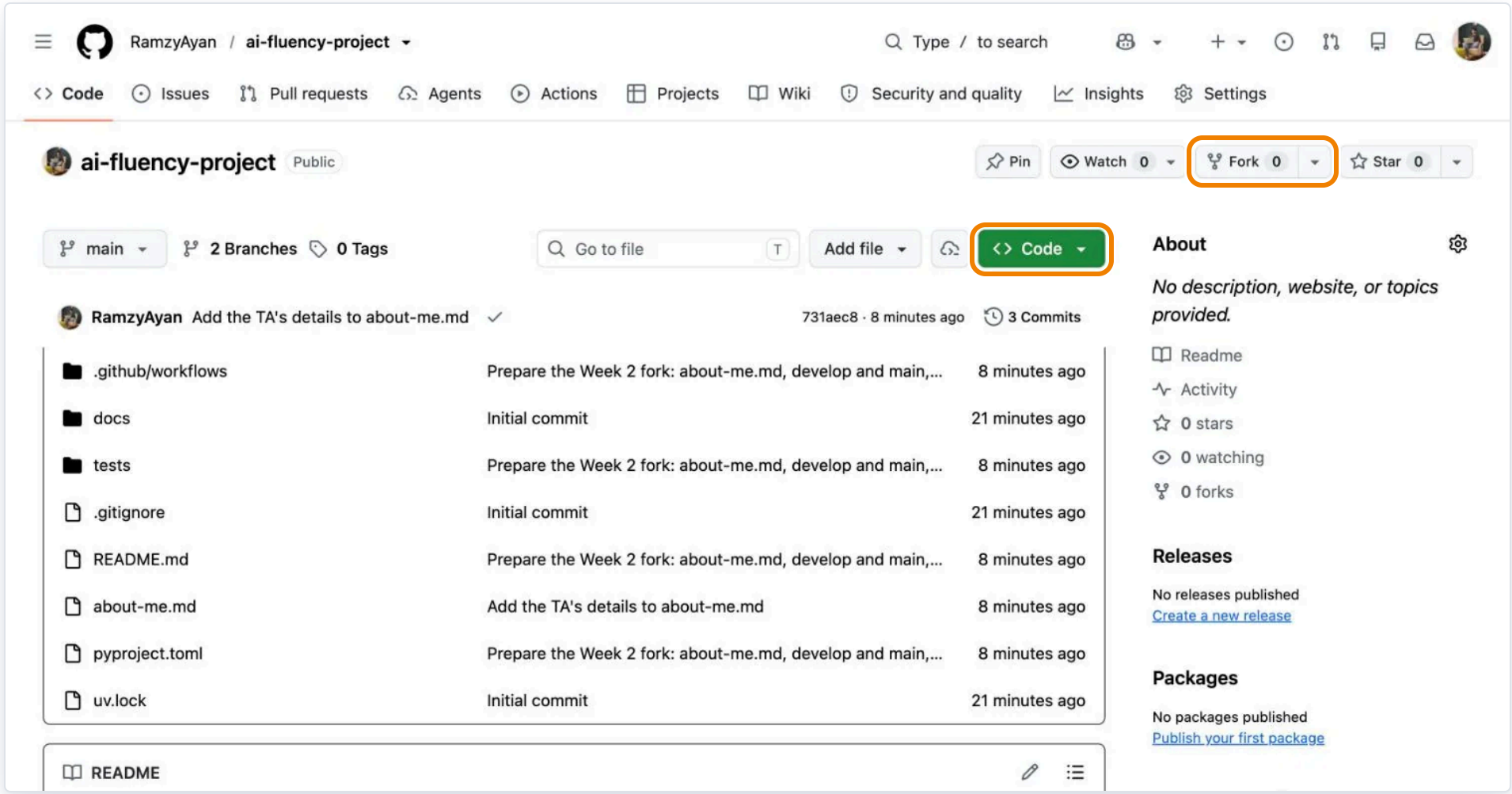
Every saved snapshot, with its message



WATCH MEET GITHUB · GITHUB · 6 OF 6

Fork and Code

Fork: your own copy · Code: its address



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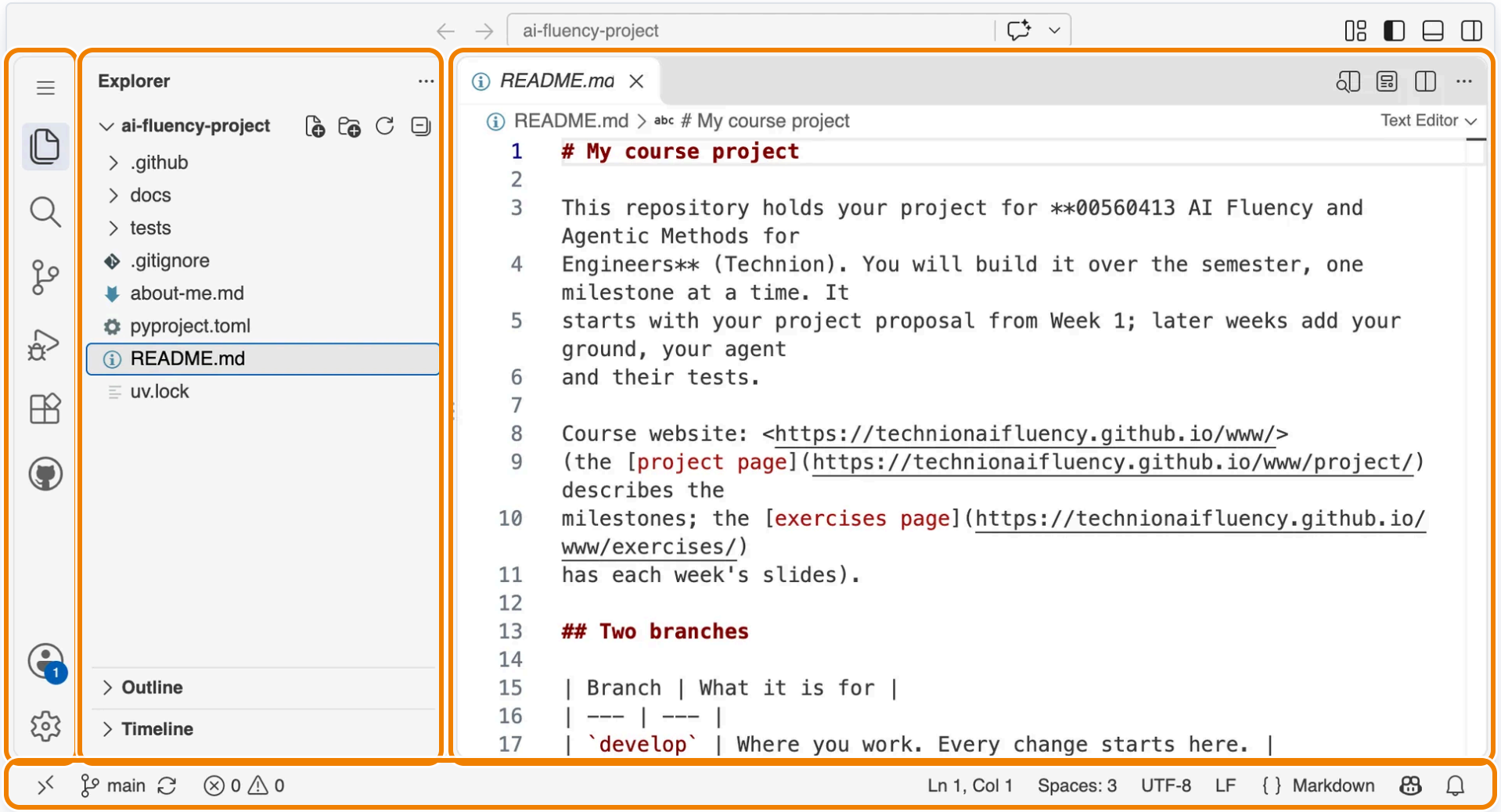
WATCH MEET VS CODE · VS CODE · 1 OF 6

The window

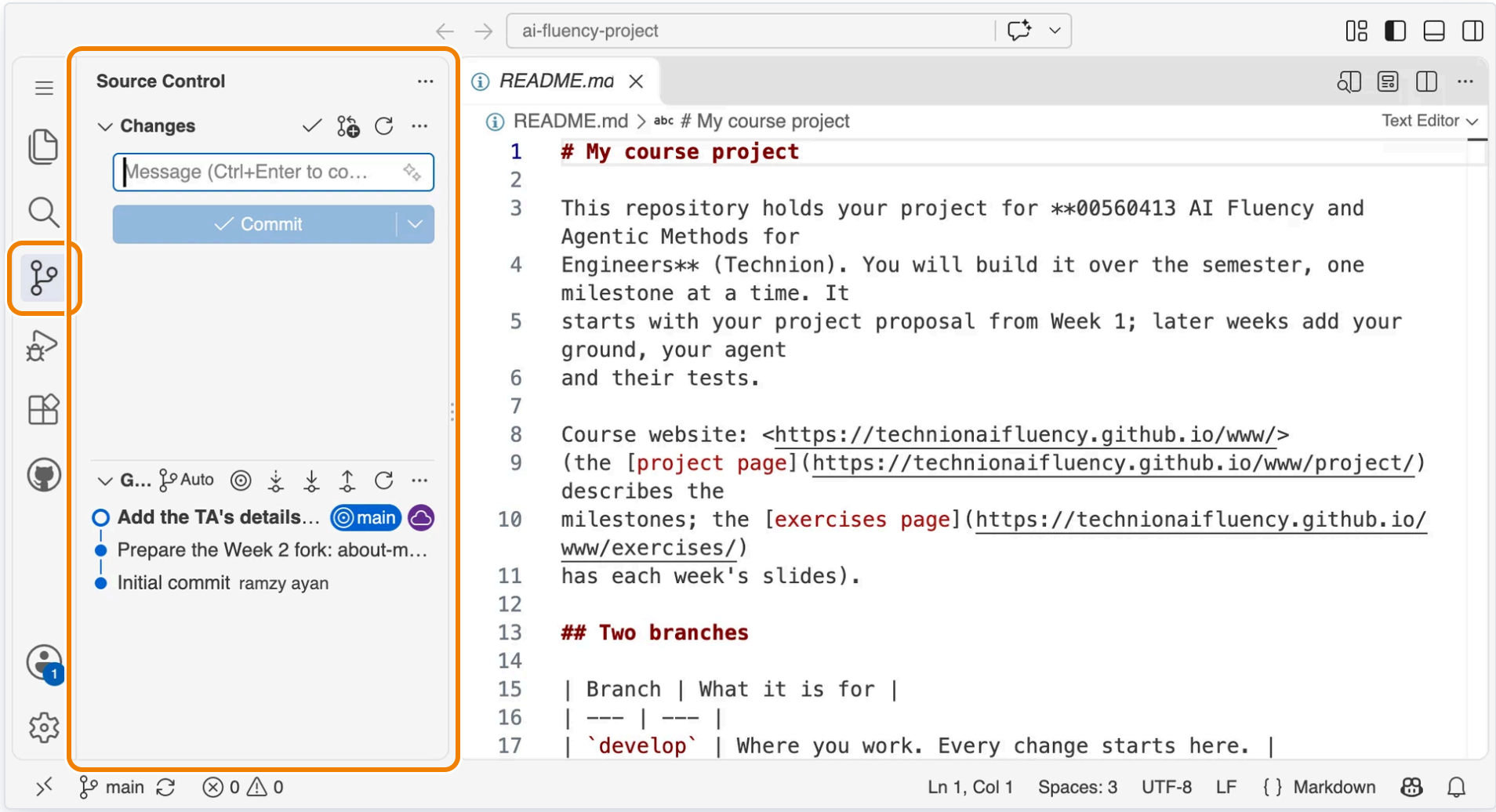
Activity bar · side bar · editor · status bar · yours may be dark



WATCH MEET VS CODE · VS CODE · 2 OF 6

Source Control

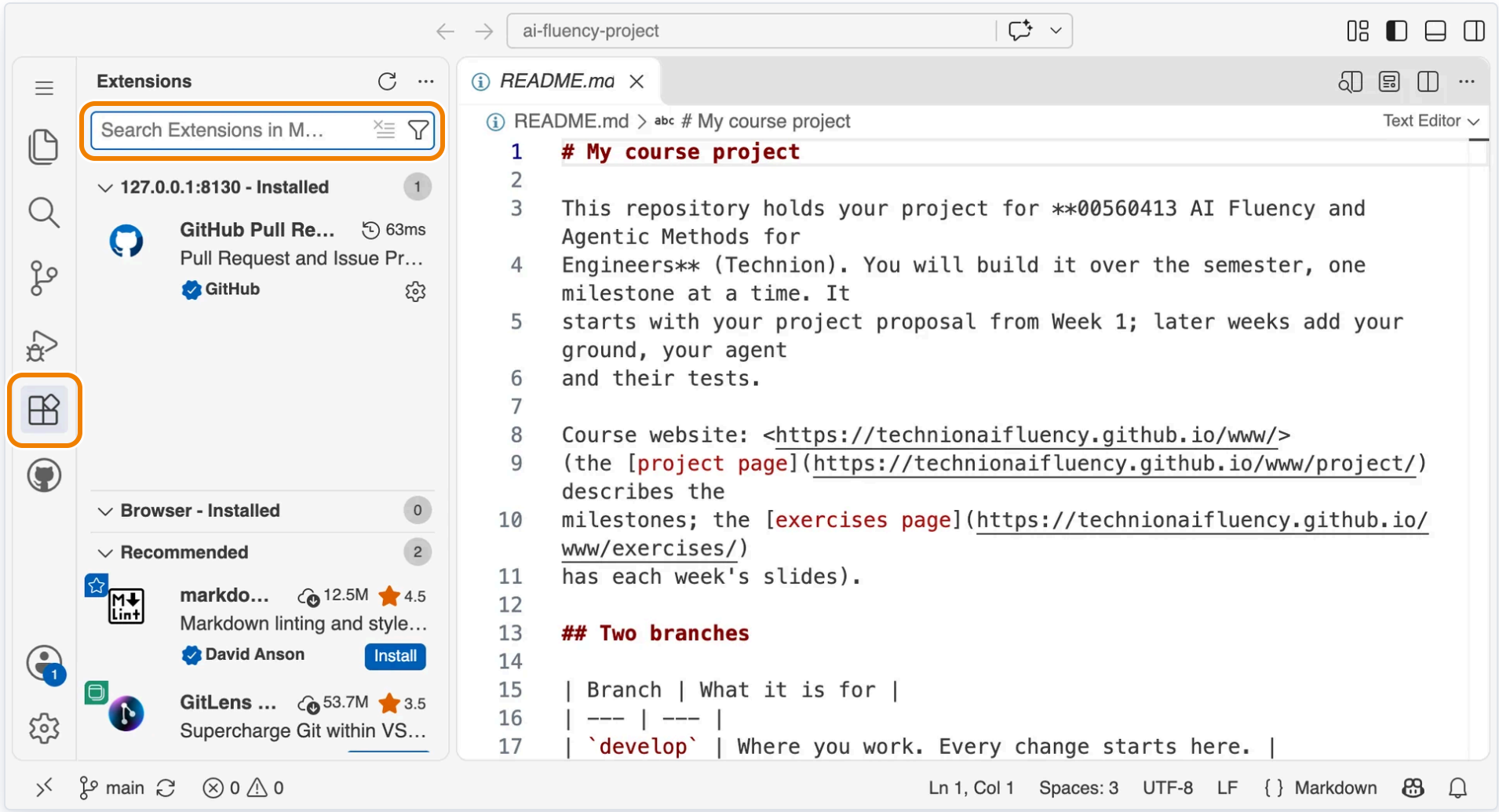
Git with buttons, and the Graph



WATCH MEET VS CODE · VS CODE · 3 OF 6

Extensions

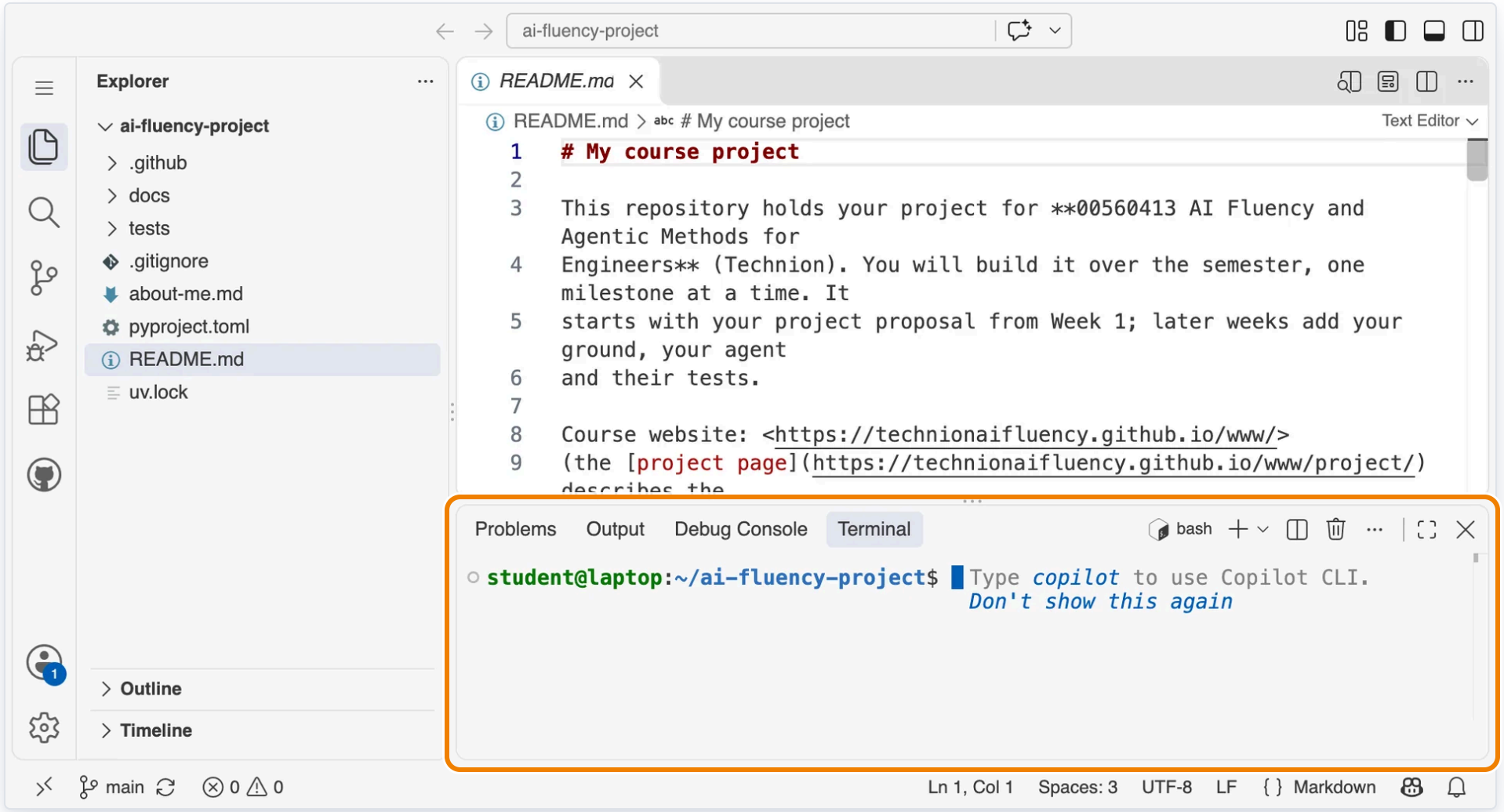
Add features: WSL, GitHub



WATCH MEET VS CODE · VS CODE · 4 OF 6

Terminal

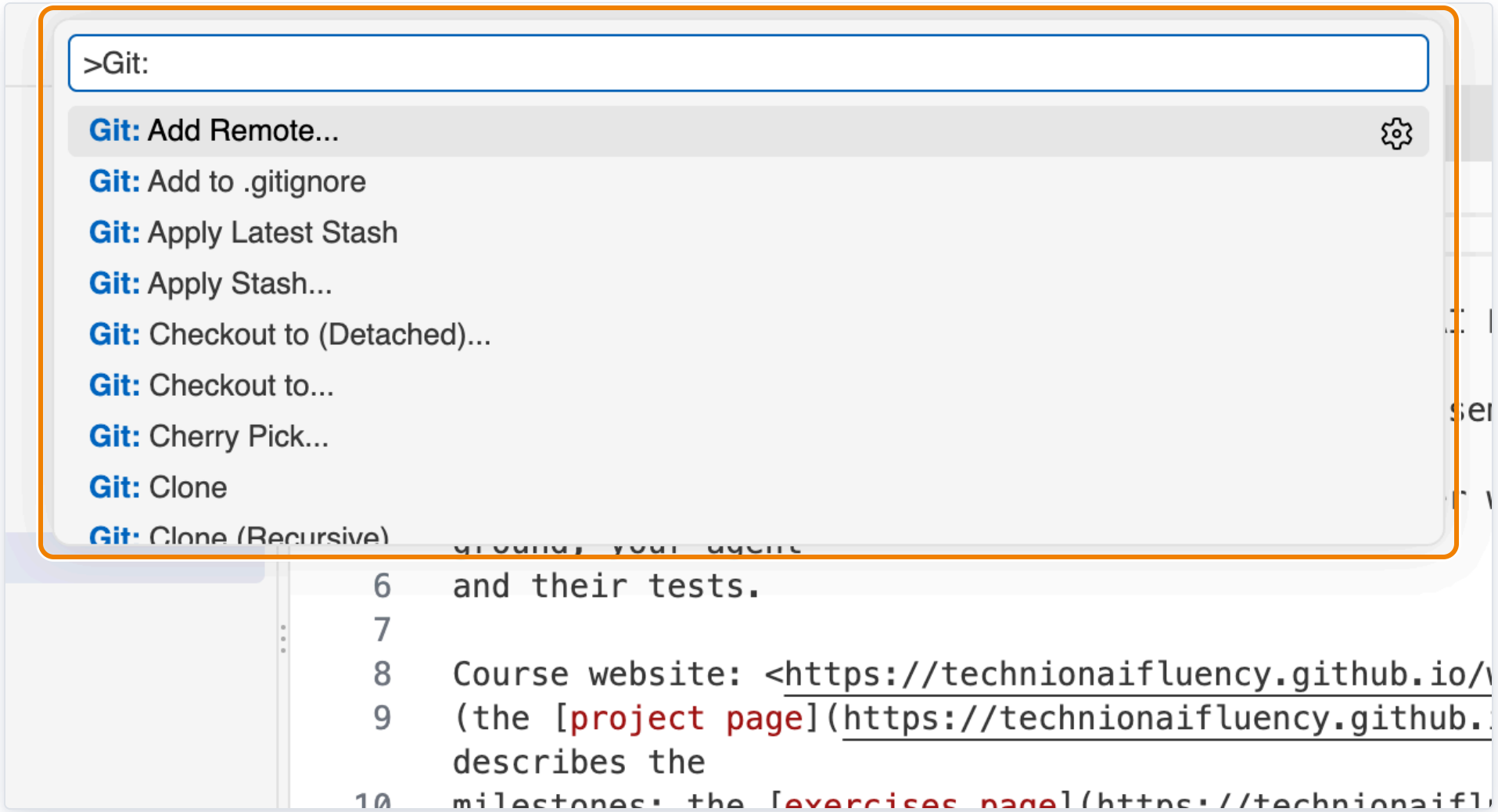
View → Terminal · the same shell, inside VS Code



WATCH MEET VS CODE · VS CODE · 5 OF 6

Command Palette

Ctrl+Shift+P (Mac: Cmd+Shift+P) · every command by name



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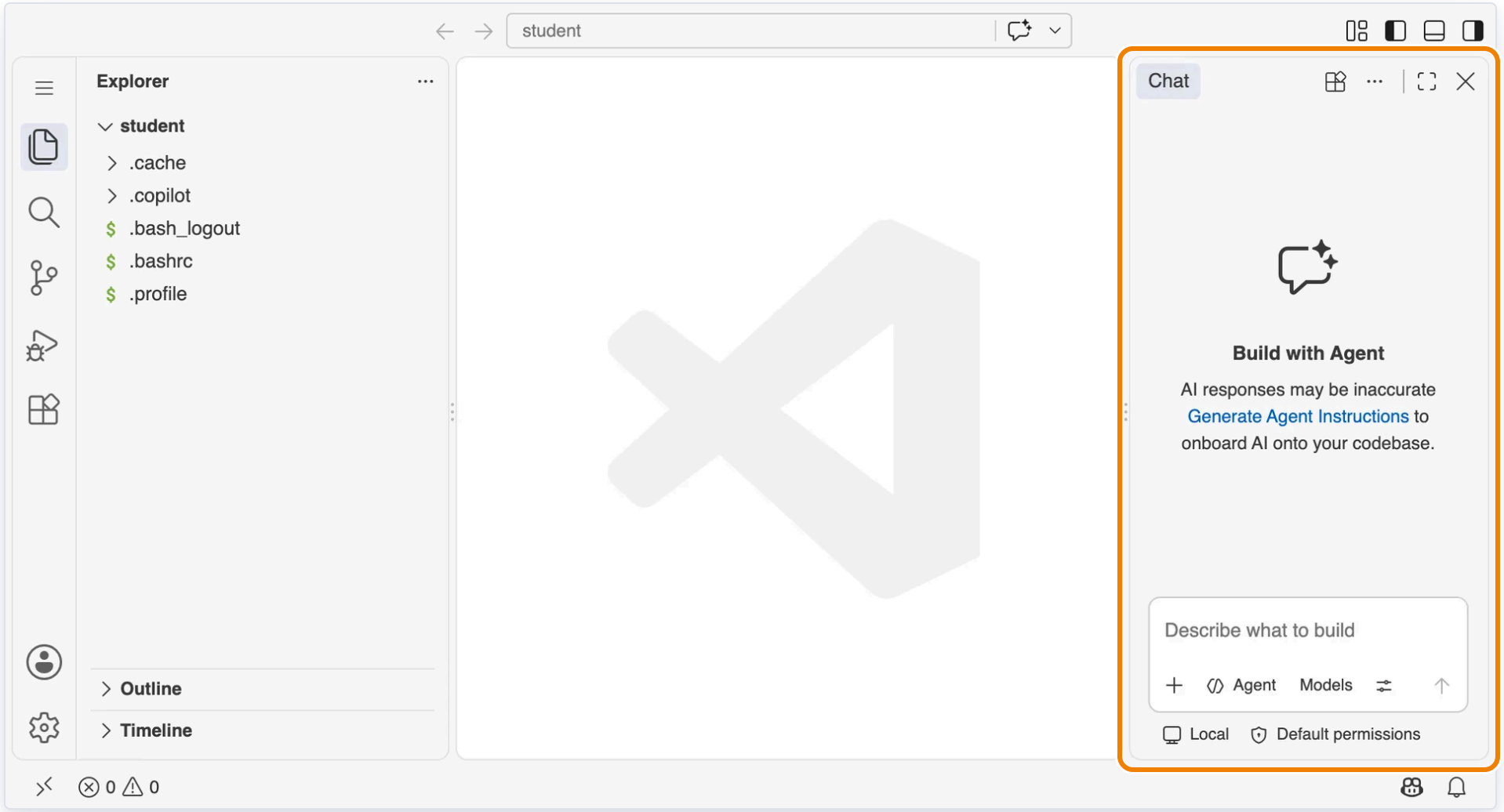
WATCH MEET VS CODE · VS CODE · 6 OF 6

Copilot lives here

Not today: close it with the X. It may open again later.



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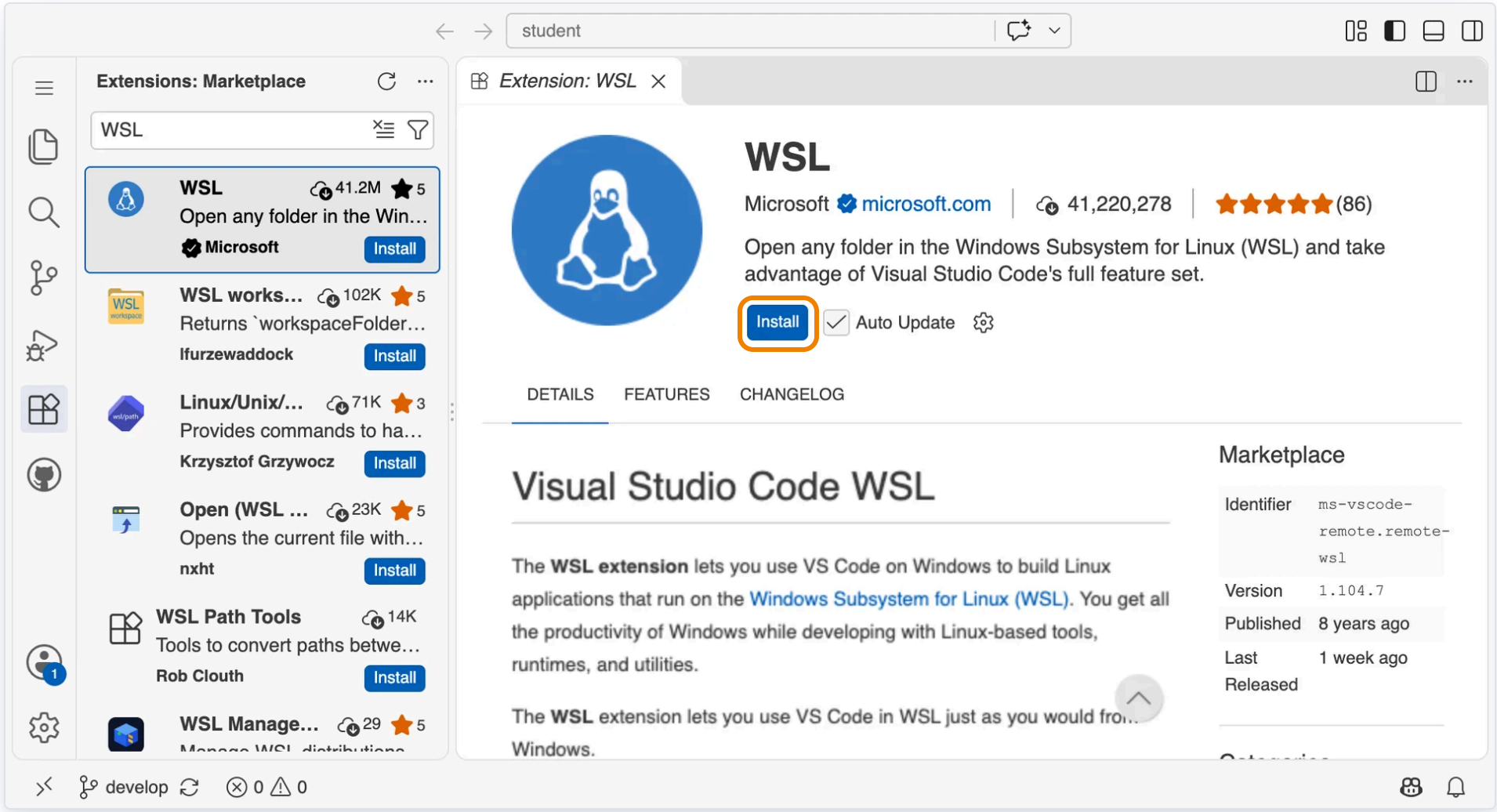
Windows, once: the WSL extension

VS Code → Extensions → WSL (Microsoft) → Install



Open a terminal

Windows: Start → Terminal · Mac: Terminal, then skip to code .

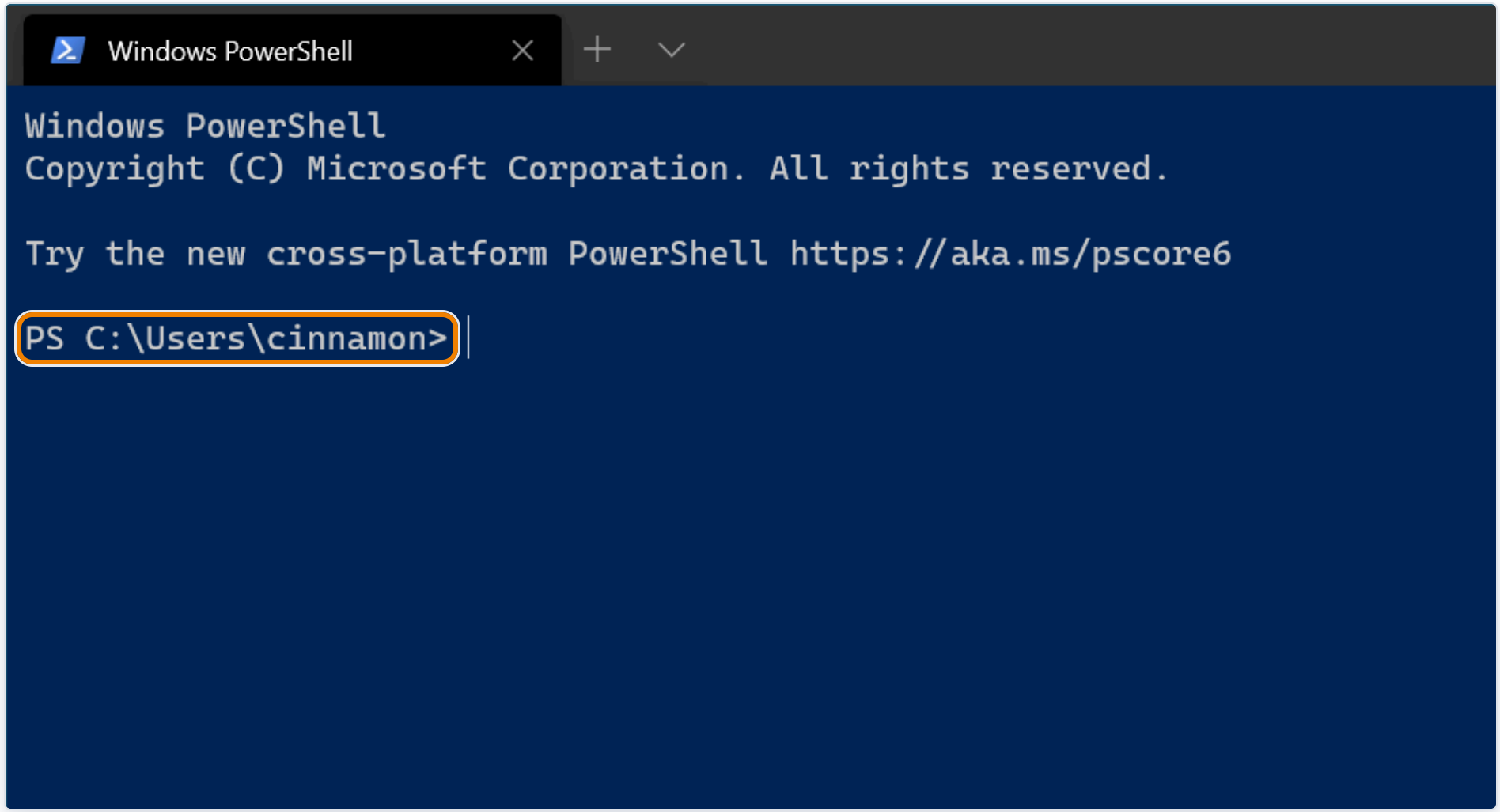


Image: Windows Terminal docs, Microsoft, CC BY 4.0 (cropped)

Type: wsl

then Enter · now you are in Ubuntu

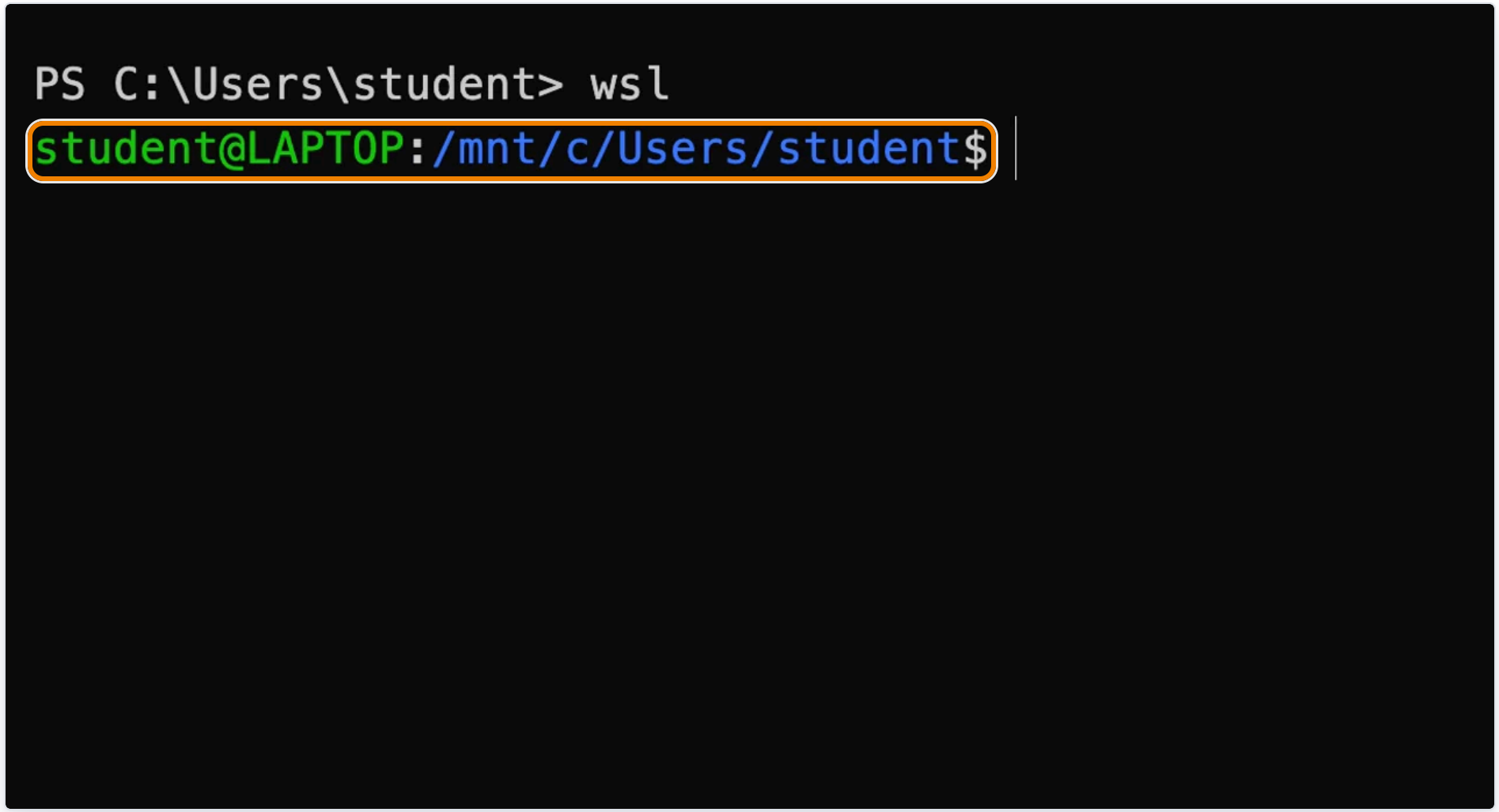


Illustration: what your terminal shows

YOUR TURN WSL → CODE . · WSL · 4 OF 10

Type: cd ~

then Enter · your Linux home

Illustration: what your terminal shows

YOUR TURN WSL → CODE . · WSL · 5 OF 10

Once: tell Git who you are

Two lines · your name · your GitHub noreply e-mail (Settings → Emails)

```
PS C:\Users\student> wsl
student@LAPTOP:/mnt/c/Users/student$ cd ~
student@LAPTOP:~$ git config --global user.name "Ramzy"
student@LAPTOP:~$ git config --global user.email "12345+ramzy@users.noreply.github.com"
student@LAPTOP:~$
```

Illustration: what your terminal shows

YOUR TURN WSL → CODE . · WSL · 6 OF 10

Type: code .

code, a space, a dot, then Enter

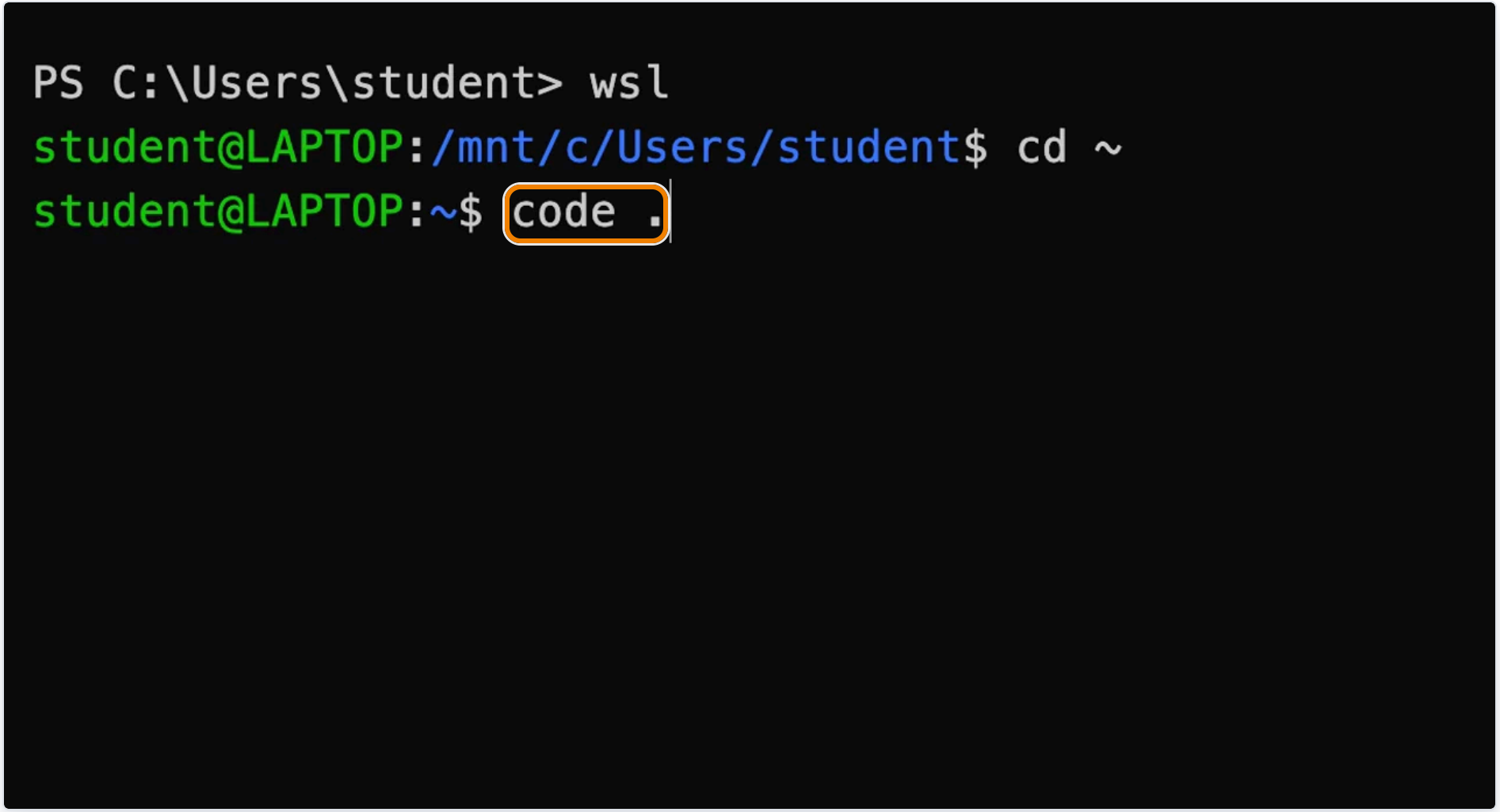


Illustration: what your terminal shows

First time: it installs

Wait: a new VS Code window opens

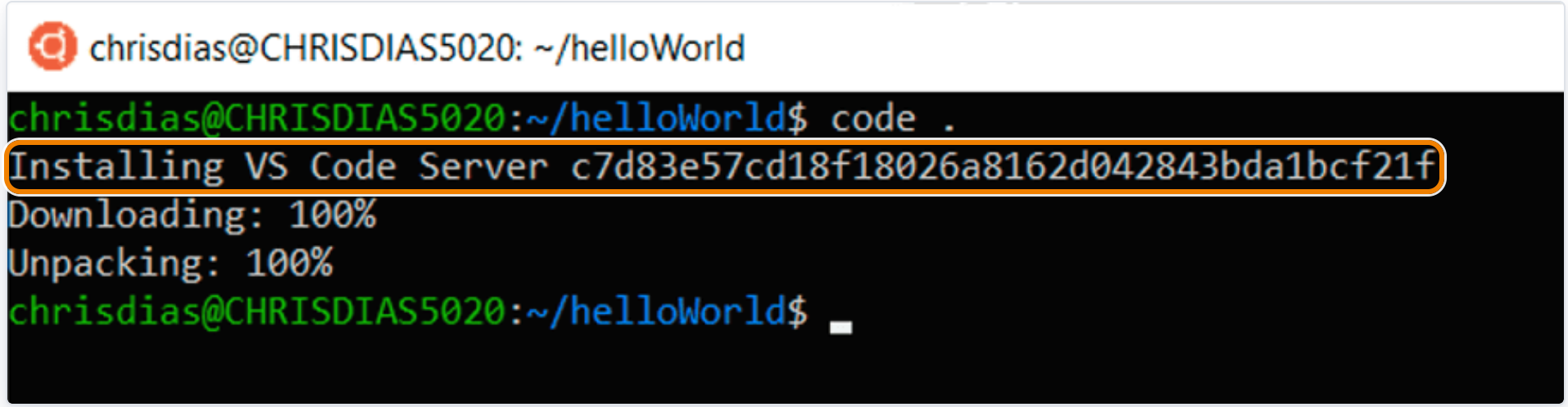


Image: VS Code docs, Microsoft, CC BY 3.0 US

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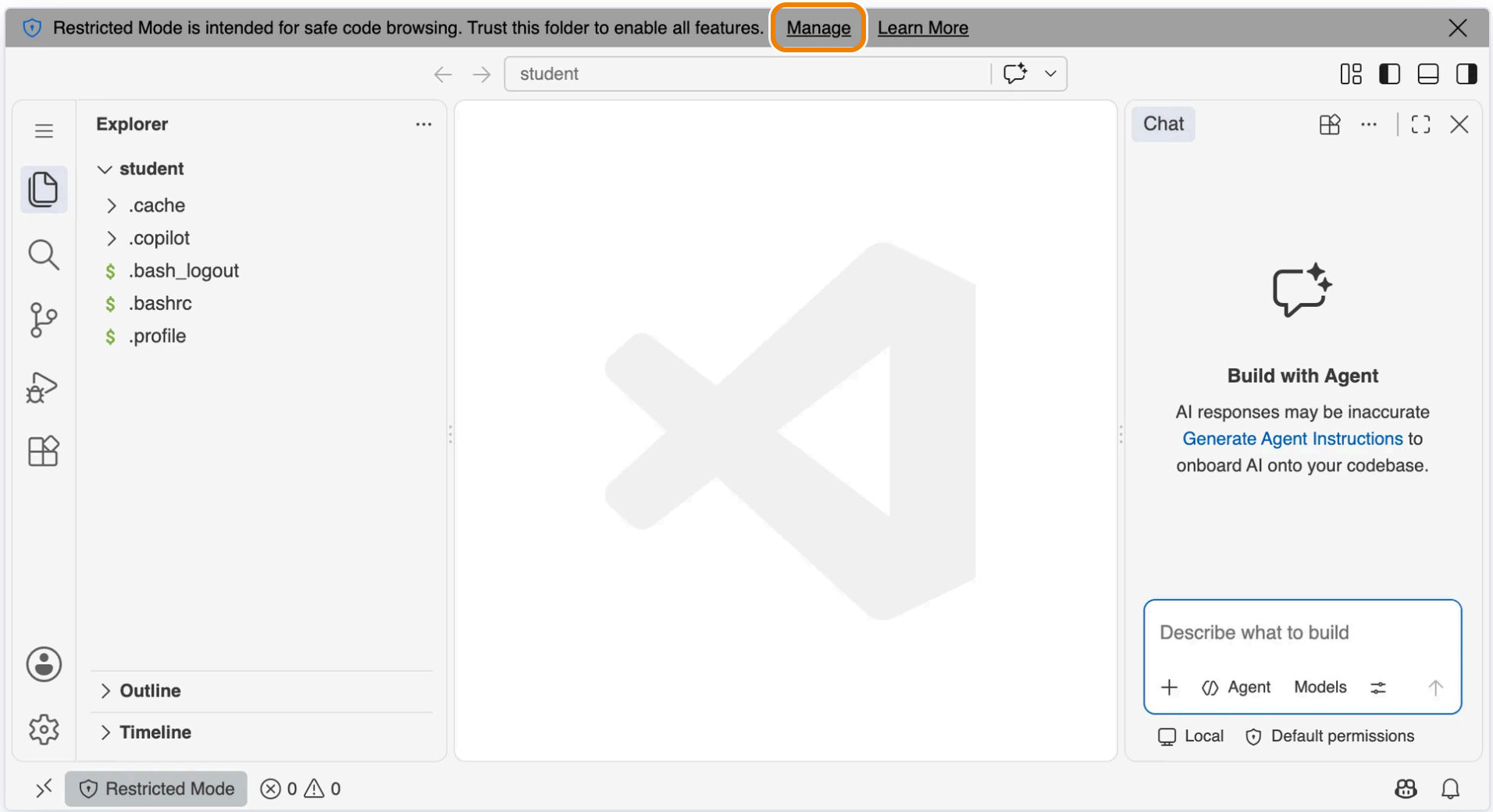
Restricted Mode? Manage



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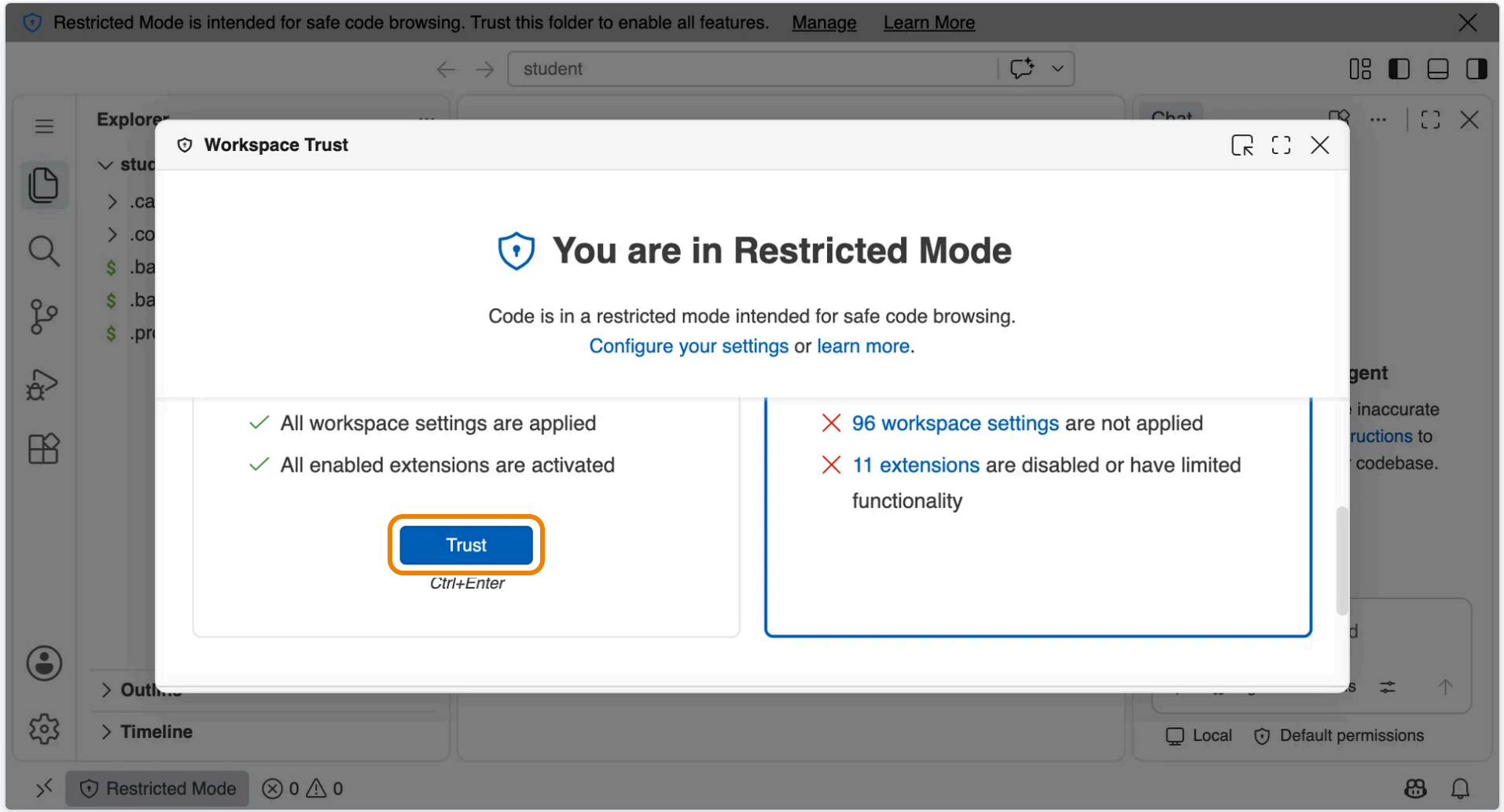
Trust


Restricted Mode is intended for safe code browsing. Trust this folder to enable all features. Manage Learn More
student
Workspace Trust
You are in Restricted Mode
Code is in a restricted mode intended for safe code browsing.
Configure your settings or learn more.
All workspace settings are applied
All enabled extensions are activated
Trust
Ctrl+Enter
96 workspace settings are not applied
11 extensions are disabled or have limited functionality
Timeline
Local
Default permissions
Restricted Mode
0 0

DONE WHEN WSL → CODE . · WSL · 10 OF 10

Bottom left: WSL: Ubuntu

Mac: no indicator, that is fine

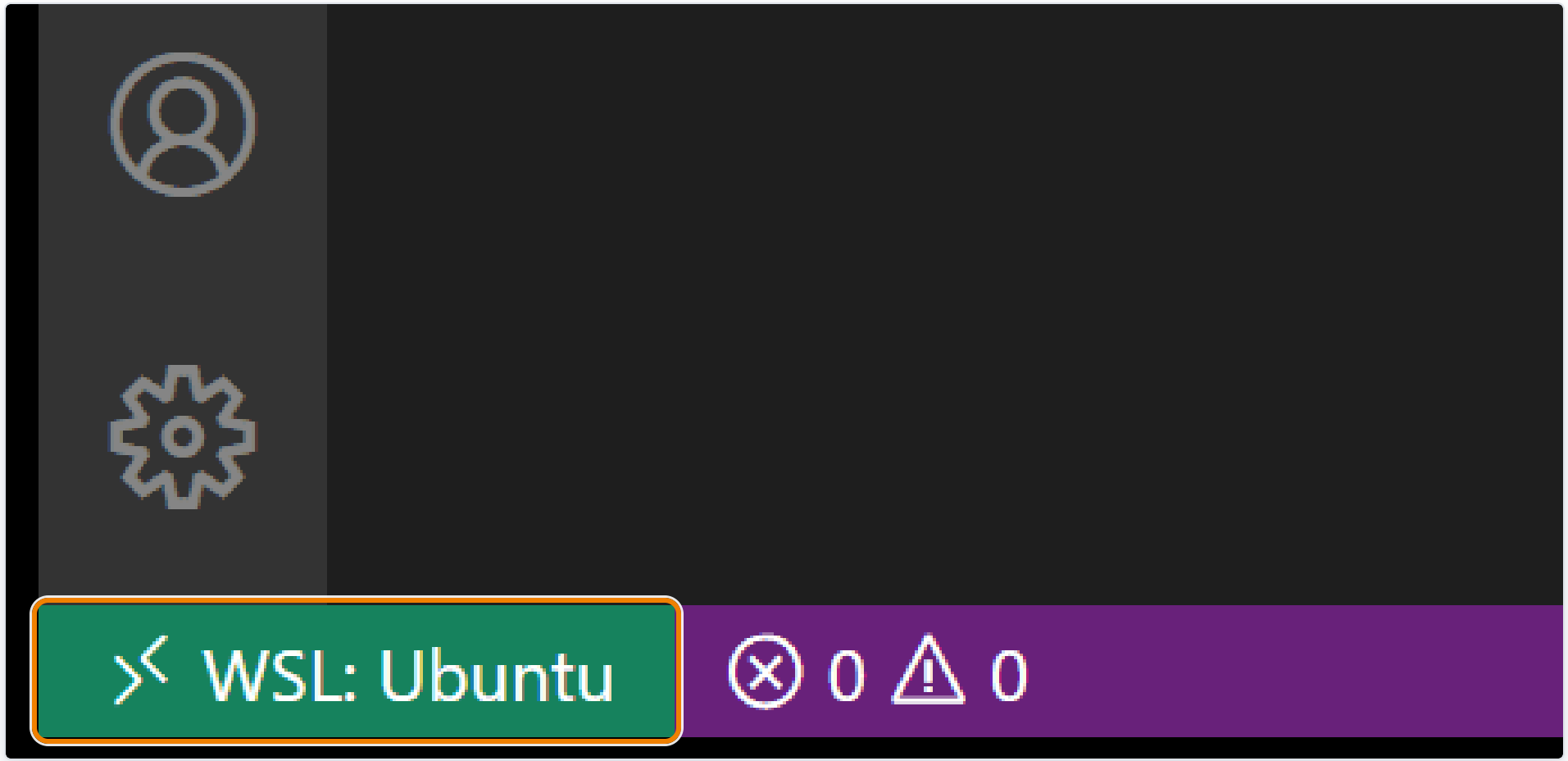
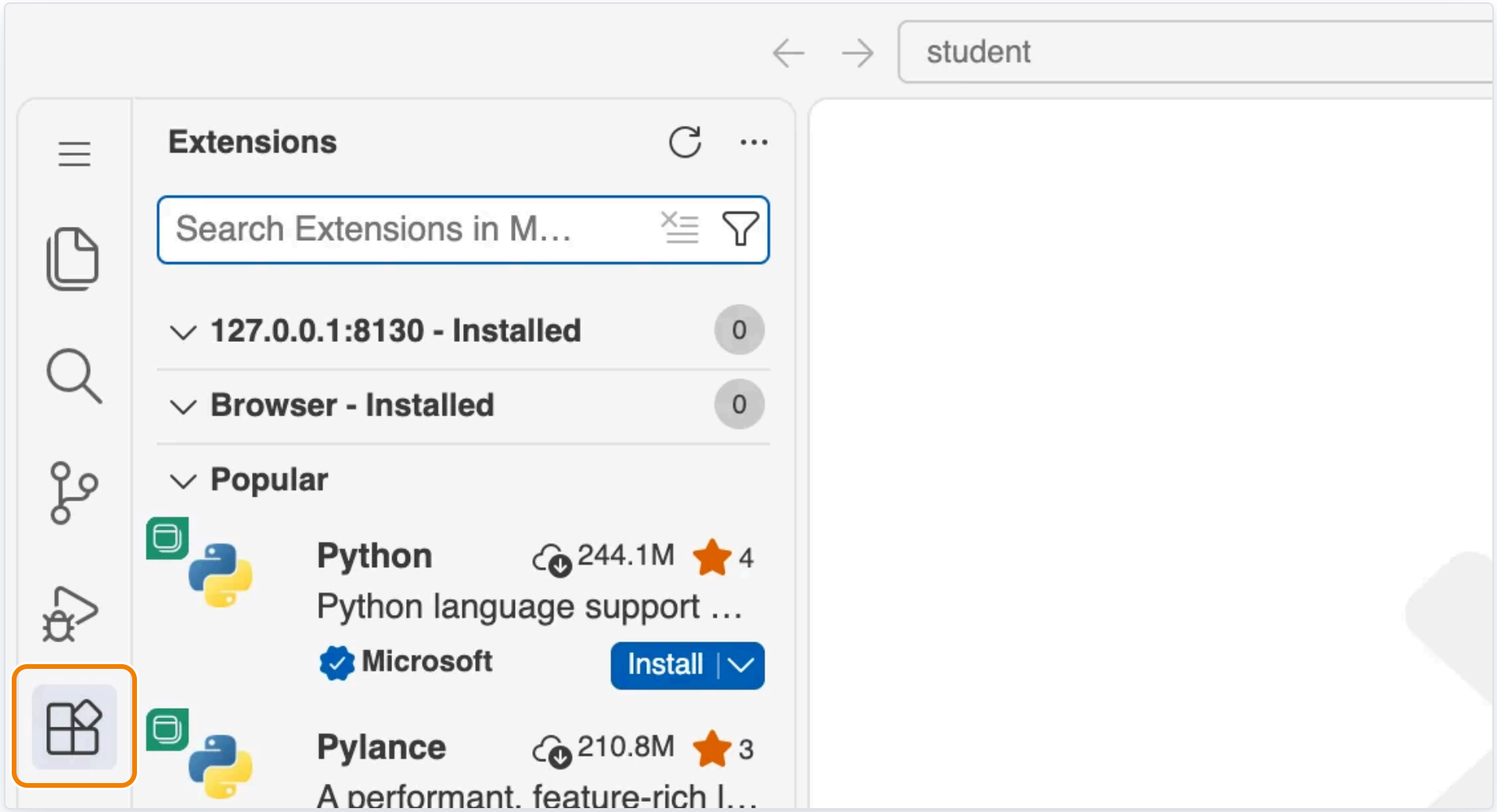


Image: VS Code docs, Microsoft, CC BY 3.0 US

YOUR TURN SIGN IN TO GITHUB · VS CODE · 1 OF 6

Extensions

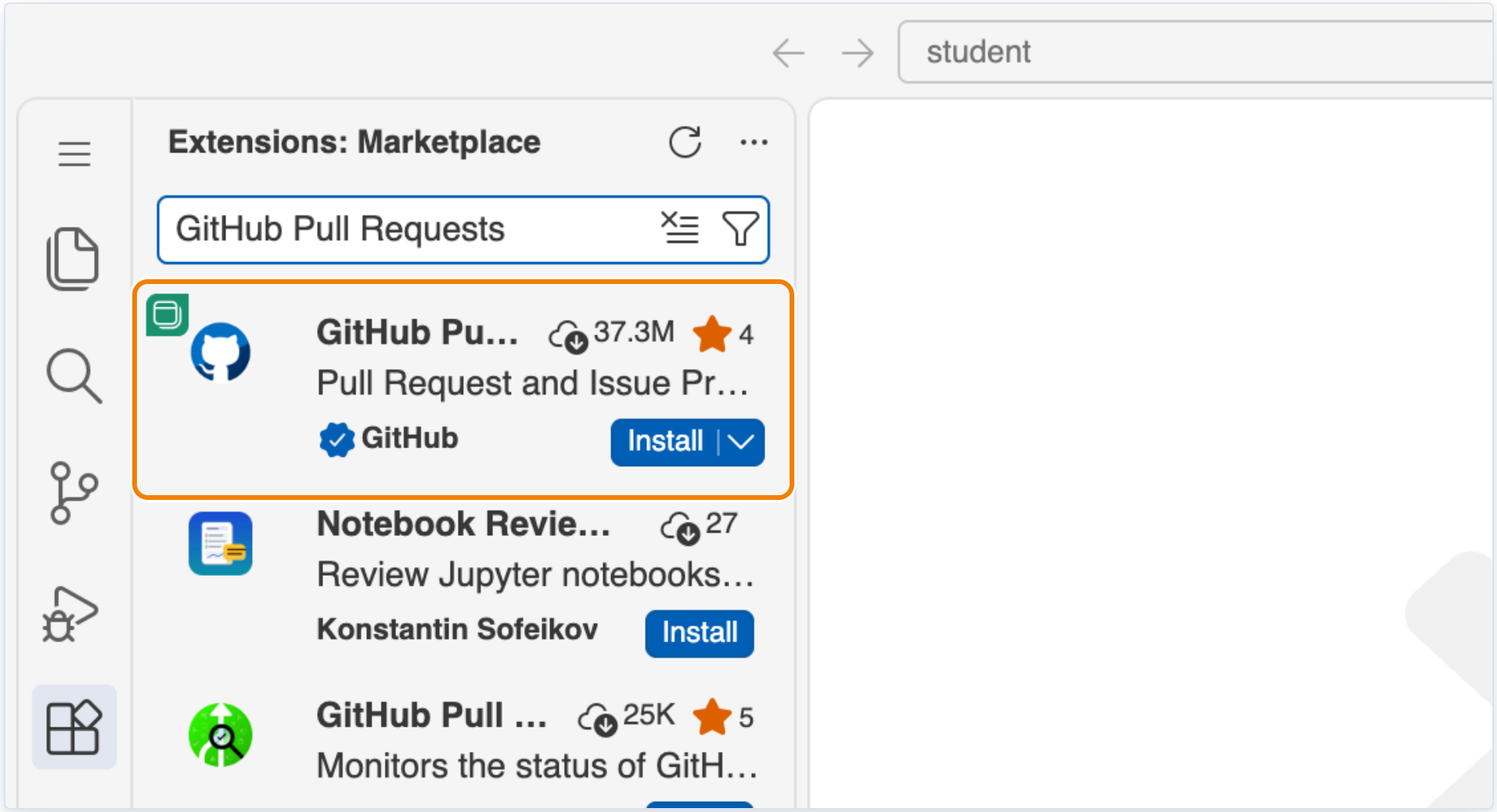
Ctrl+Shift+X (Mac: Cmd+Shift+X)



YOUR TURN SIGN IN TO GITHUB · VS CODE · 2 OF 6

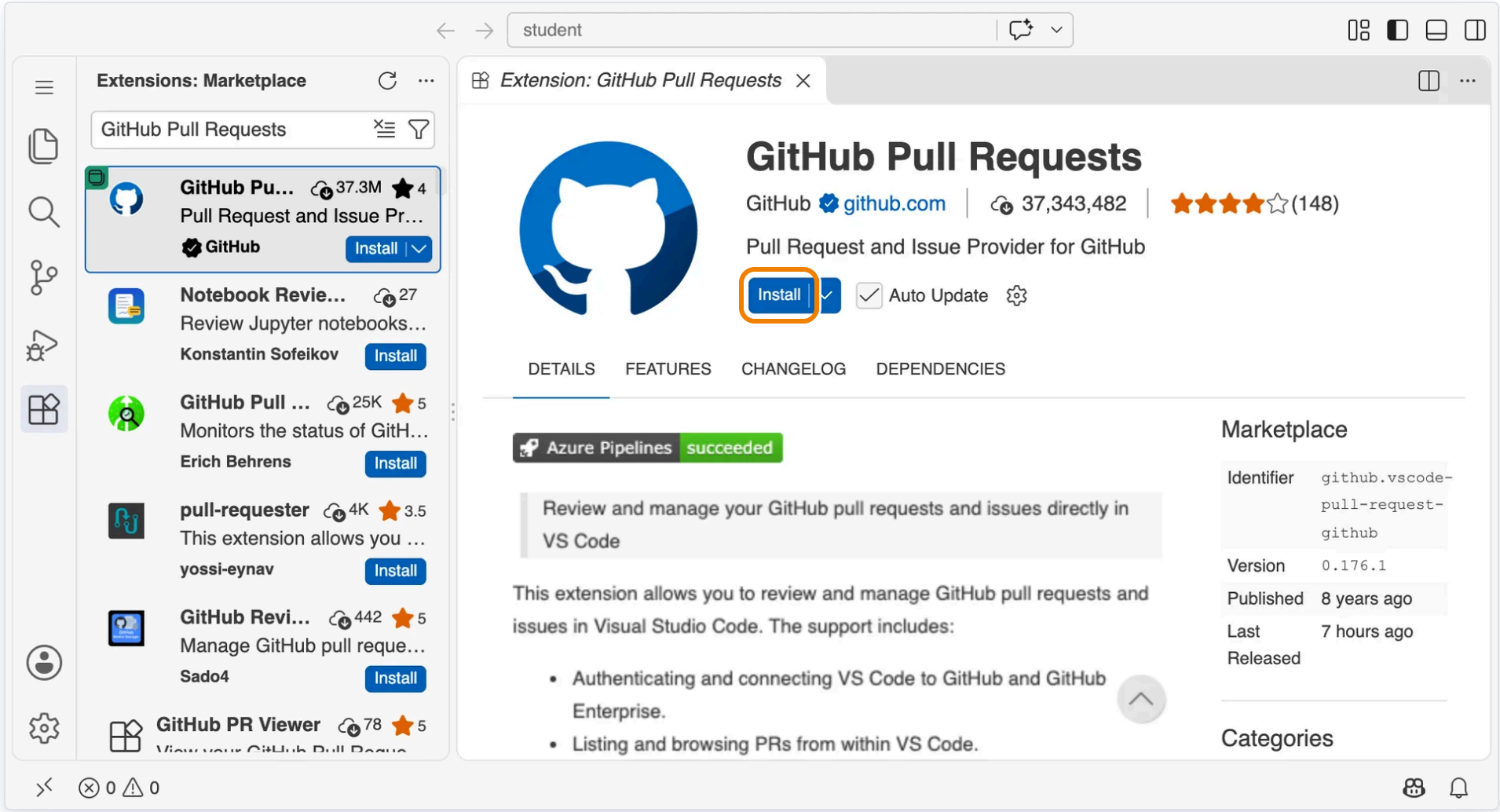
Search: GitHub Pull Requests

Only for the sign-in · publisher: GitHub, with the blue tick



YOUR TURN SIGN IN TO GITHUB · VS CODE · 3 OF 6

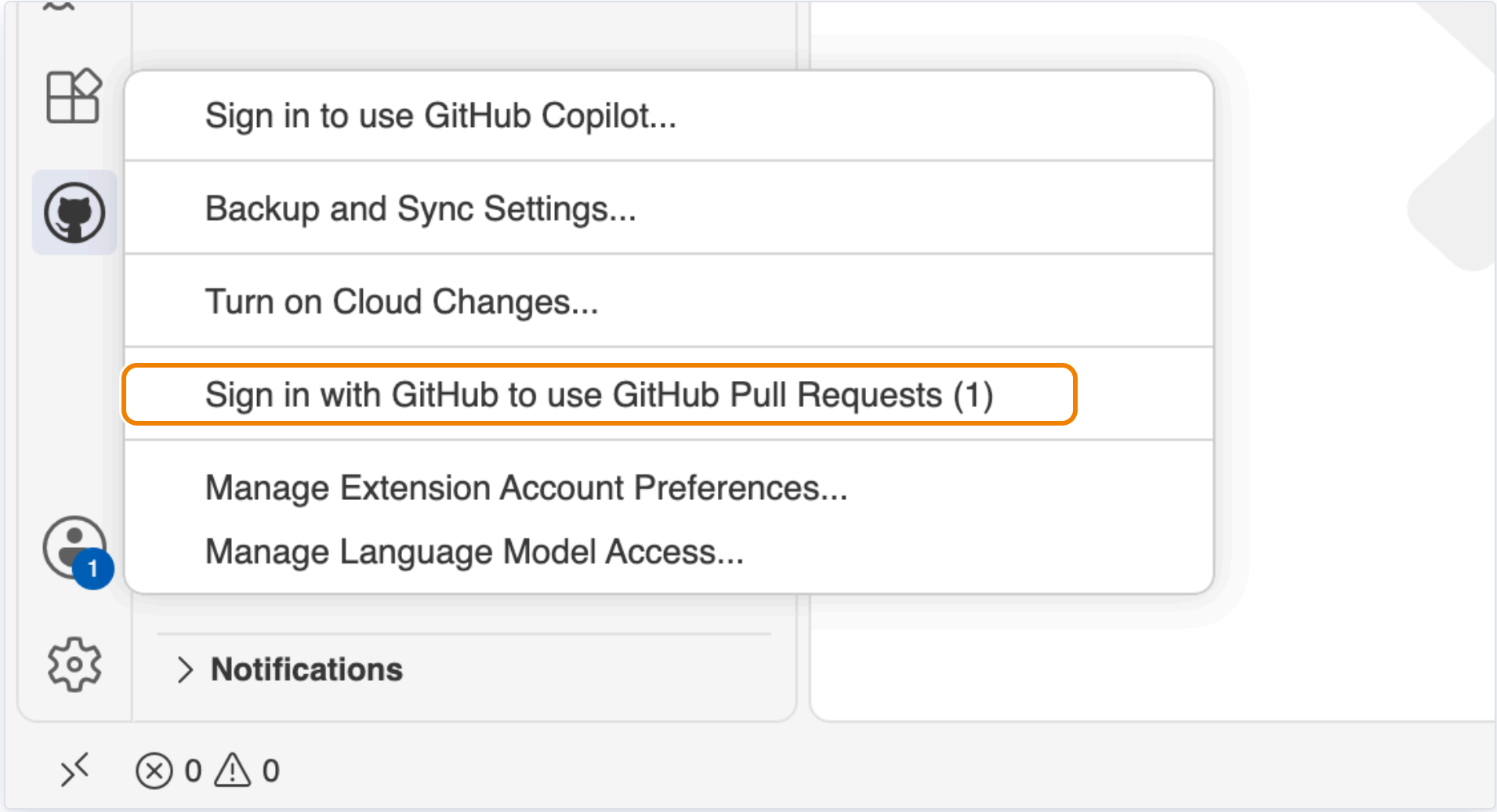
Install



YOUR TURN SIGN IN TO GITHUB · VS CODE · 4 OF 6

Accounts → Sign in with GitHub

The person icon with a 1, bottom left



YOUR TURN SIGN IN TO GITHUB · VS CODE · 5 OF 6

Continue in the browser

WHAT YOU WILL SEE

Your browser opens GitHub: Continue (and Authorize, the first time), then go back to VS Code.

VS CODE

10:38 Student package ✓
GITHUB

10:46 Meet GitHub ✓
GITHUB

10:51 Meet VS Code ✓
VS CODE

10:55 wsl → code . ✓
WSL

11:02 Sign in to GitHub
VS CODE

11:07 Fork
GITHUB

11:12 Clone
VS CODE

11:18 Switch to develop
VS CODE

11:24 Your details
VS CODE

----- 11:34 BREAK -----

11:44 Push develop
VS CODE → GITHUB

11:50 Merge: a conflict
VS CODE → GITHUB

12:07 Peer review
GITHUB

DONE WHEN SIGN IN TO GITHUB · VS CODE · 6 OF 6

Signed in

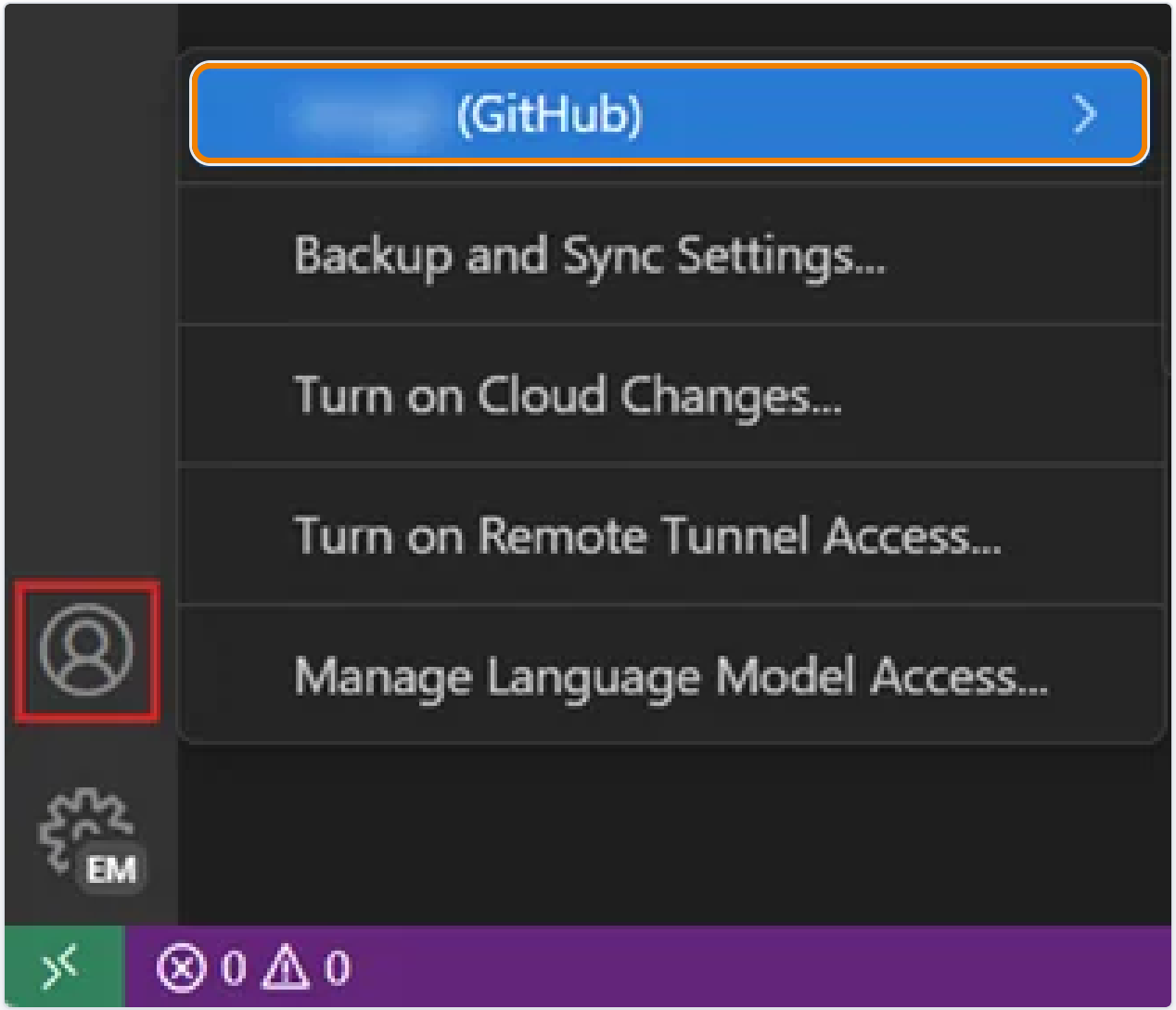


Image: VS Code docs, Microsoft, CC BY 3.0 US (cropped)

GITHUB

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VS CODE → GITHUB

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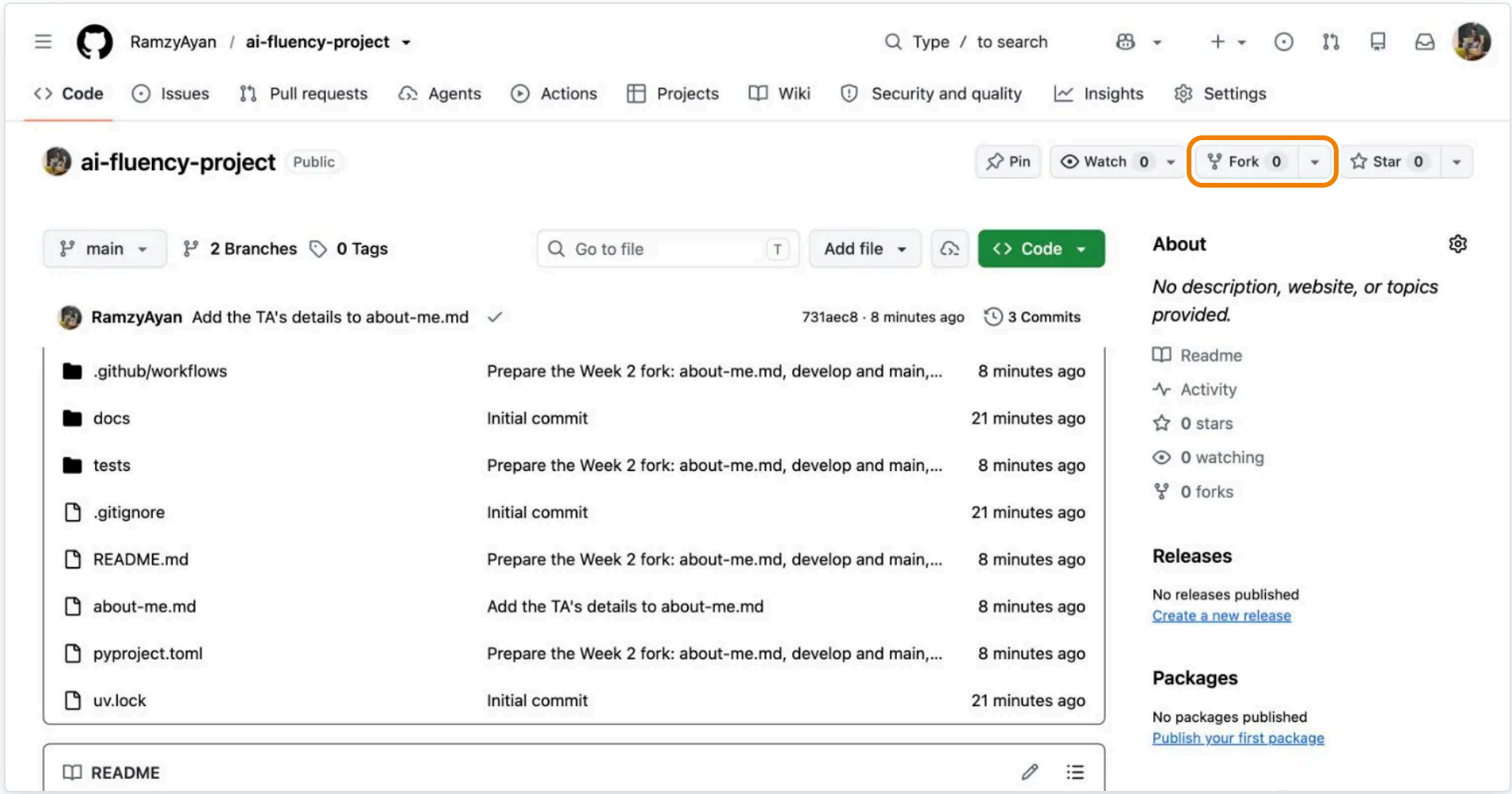
GITHUB

YOUR TURN FORK · GITHUB · 1 OF 4

Open the repo → Fork

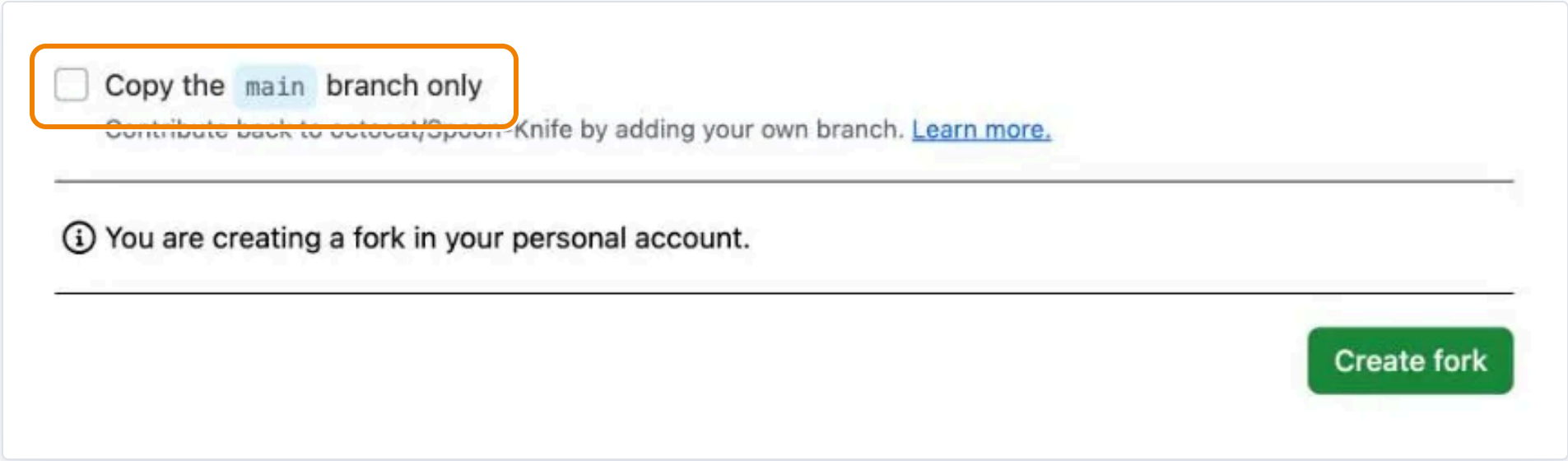
github.com/RamzyAyan/ai-fluency-project



YOUR TURN FORK · GITHUB · 2 OF 4

Untick: Copy the main branch only

GitHub ticks it: click to clear it, or your copy gets only main

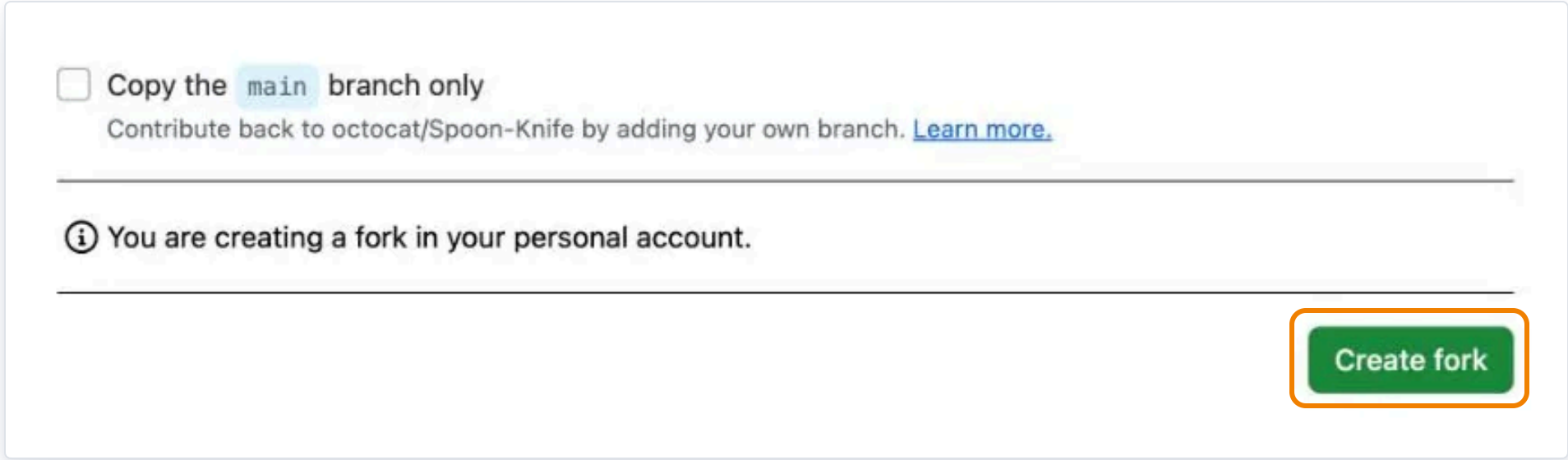


Stand-in: GitHub's fork form on its demo repository octocat/Spoon-Knife; yours names ai-fluency-project

YOUR TURN FORK · GITHUB · 3 OF 4

Create fork

Keep the name: ai-fluency-project

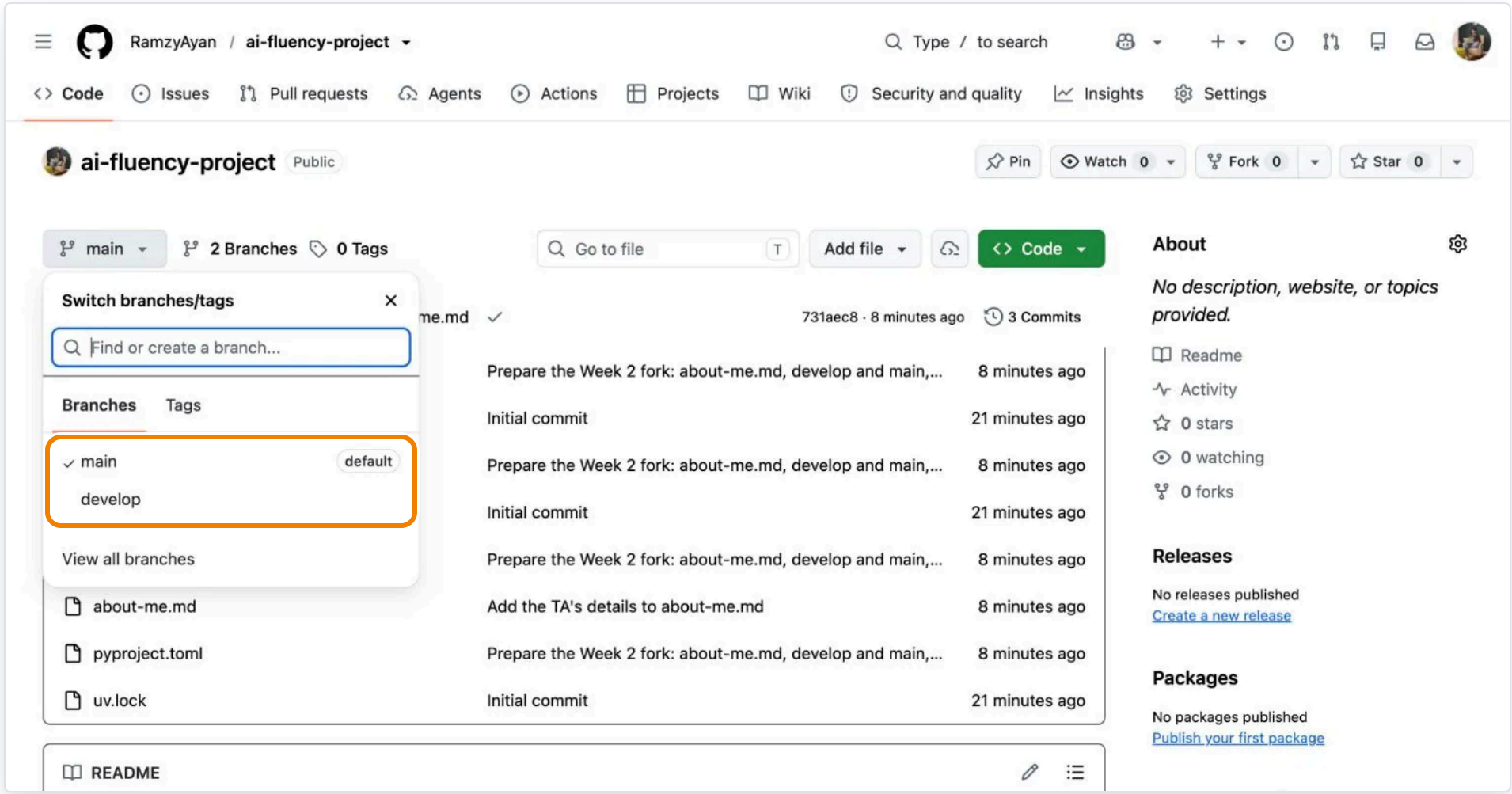


Stand-in: GitHub's fork form on its demo repository octocat/Spoon-Knife; yours names ai-fluency-project

DONE WHEN FORK · GITHUB · 4 OF 4

Your copy: 2 branches

Top left: <you>/ai-fluency-project, forked from RamzyAyan/ai-fluency-project



Shown on the original repository; your fork shows the same menu under your name

BACKUP FORK · GITHUB

Only main? Add develop

View all branches → New branch → source: the original's develop

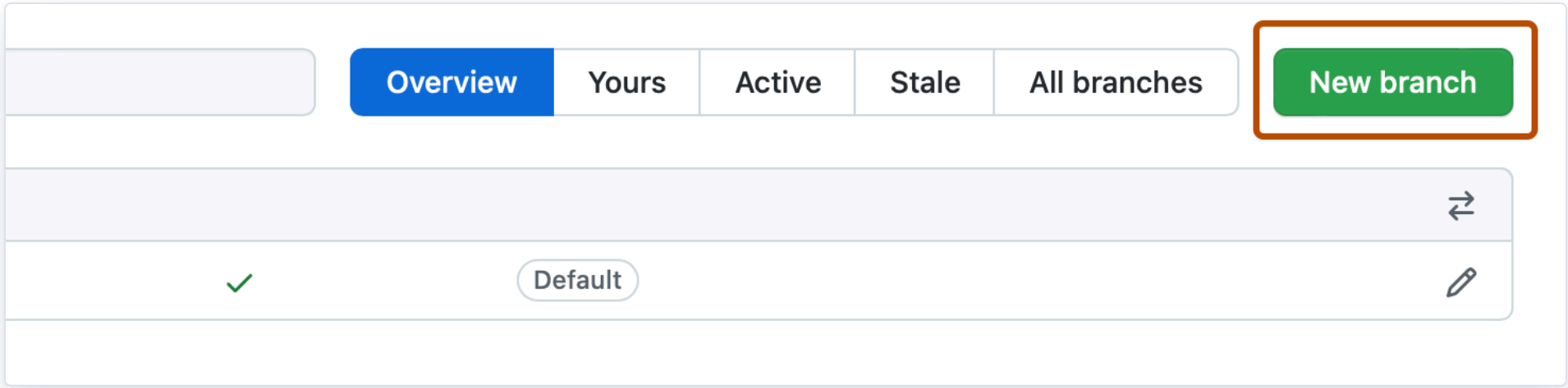


Image: GitHub Docs, CC BY 4.0 (cropped)

GITHUB

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VS CODE

11:07 Fork ✓
GITHUB

11:12 Clone
VS CODE
●○○○○○

11:18 Switch to develop
VS CODE

11:24 Your details
VS CODE

----- 11:34 BREAK -----

11:44 Push develop
VS CODE → GITHUB

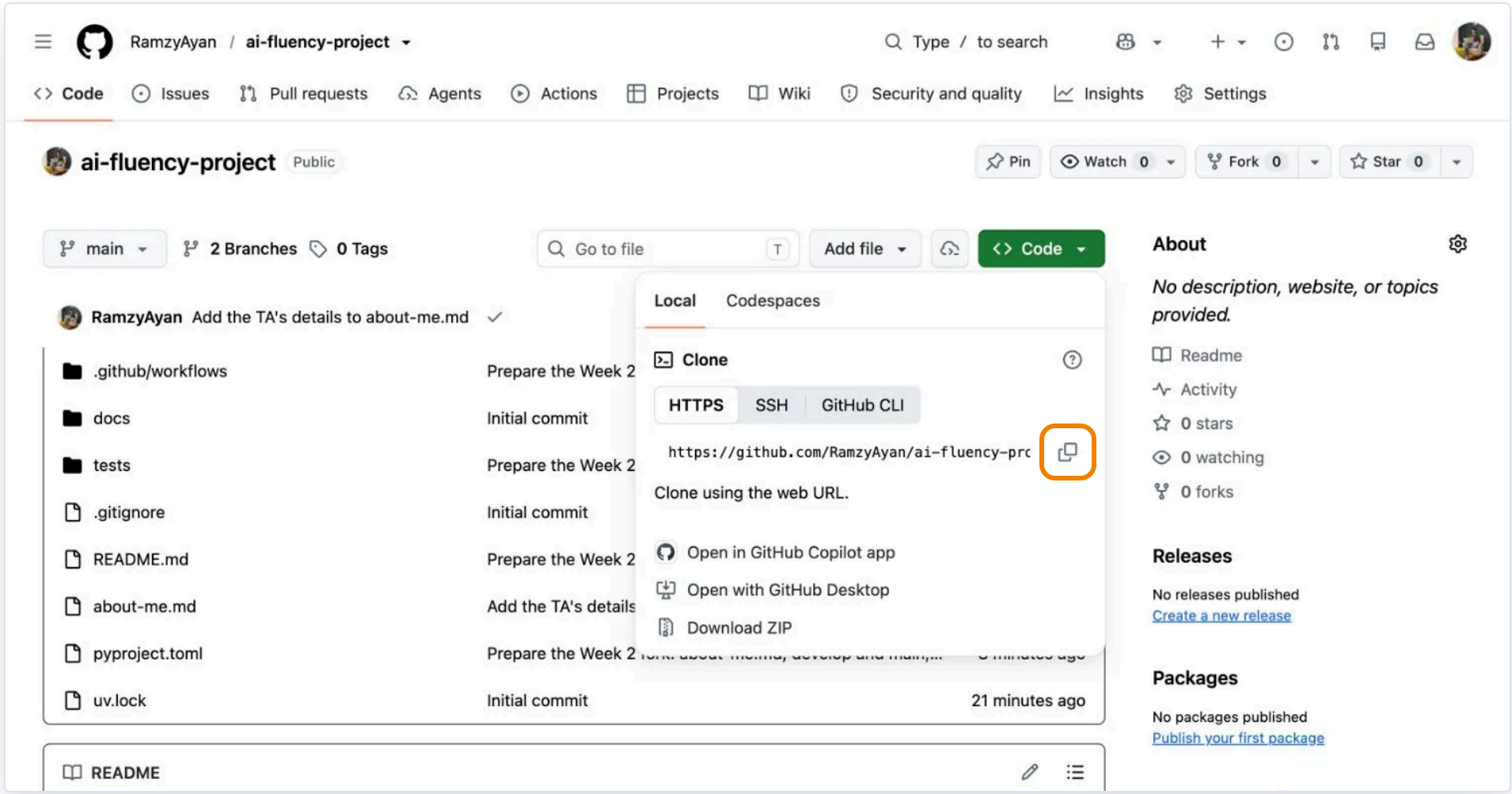
11:50 Merge: a conflict
VS CODE → GITHUB

12:07 Peer review
GITHUB

YOUR TURN CLONE · GITHUB · 1 OF 6

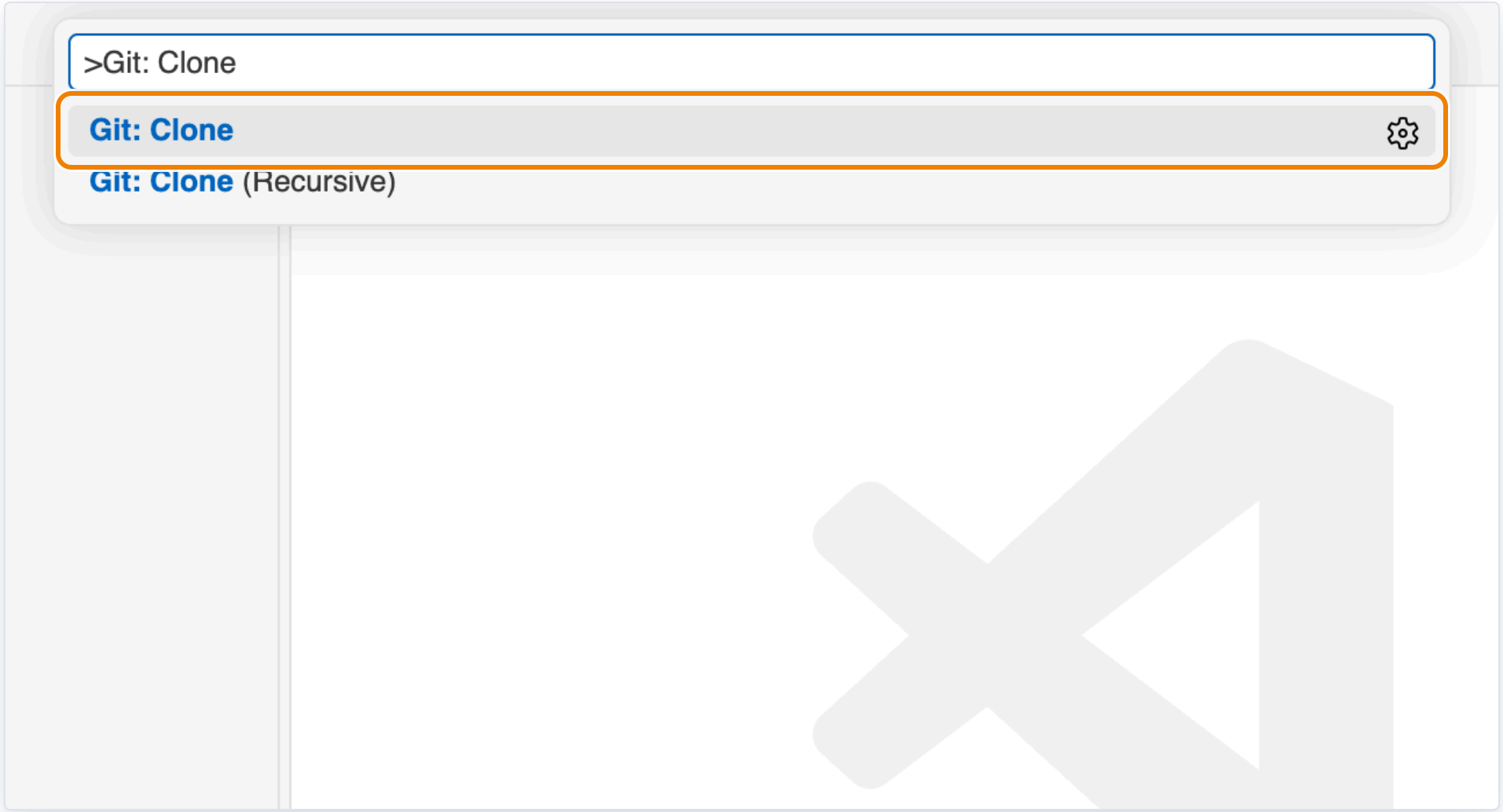
Code → copy the URL

On YOUR fork: github.com/<you>/ai-fluency-project



YOUR TURN CLONE · VS CODE · 2 OF 6

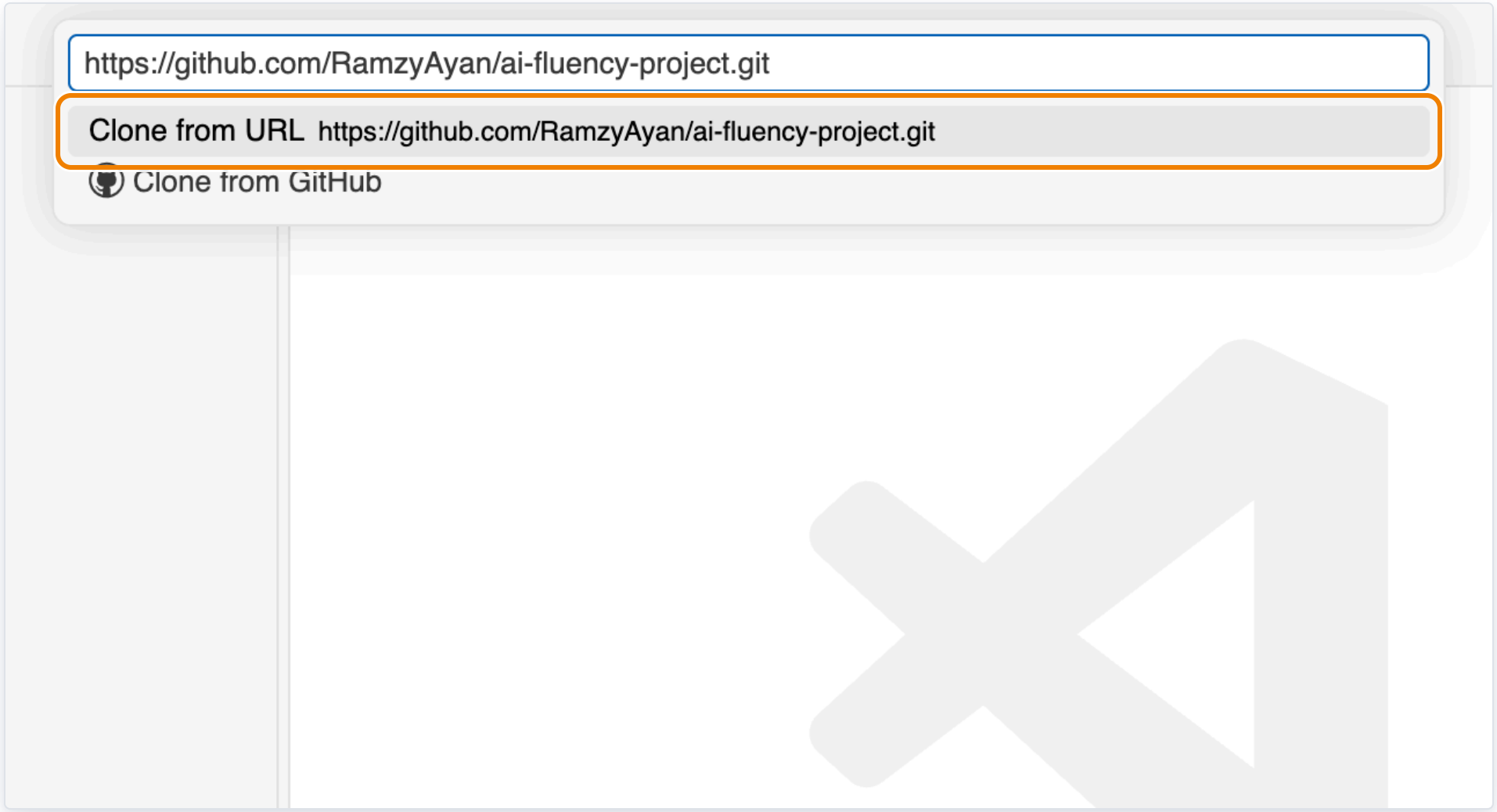
Ctrl+Shift+P → Git: Clone



YOUR TURN CLONE · VS CODE · 3 OF 6

Paste → Clone from URL

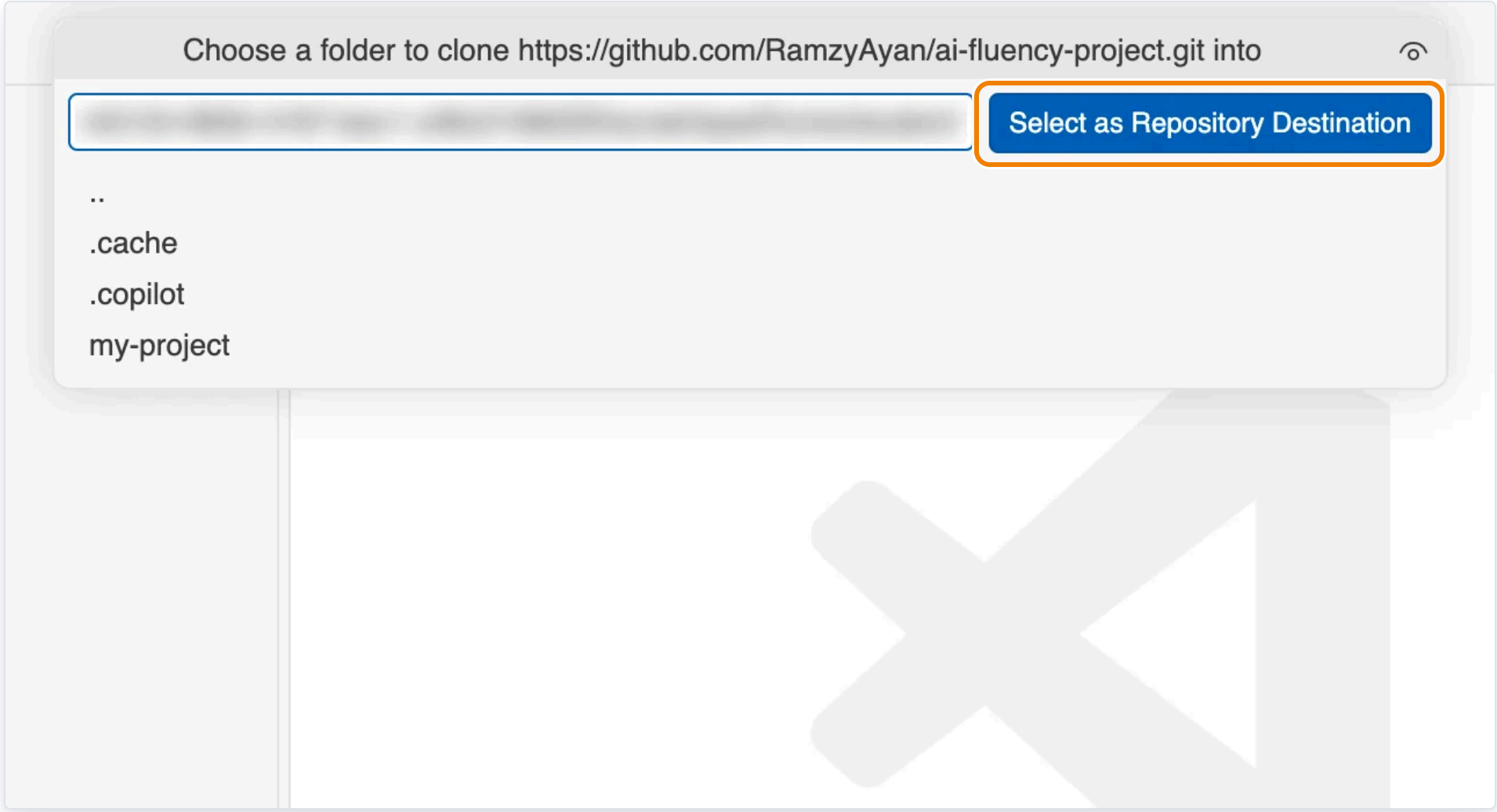
The URL has your name, not RamzyAyan



YOUR TURN CLONE · VS CODE · 4 OF 6

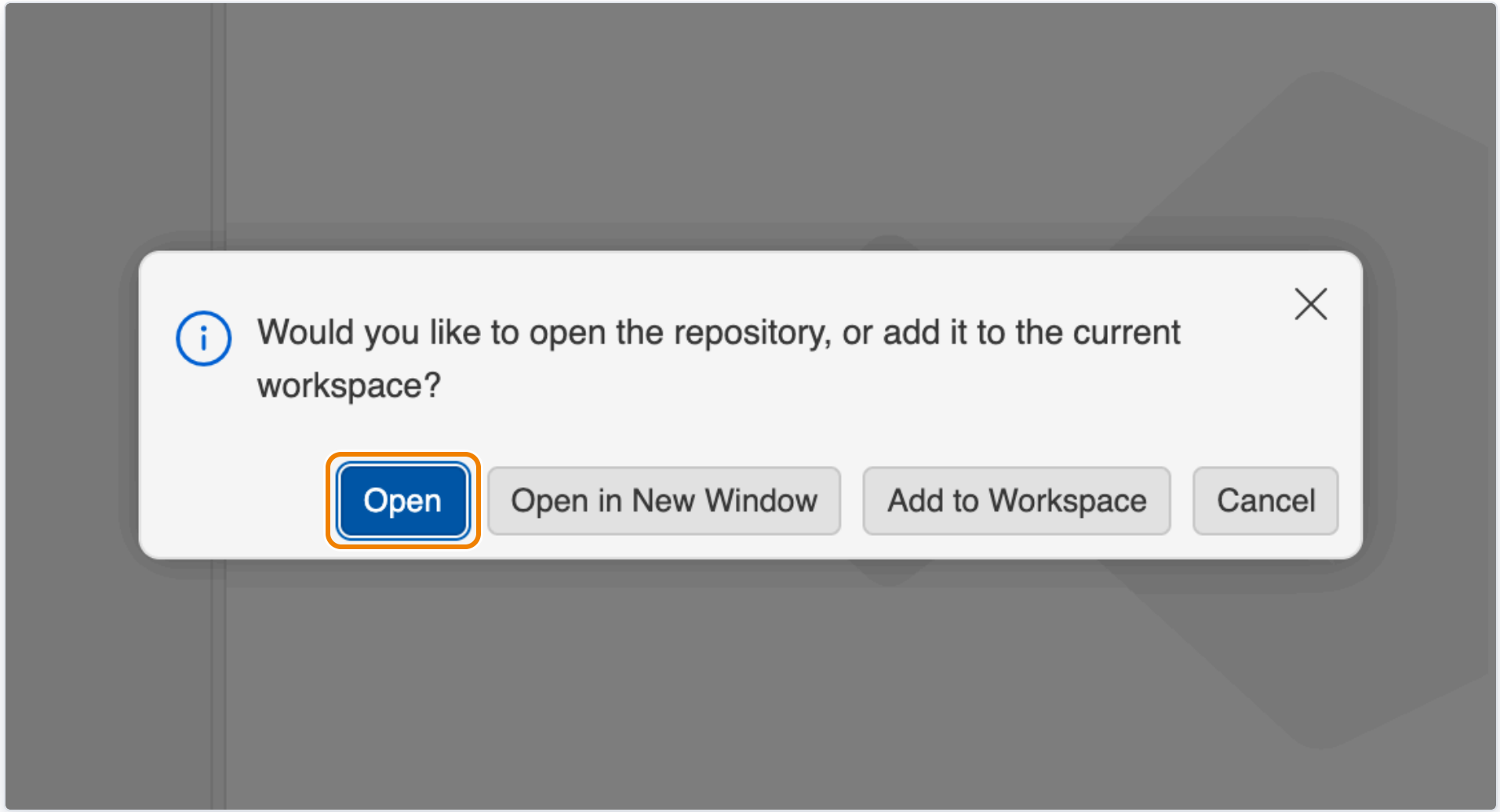
Pick your home folder

Select as Repository Destination · on WSL the box reads /home/<you>/



YOUR TURN CLONE · VS CODE · 5 OF 6

Open



VS CODE

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VS CODE

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VS CODE

----- 11:34 BREAK ----------

11:44 Push develop
VS CODE → GITHUB

11:50 Merge: a conflict
VS CODE → GITHUB

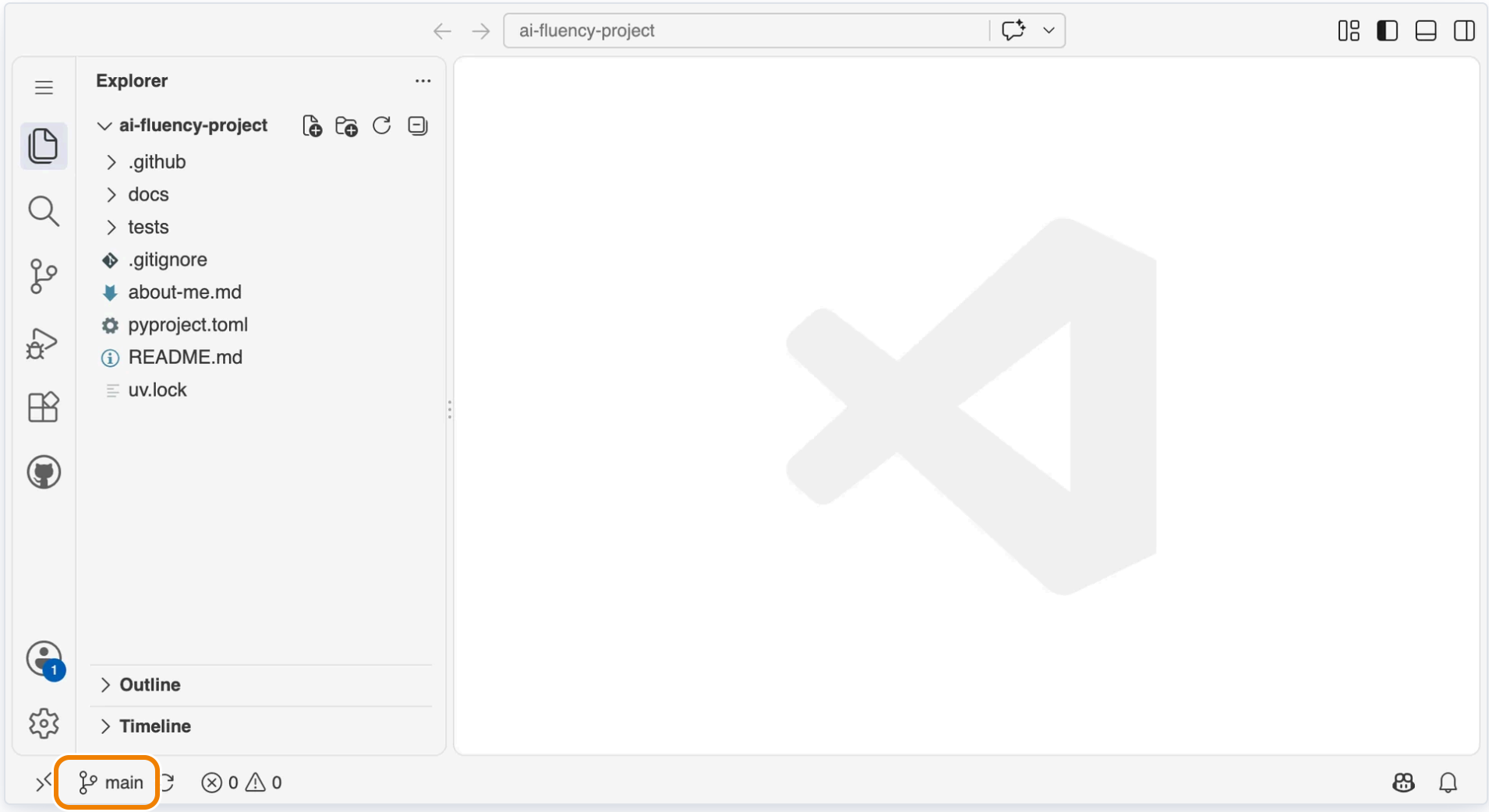
12:07 Peer review
GITHUB

ON BRANCH
main

DONE WHEN CLONE · VS CODE · 6 OF 6

Status bar: main

about-me.md, docs, tests and README.md in the Explorer



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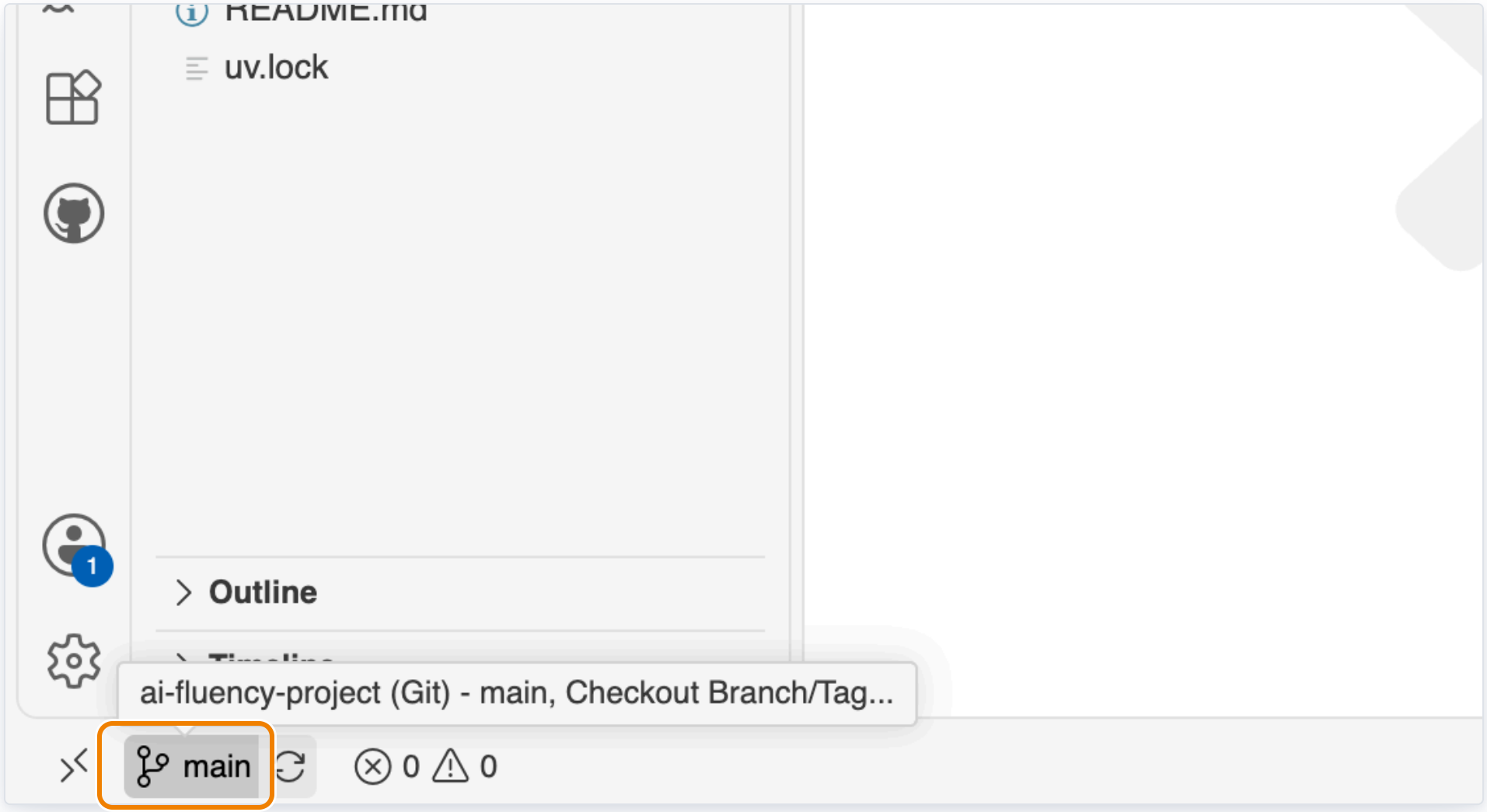
GITHUB

ON BRANCH
main

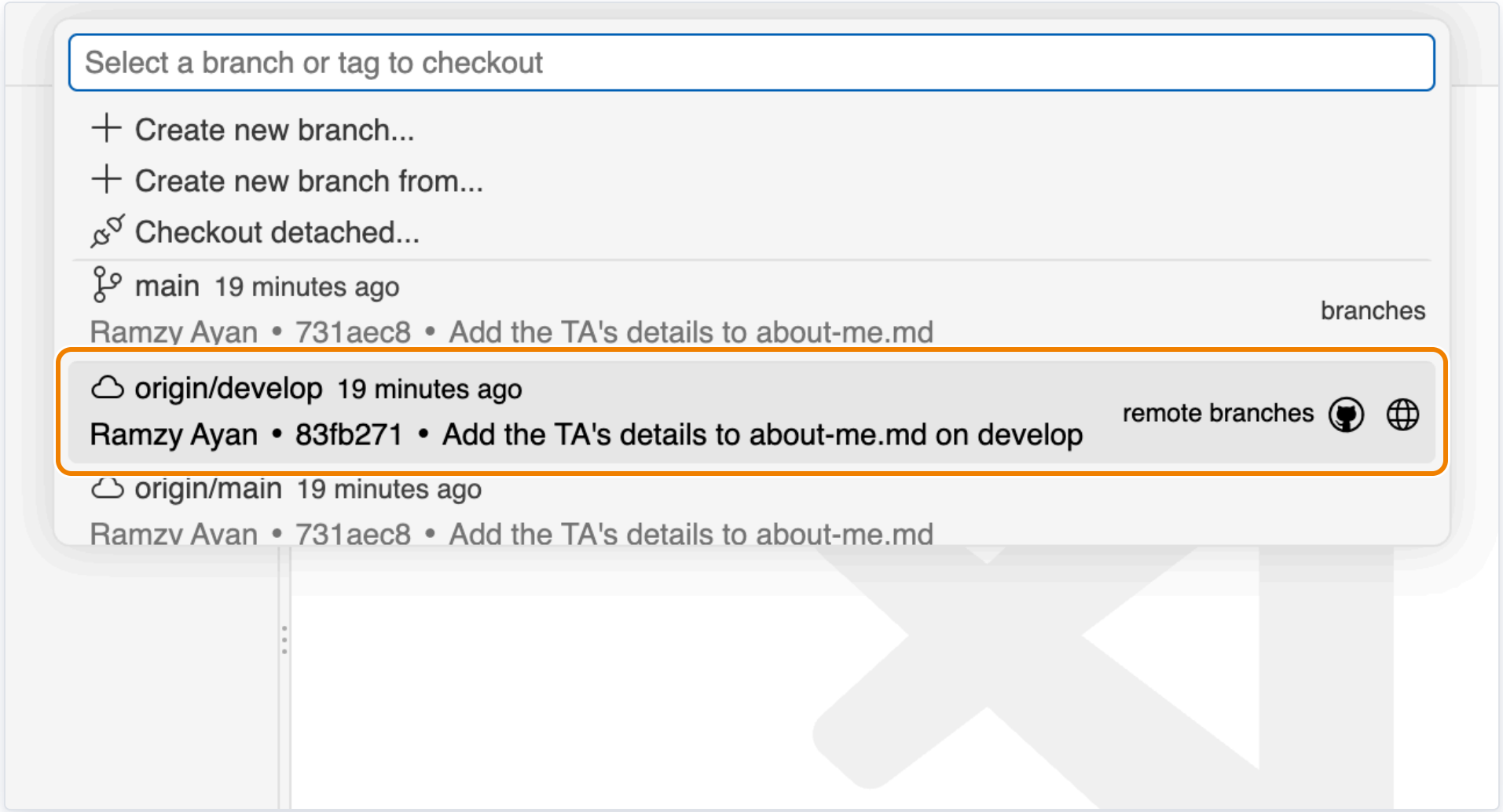
YOUR TURN SWITCH TO DEVELOP · VS CODE · 1 OF 6

Click main



YOUR TURN SWITCH TO DEVELOP · VS CODE · 2 OF 6

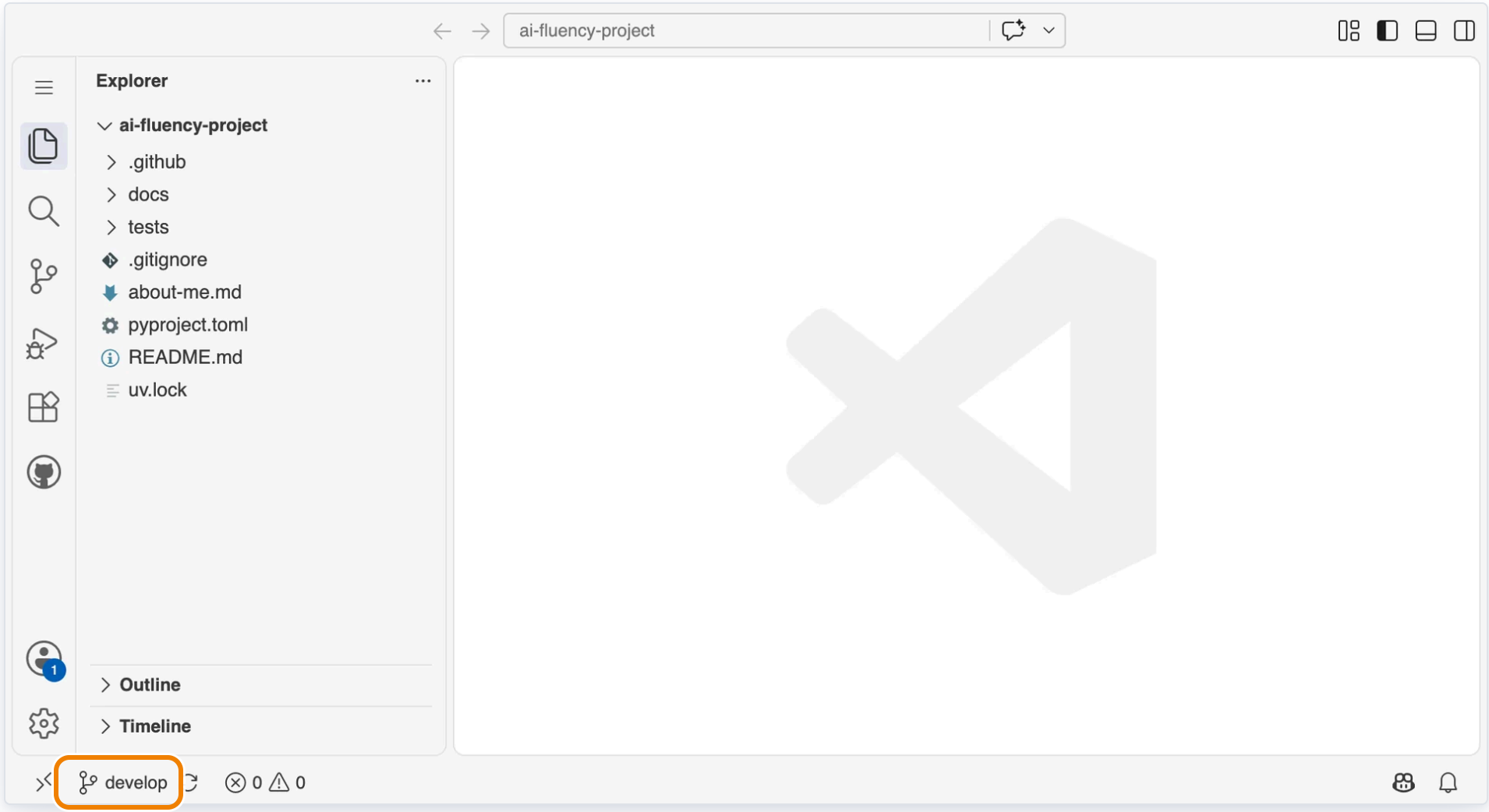
Pick origin/develop



DONE WHEN SWITCH TO DEVELOP · VS CODE · 3 OF 6

Now: develop

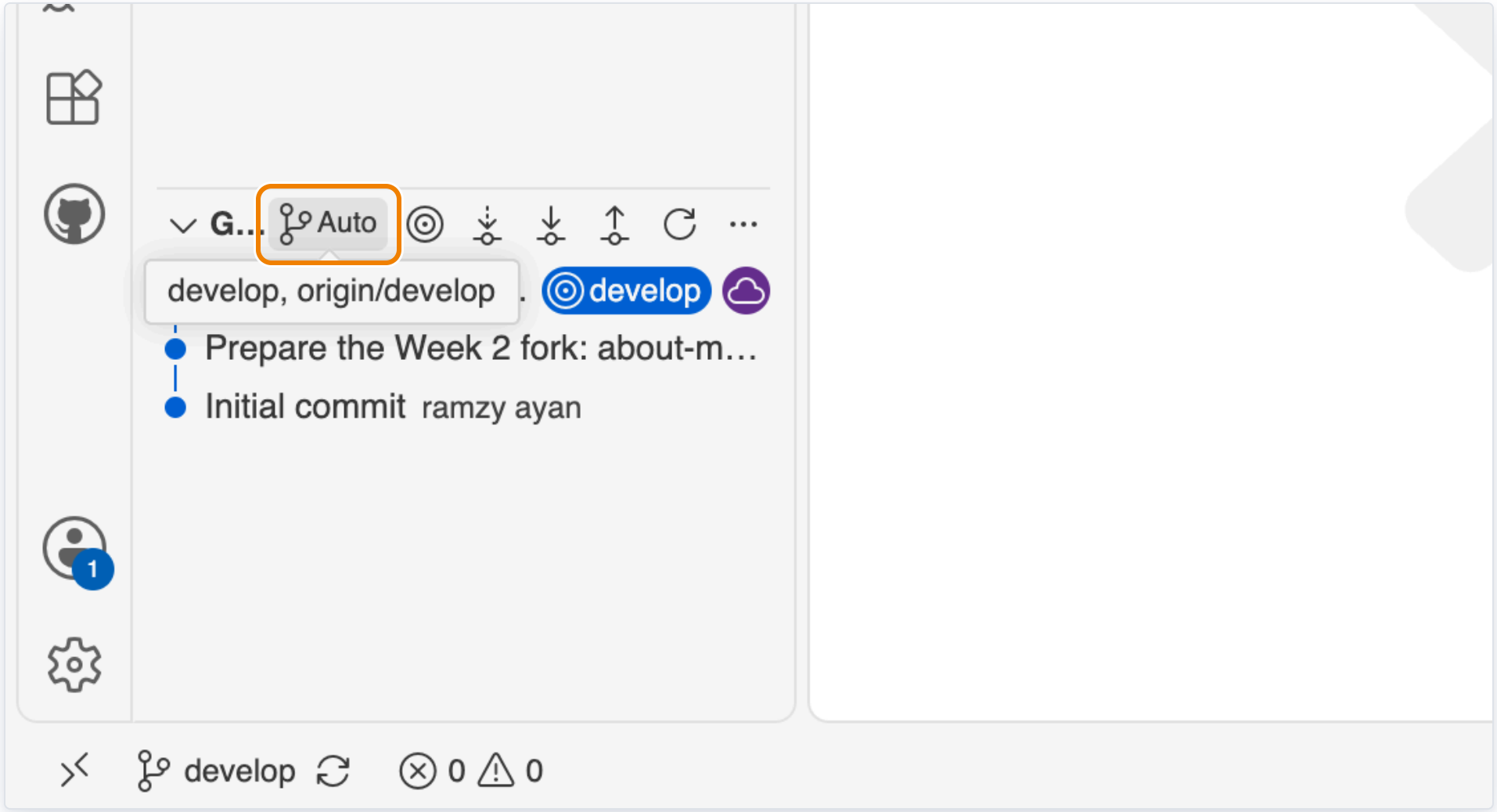
The status bar says develop



YOUR TURN SWITCH TO DEVELOP · VS CODE · 4 OF 6

Graph: click Auto

Source Control (Ctrl+Shift+G) → Graph, at the bottom → Auto



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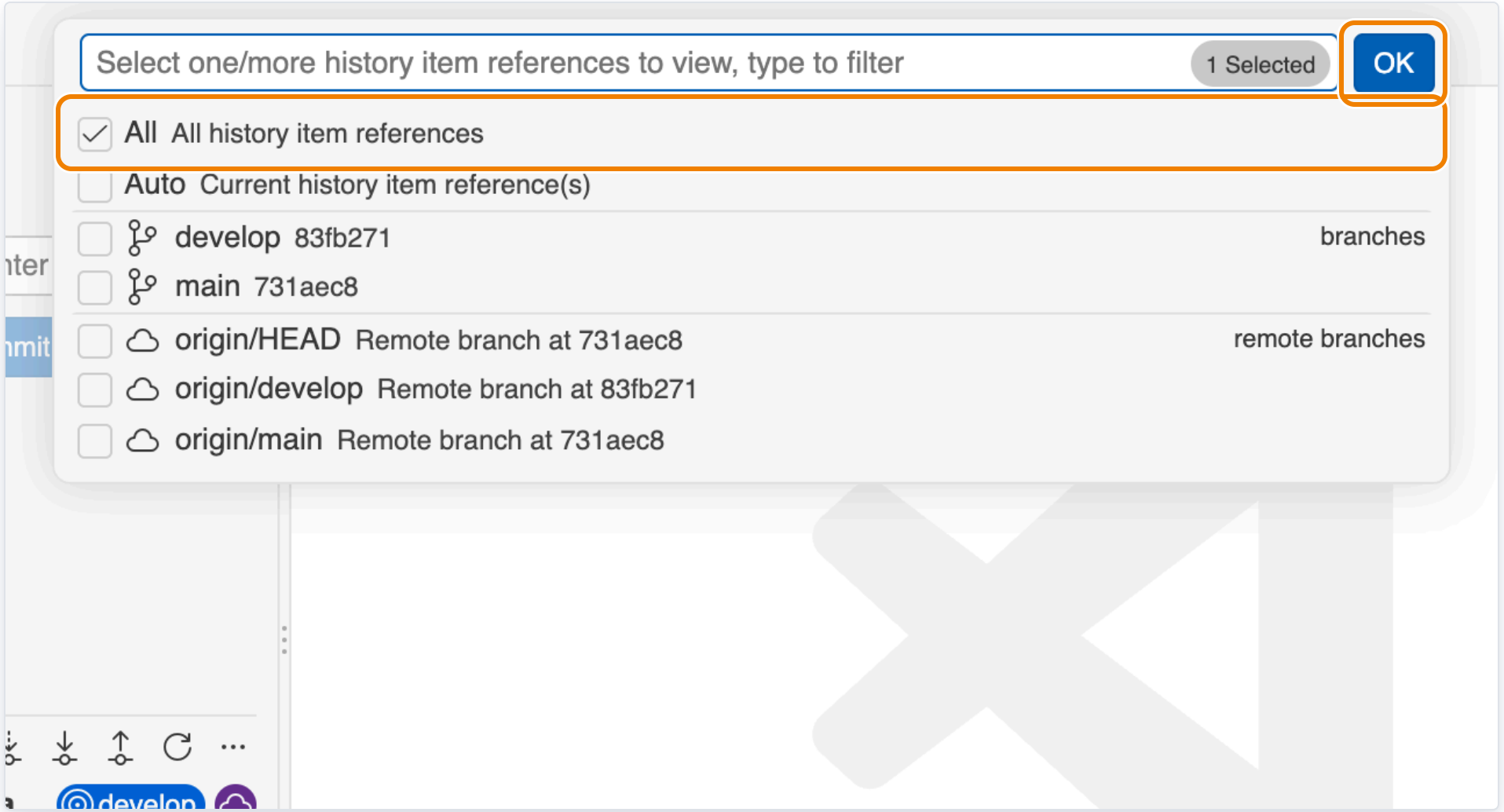
12:07 Peer review
GITHUB

ON BRANCH
develop

YOUR TURN SWITCH TO DEVELOP · VS CODE · 5 OF 6

Tick All → OK



VS CODE

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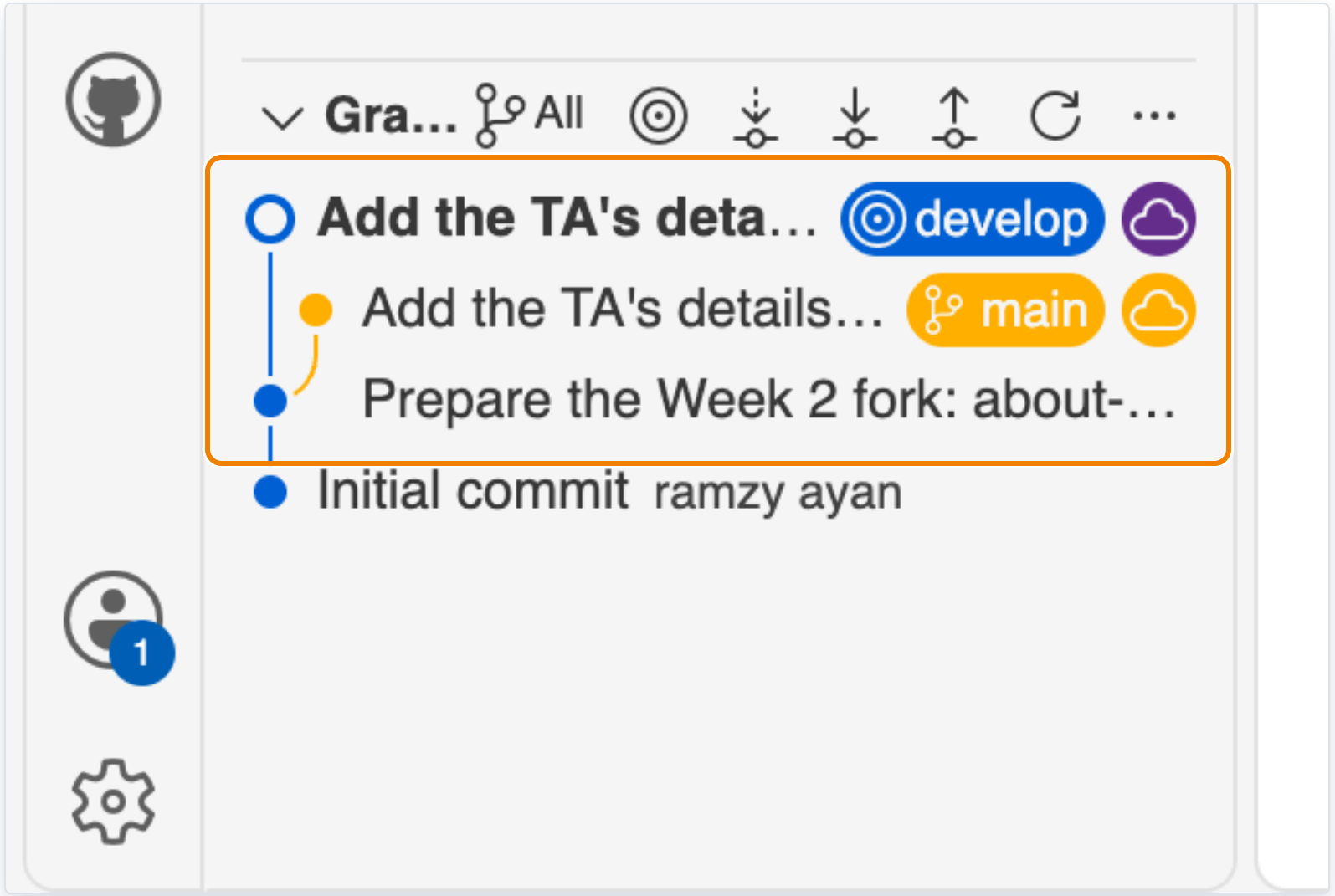
GITHUB

ON BRANCH
develop

DONE WHEN SWITCH TO DEVELOP · VS CODE · 6 OF 6

Graph: two lines

main and develop split after 'Prepare the Week 2 fork'



VS CODE

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----- 11:34 BREAK -----

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VS CODE → GITHUB

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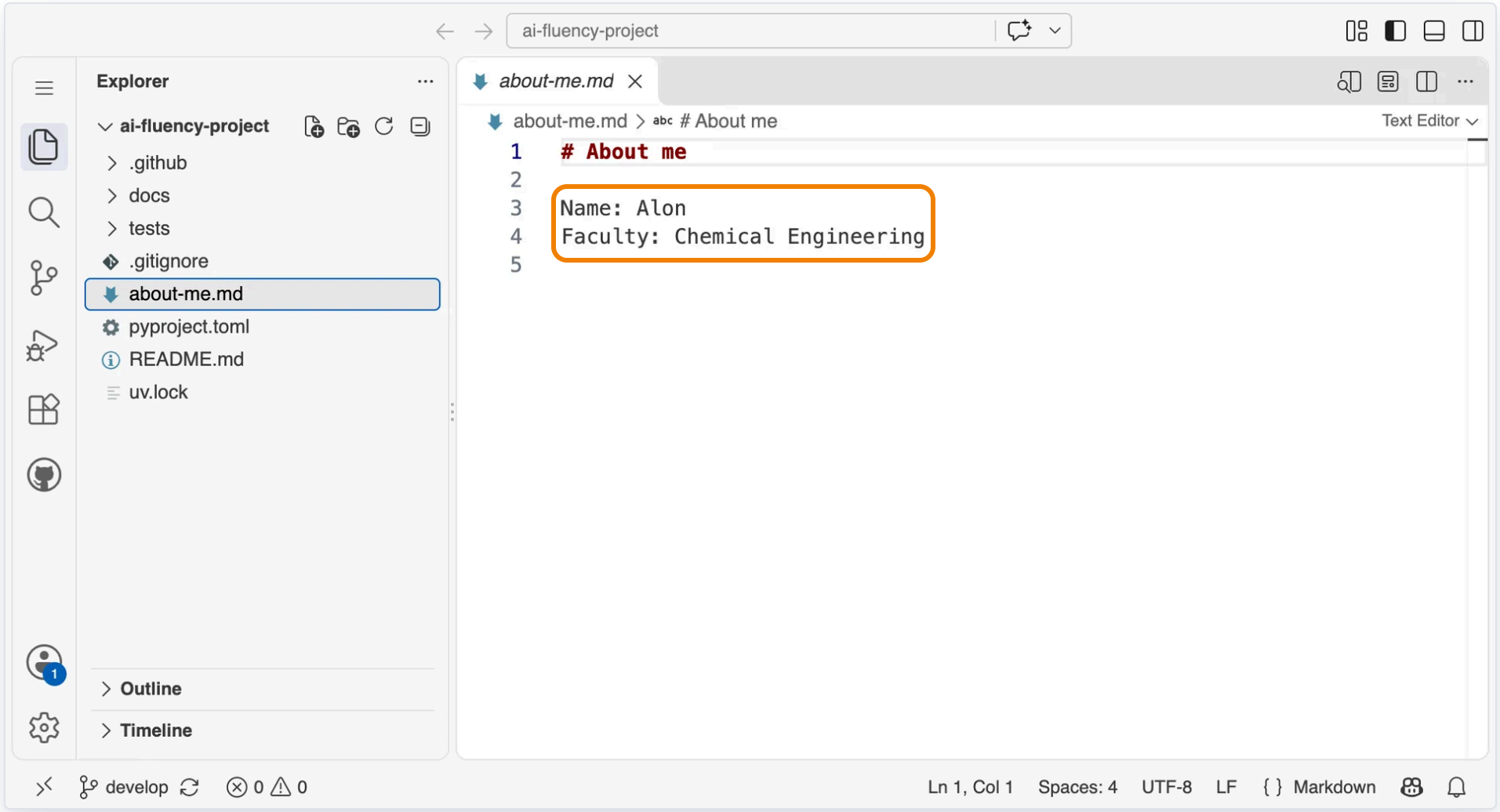
12:07 Peer review
GITHUB

ON BRANCH
develop

Open about-me.md

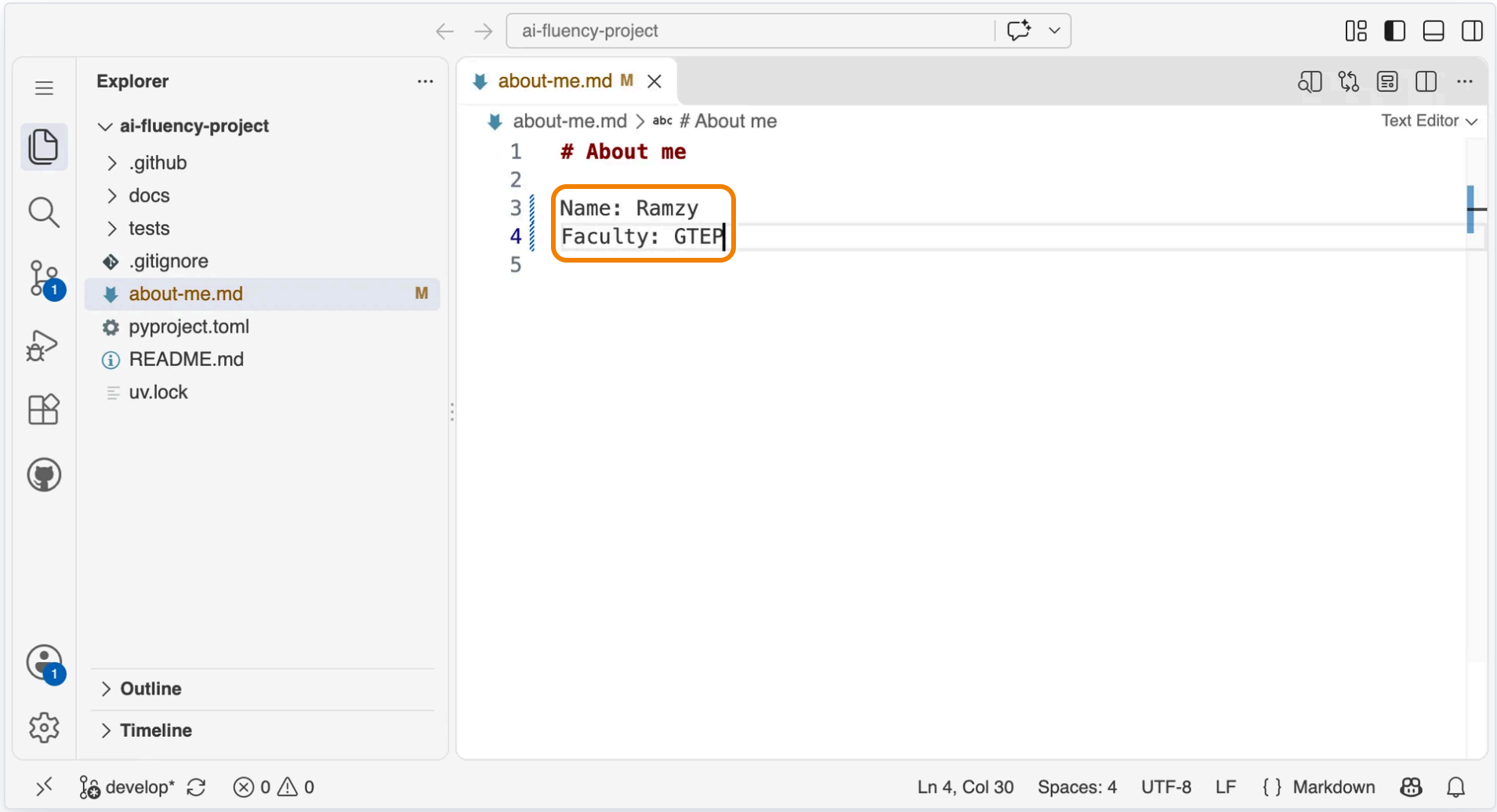
Explorer (Ctrl+Shift+E) → about-me.md · Alon's details



YOUR TURN YOUR DETAILS · VS CODE · 2 OF 8

Your name, your faculty · Ctrl+S

Change the two lines · Mac: Cmd+S



VS CODE

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GITHUB

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VS CODE

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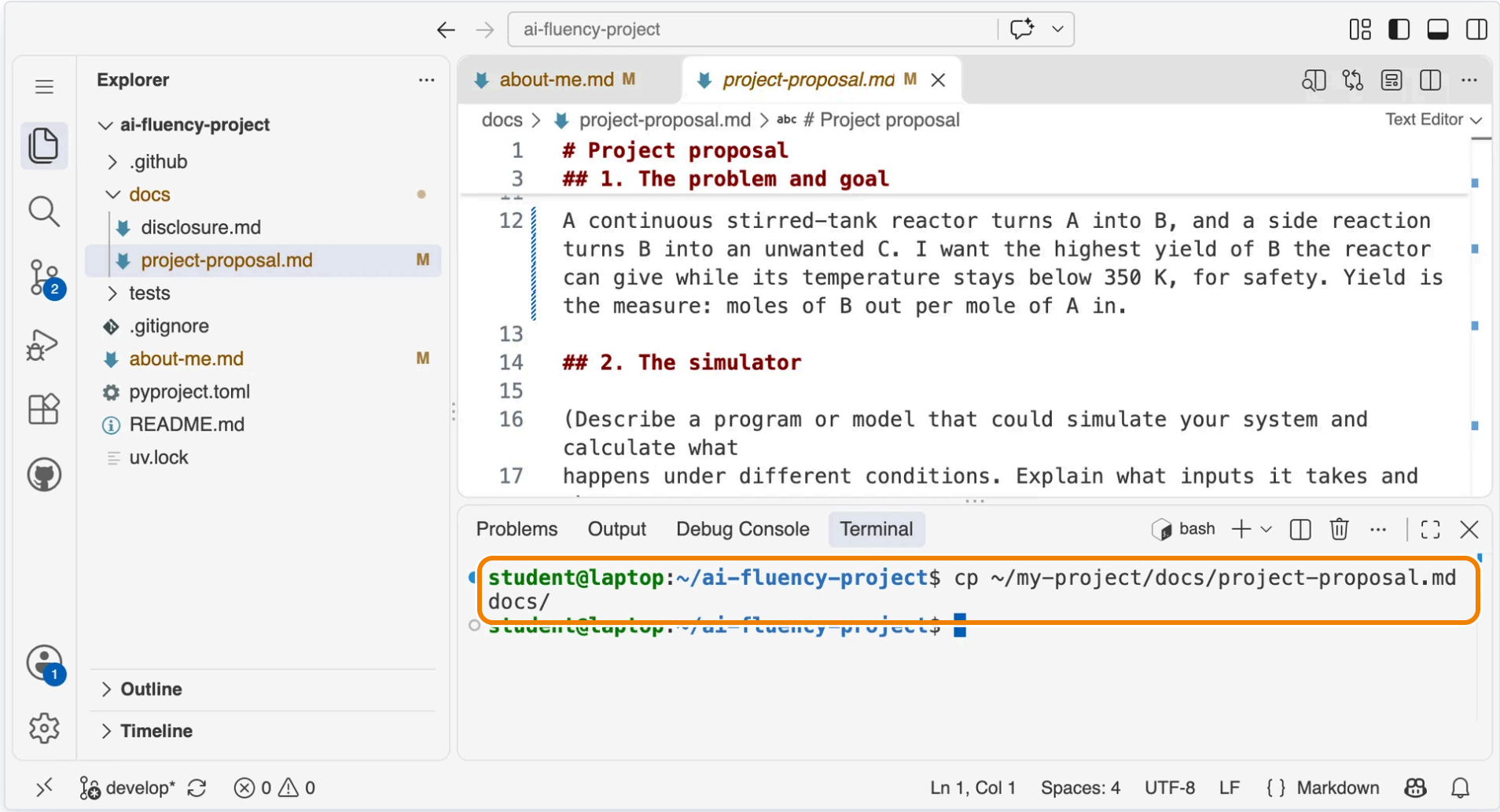
12:07 Peer review
GITHUB

ON BRANCH
develop

Bring in your Week 1 proposal

Ctrl+` → cp ~/my-project/docs/project-proposal.md docs/



VS CODE

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VS CODE

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VS CODE
●●●◐○○○○

----- 11:34 BREAK -----

11:44 Push develop
VS CODE → GITHUB

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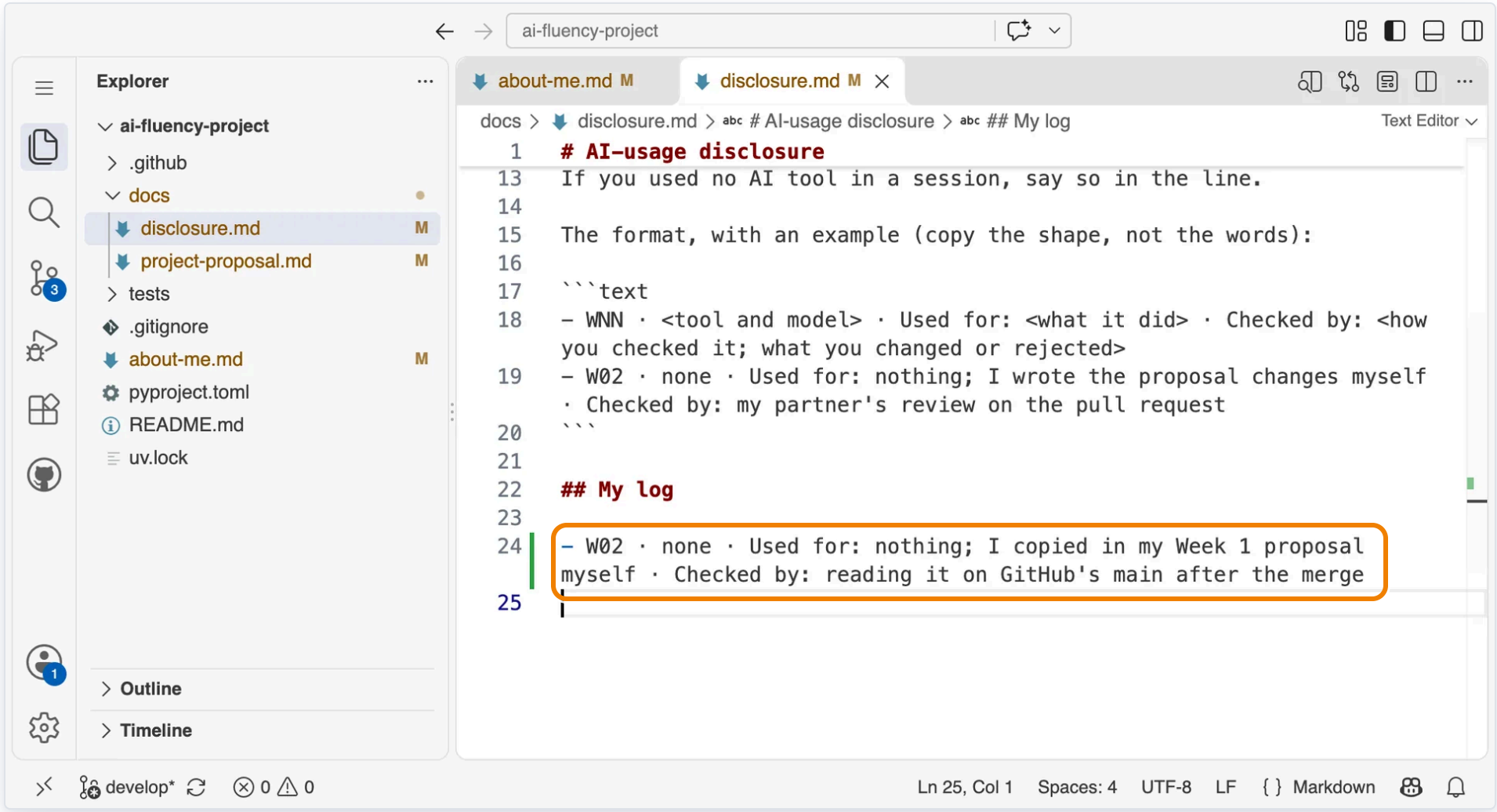
VS CODE → GITHUB

12:07 Peer review
GITHUB

ON BRANCH
develop

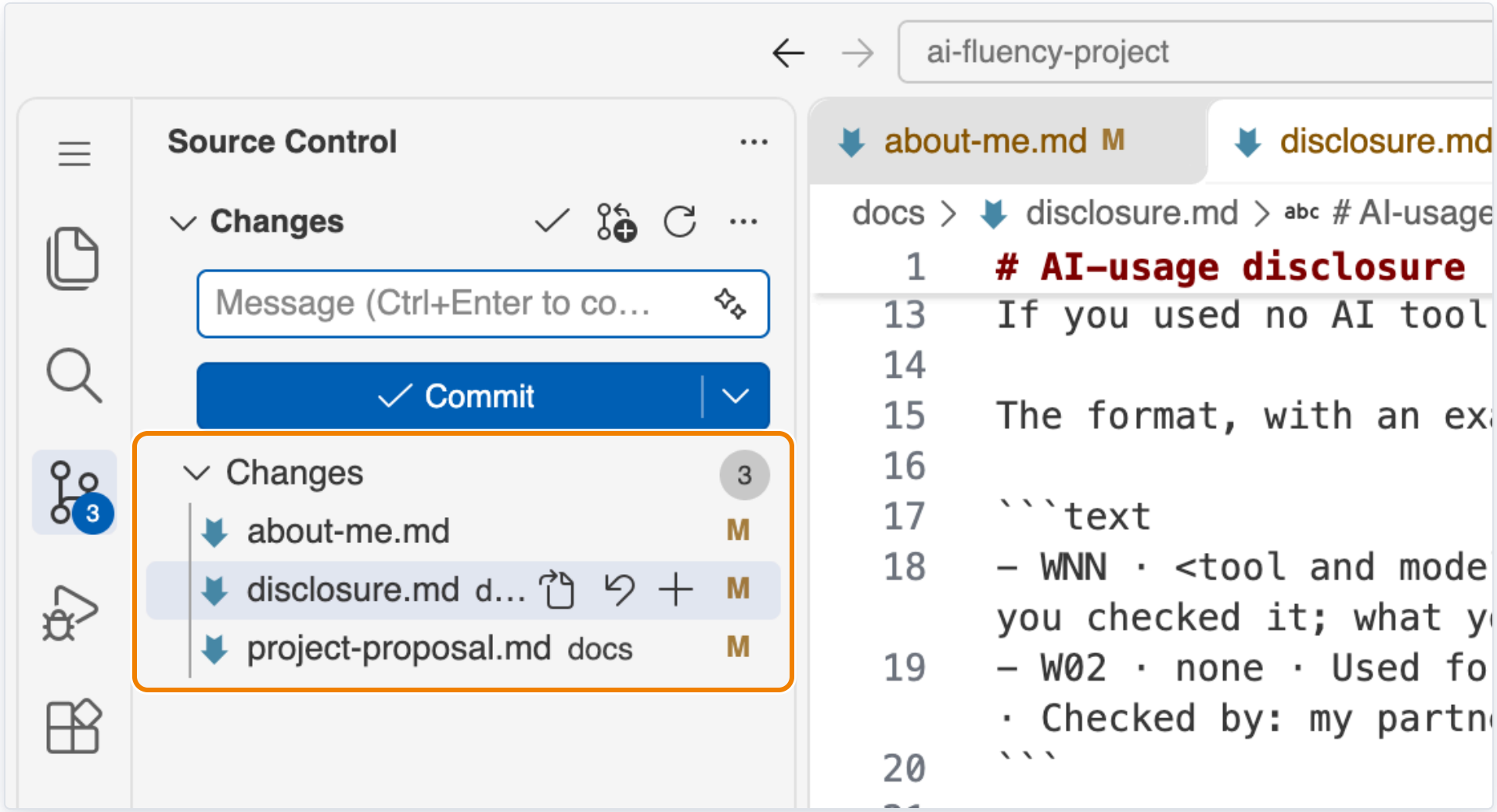
Add your W02 line · Ctrl+S

docs/disclosure.md: a new last line, under My log



Source Control: 3 changes

Ctrl+Shift+G (Mac: Control+Shift+G)



VS CODE

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12:07 Peer review
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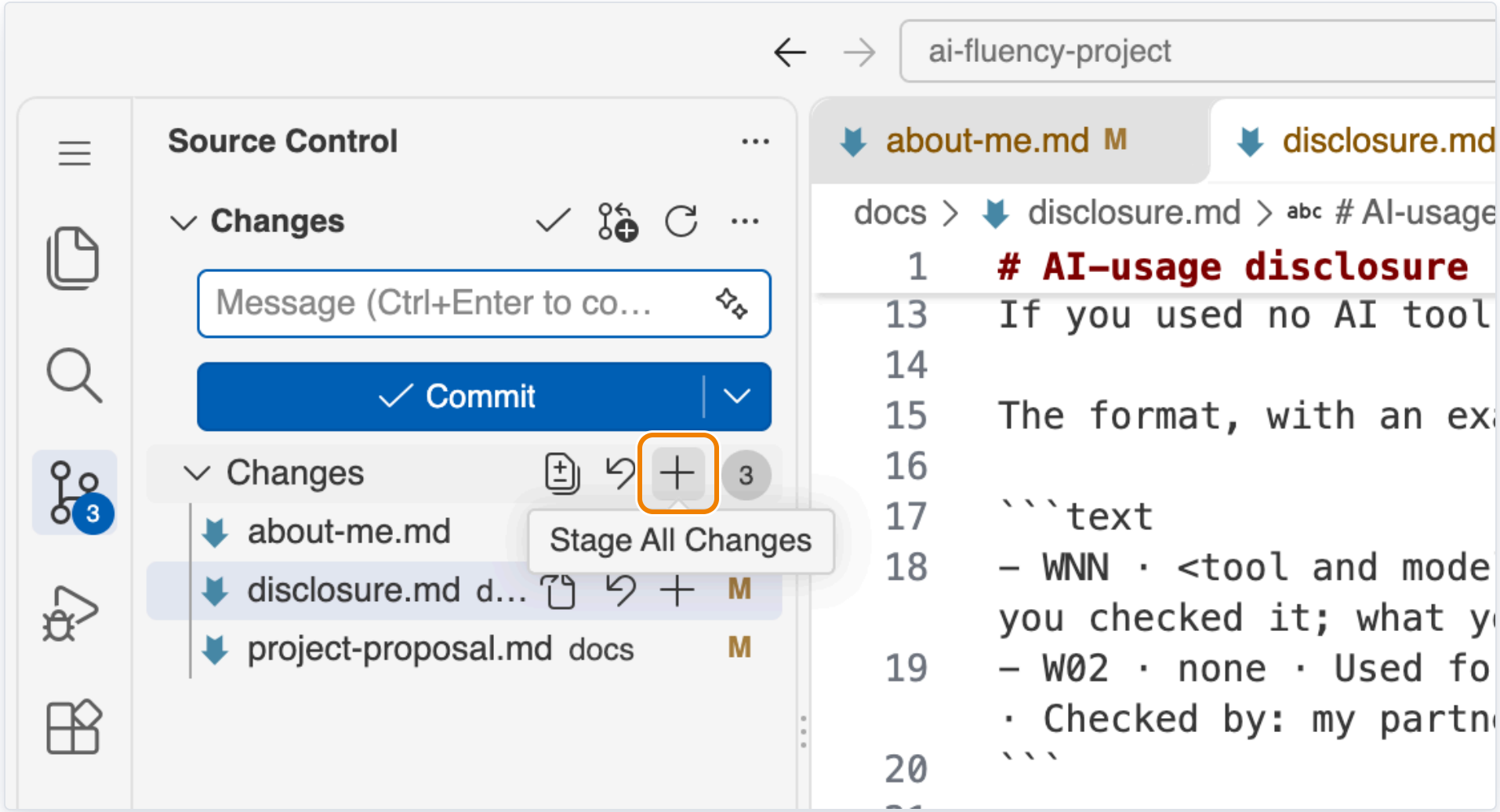
ON BRANCH
develop

+ Stage All Changes



VS CODE

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VS CODE
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VS CODE
11:24 **Your details**
VS CODE
----- 11:34 BREAK -----
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VS CODE → GITHUB
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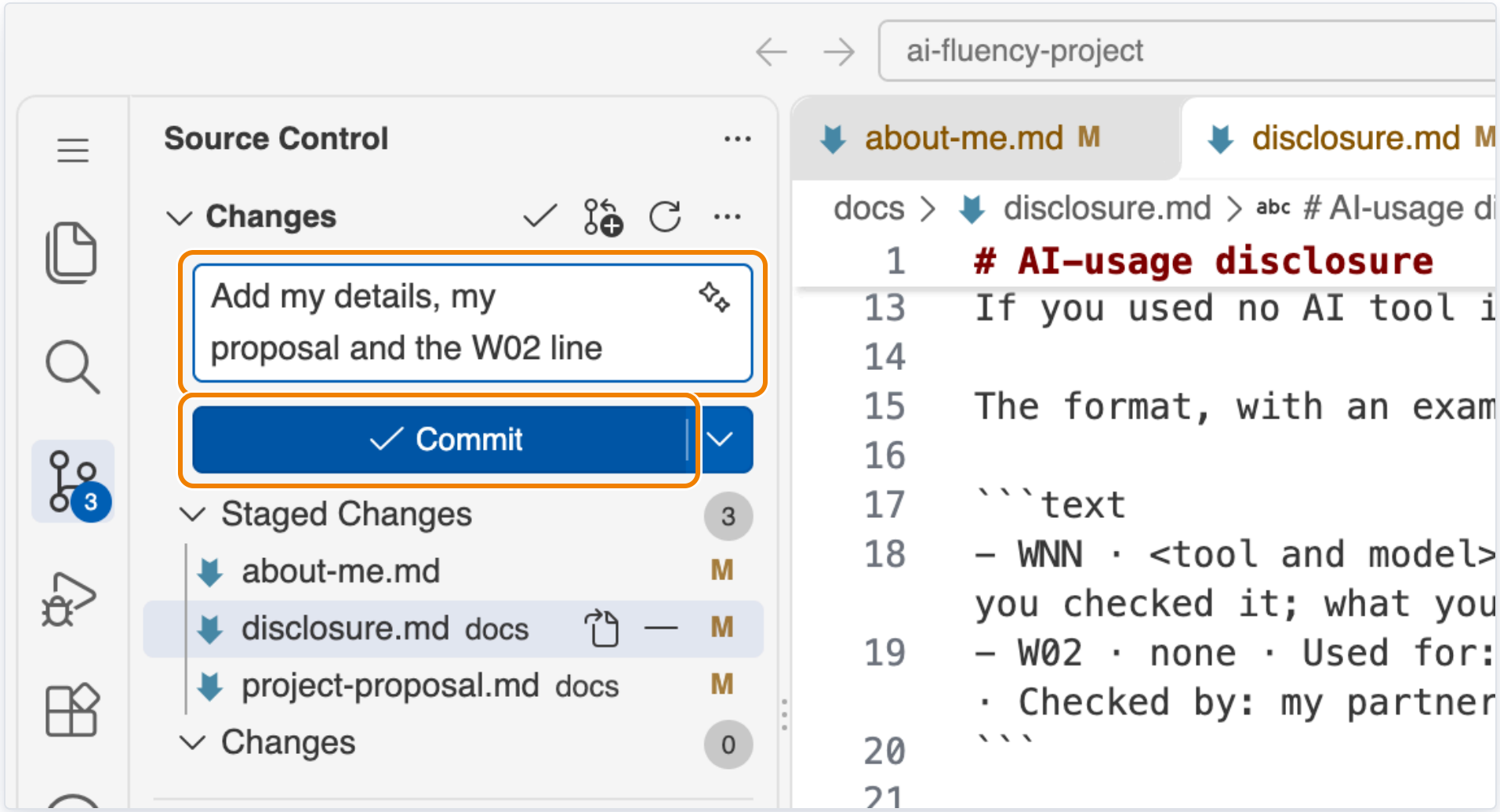
GITHUB

ON BRANCH
develop

Message → Commit

Add my details, my proposal and the W02 line



VS CODE

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- 11:18 Switch to develop ✓ — VS CODE
- **11:24 Your details** — VS CODE
- ----- 11:34 BREAK -----
- 11:44 Push develop — VS CODE → GITHUB
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- 12:07 Peer review — GITHUB

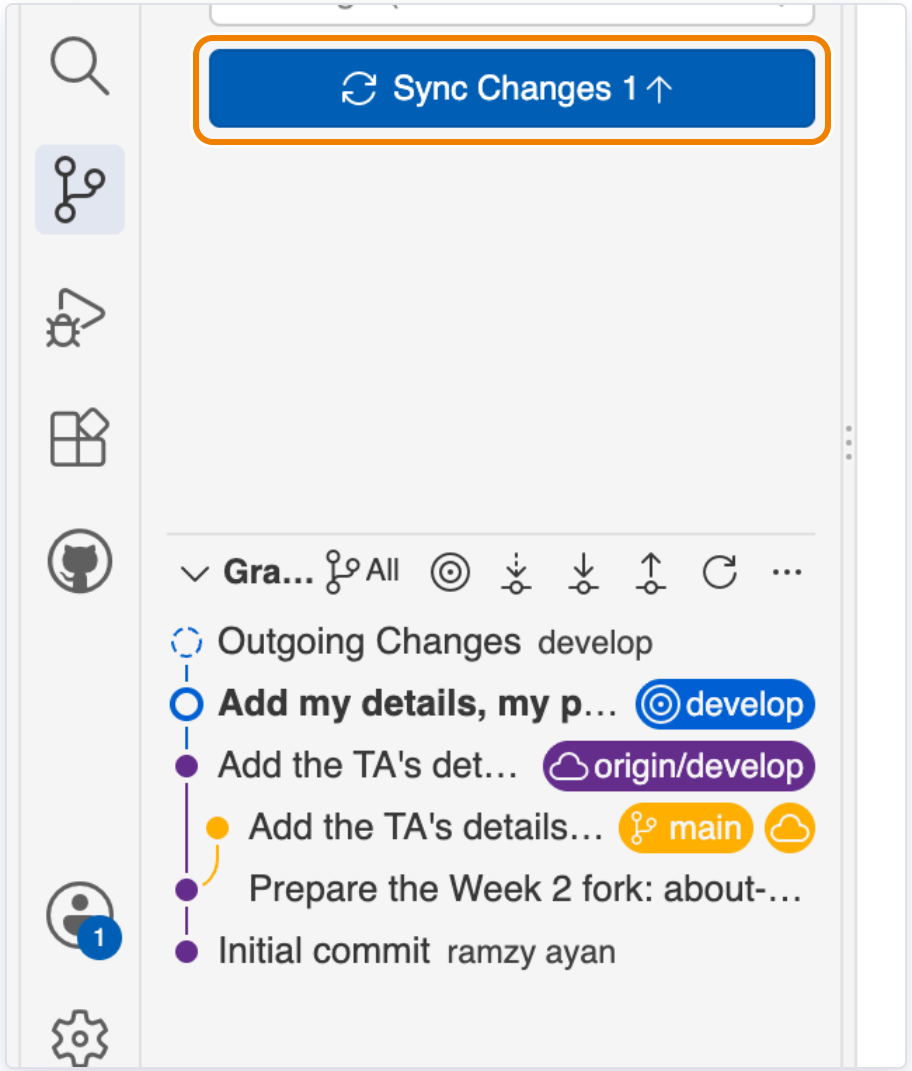
ON BRANCH

develop

DONE WHEN YOUR DETAILS · VS CODE · 8 OF 8

Saved here, not on GitHub yet

Sync Changes 1↑



BREAK · 10 MIN

Back at 11:44

11:34 11:44

Next: **Push develop** · VS Code → GitHub

10:38 Student package ✓
GITHUB
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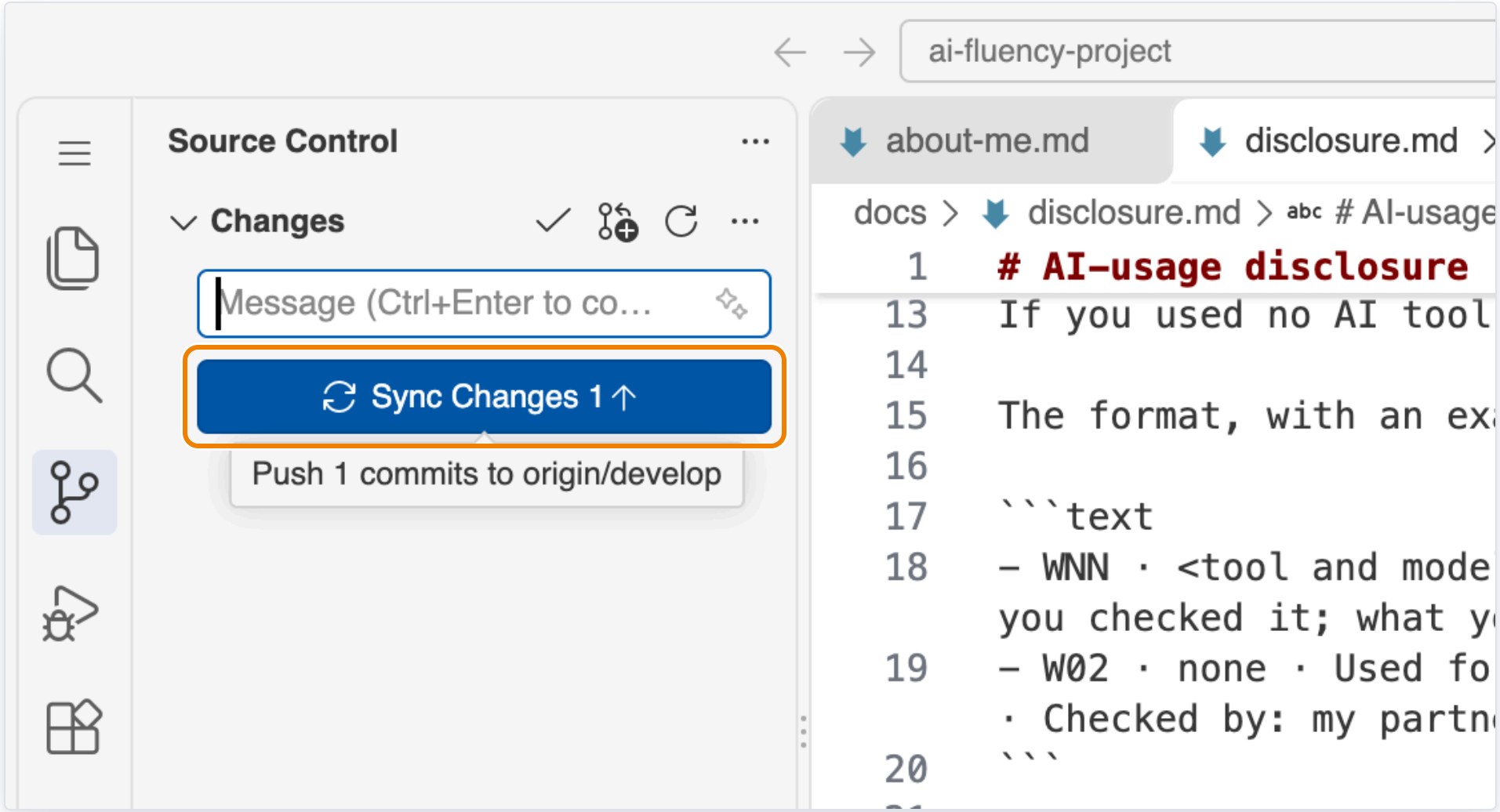
GITHUB

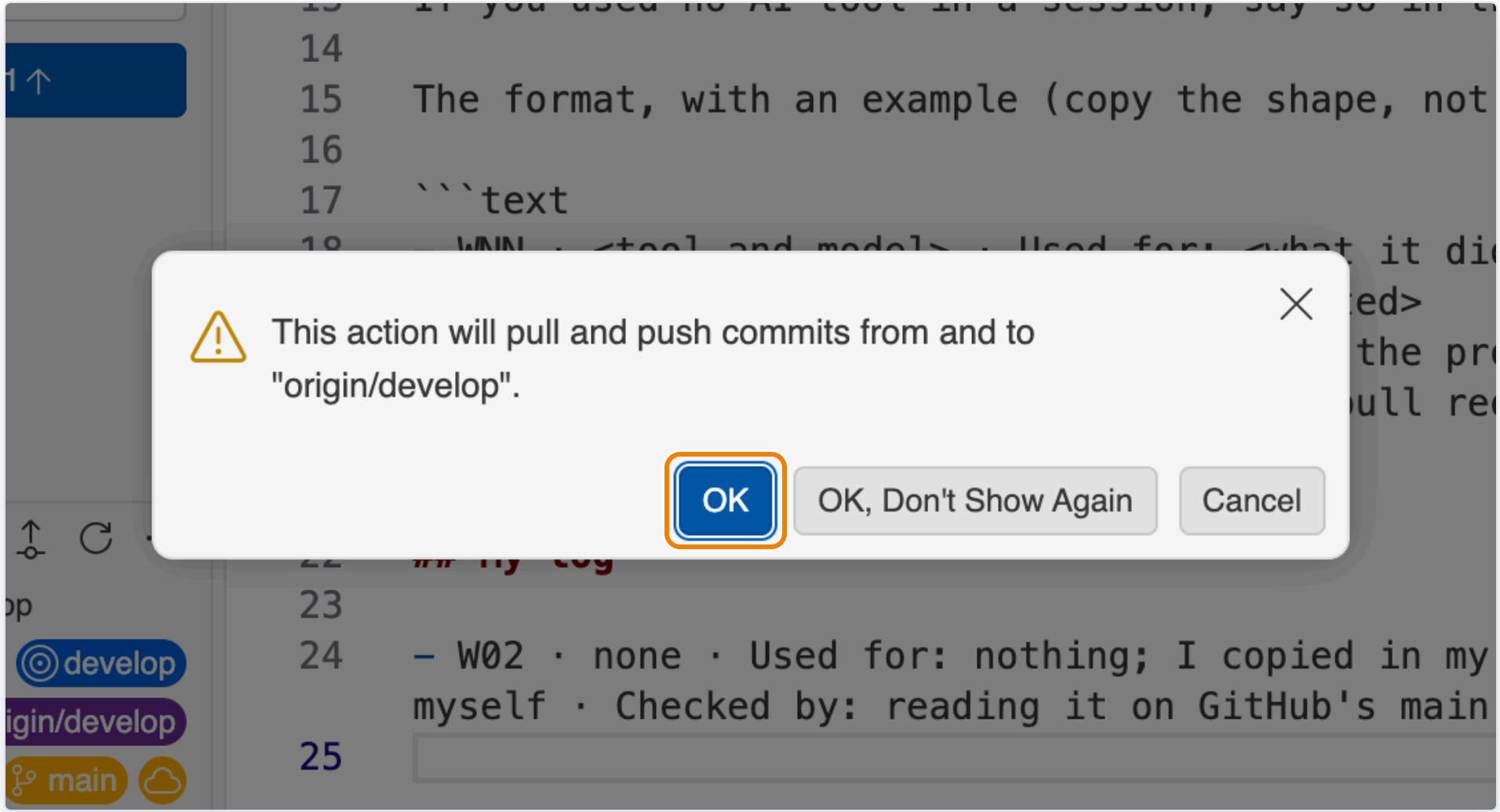
ON BRANCH
develop

YOUR TURN PUSH DEVELOP · VS CODE → GITHUB · 1 OF 5

Sync Changes 1 ↑



OK



VS CODE → GITHUB

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ON BRANCH develop

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ON BRANCH
develop

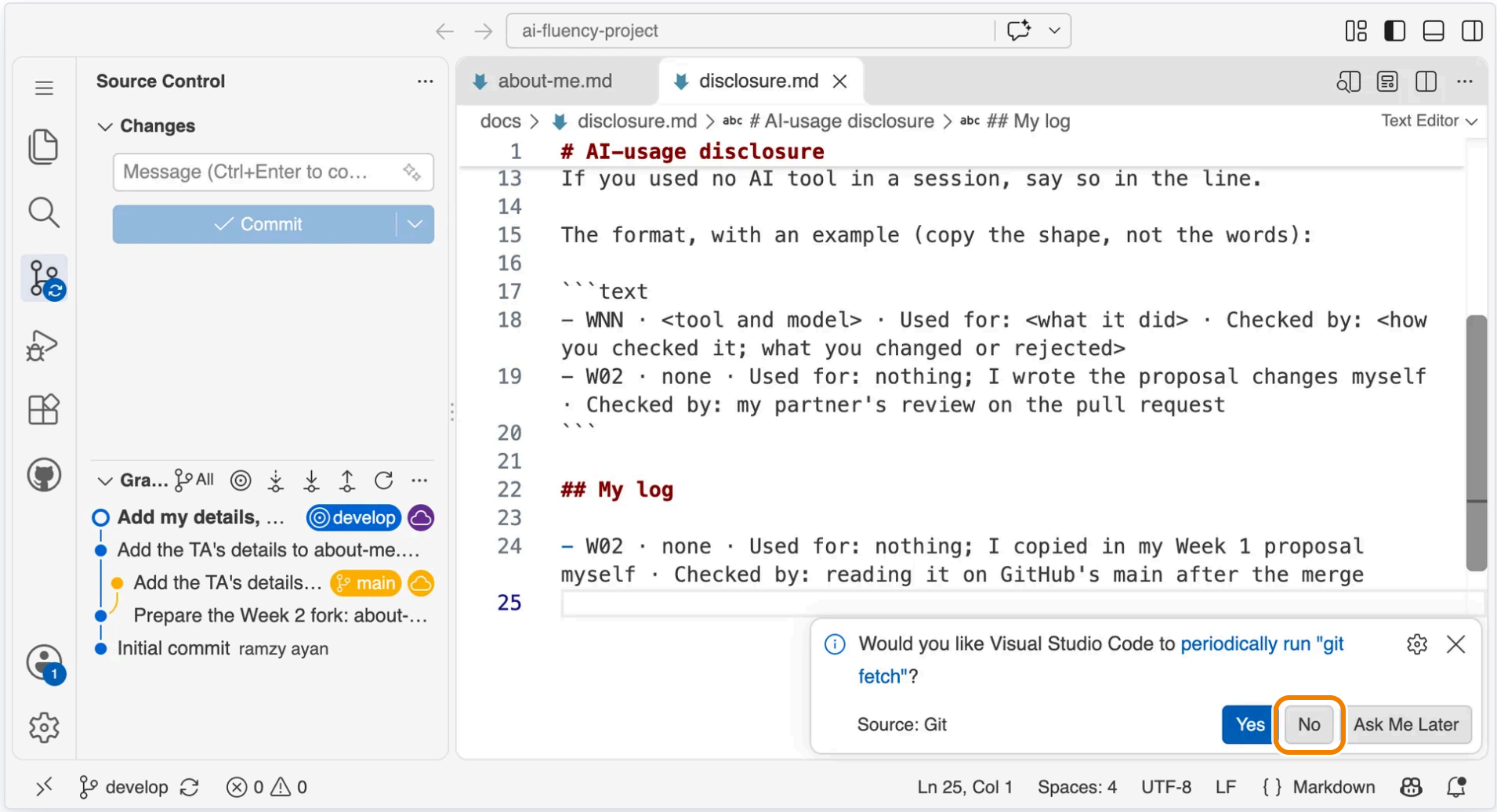
BACKUP PUSH DEVELOP · VS CODE → GITHUB

Asked to sign in? Allow

YOUR TURN PUSH DEVELOP · VS CODE → GITHUB · 3 OF 5

Run git fetch periodically? No

Not needed today



GITHUB

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WSL

11:02 Sign in to GitHub ✓
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GITHUB

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VS CODE

11:18 Switch to develop ✓
VS CODE

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VS CODE

----- 11:34 BREAK -----

11:44 Push develop
VS CODE → GITHUB

11:50 Merge: a conflict
VS CODE → GITHUB

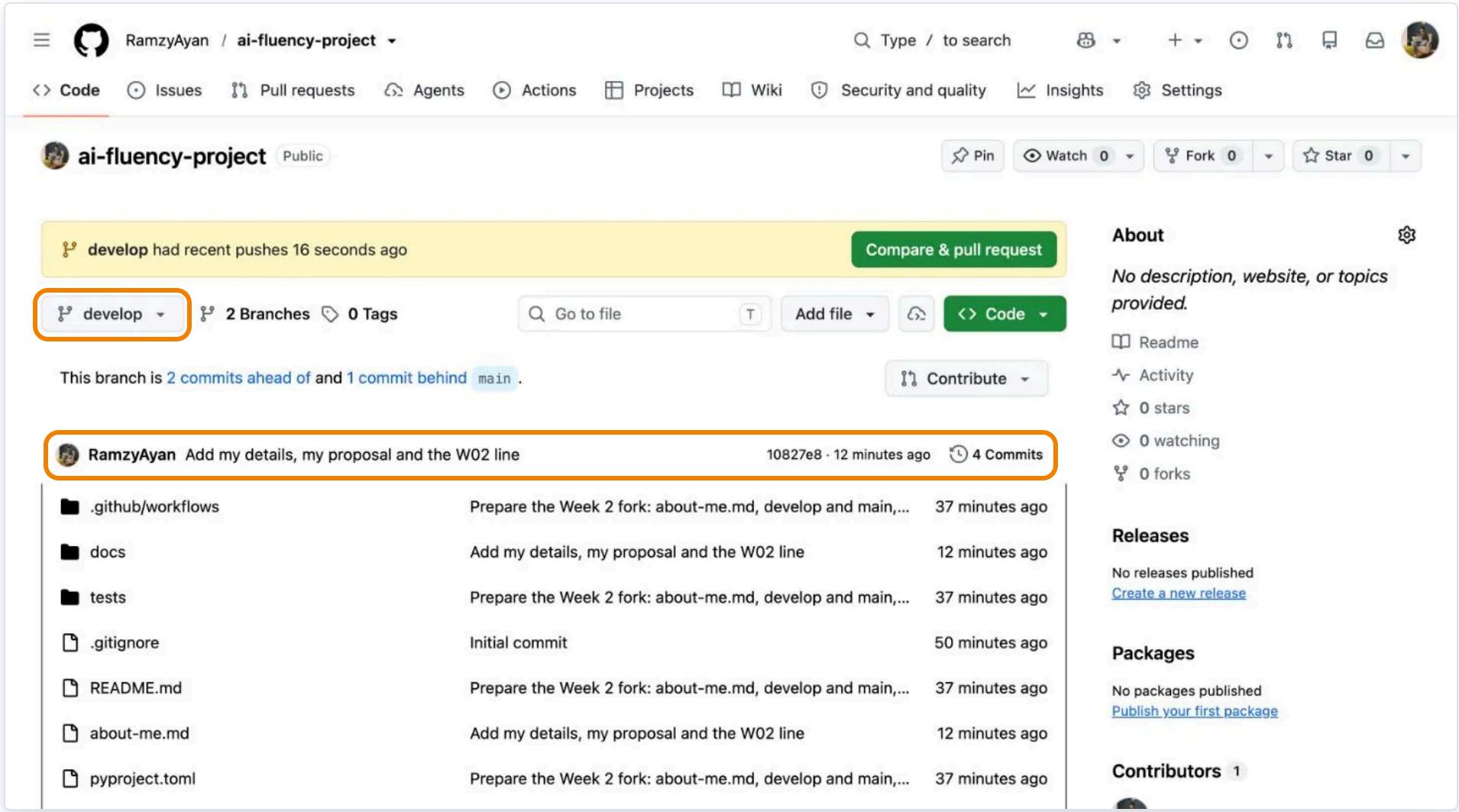
12:07 Peer review
GITHUB

ON BRANCH
develop

DONE WHEN PUSH DEVELOP · GITHUB · 4 OF 5

On GitHub: develop has it

Browser: your fork → refresh → branch menu: develop · ignore the yellow banner



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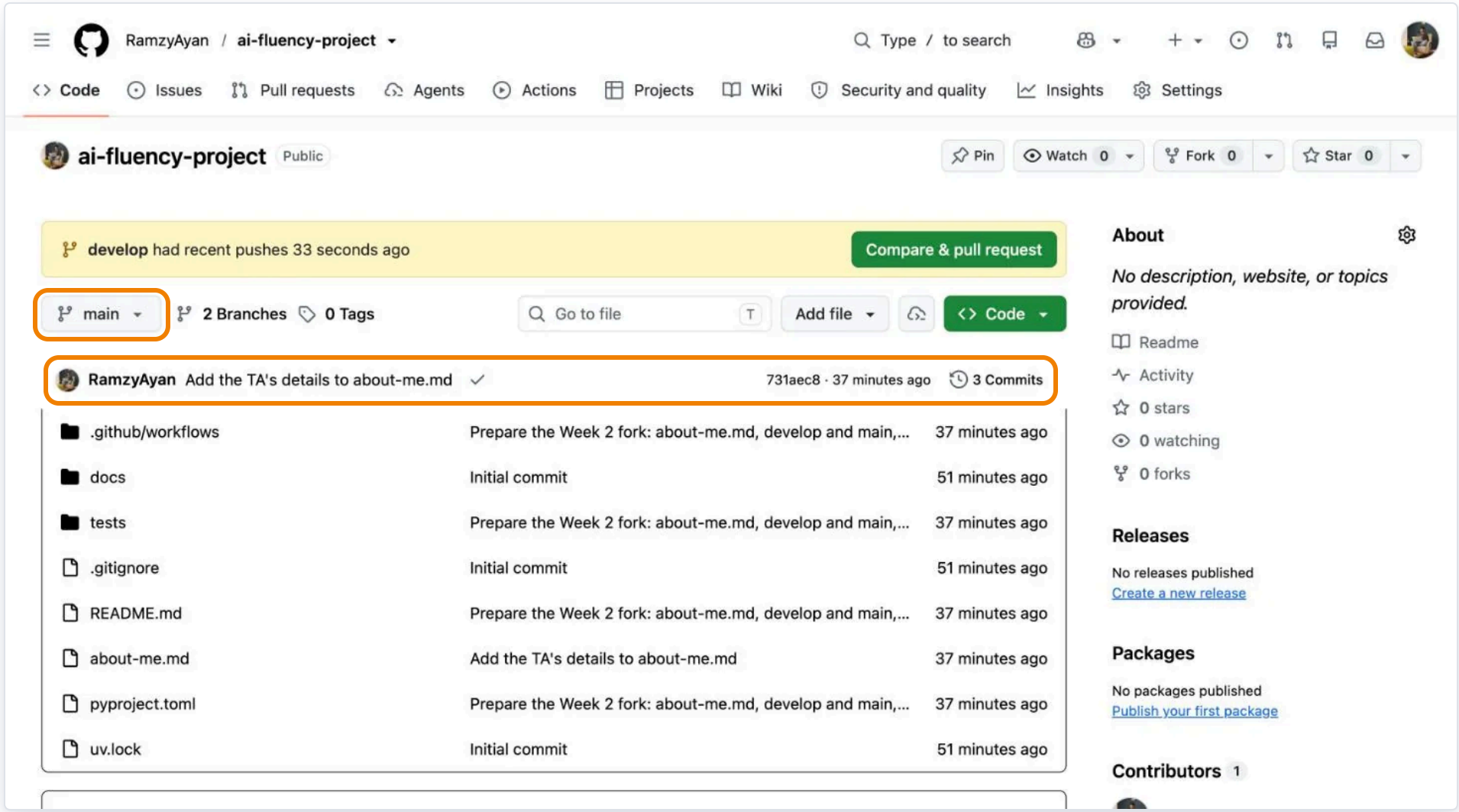
ON BRANCH
develop

DONE WHEN PUSH DEVELOP · GITHUB · 5 OF 5

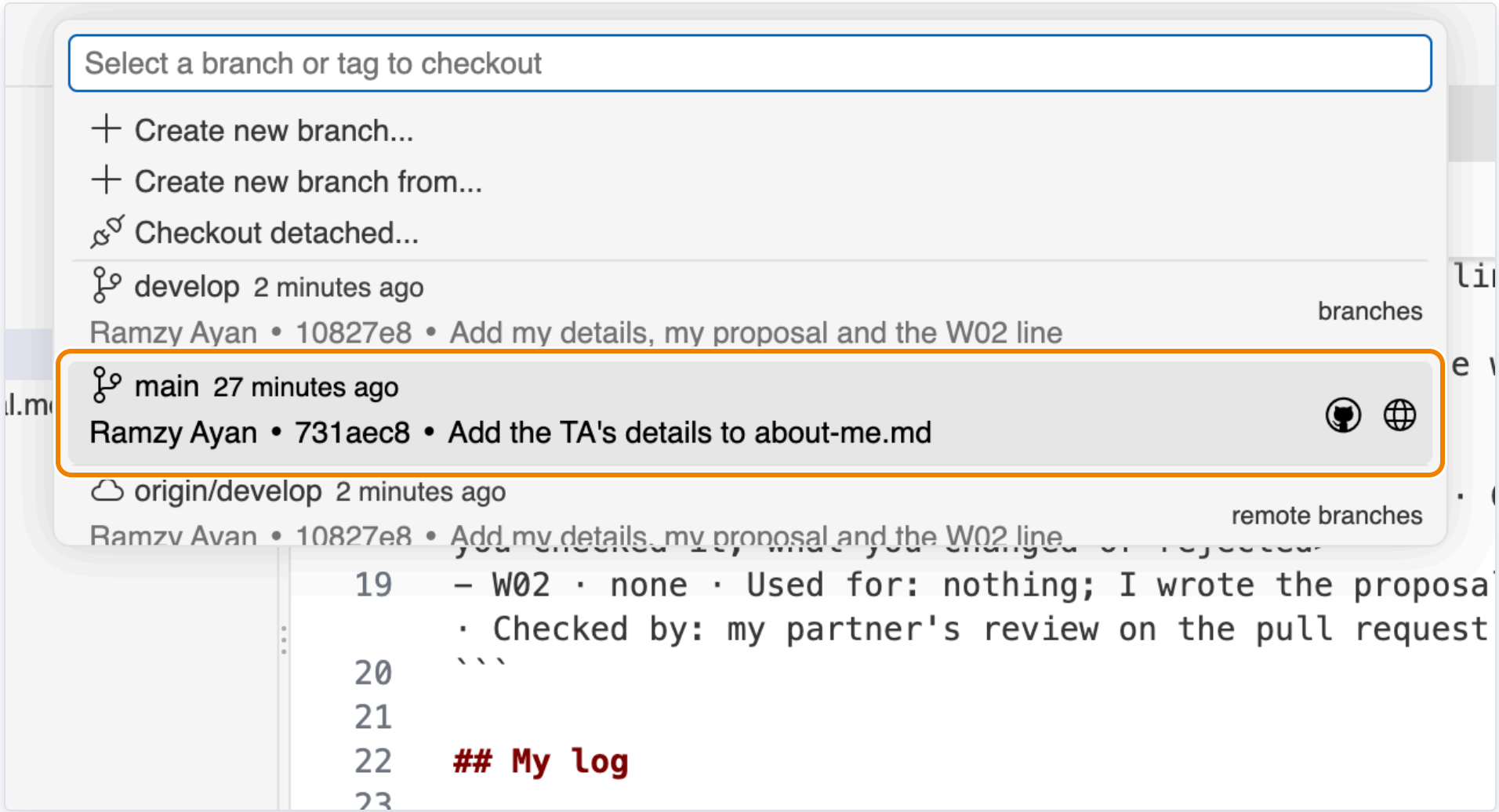
main does not

Branch menu: main



Switch to main

Click develop in the status bar → main (not origin/main)



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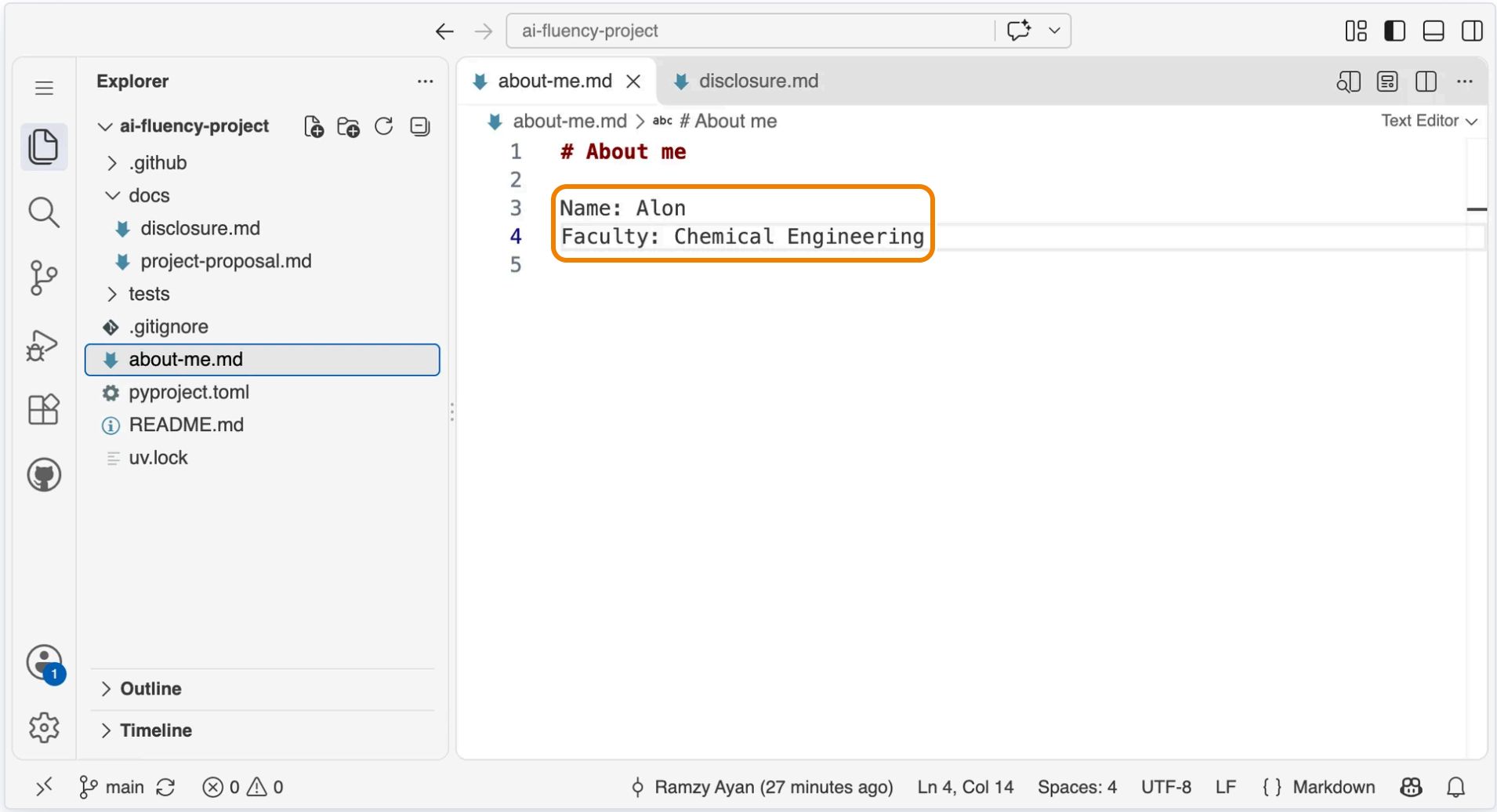
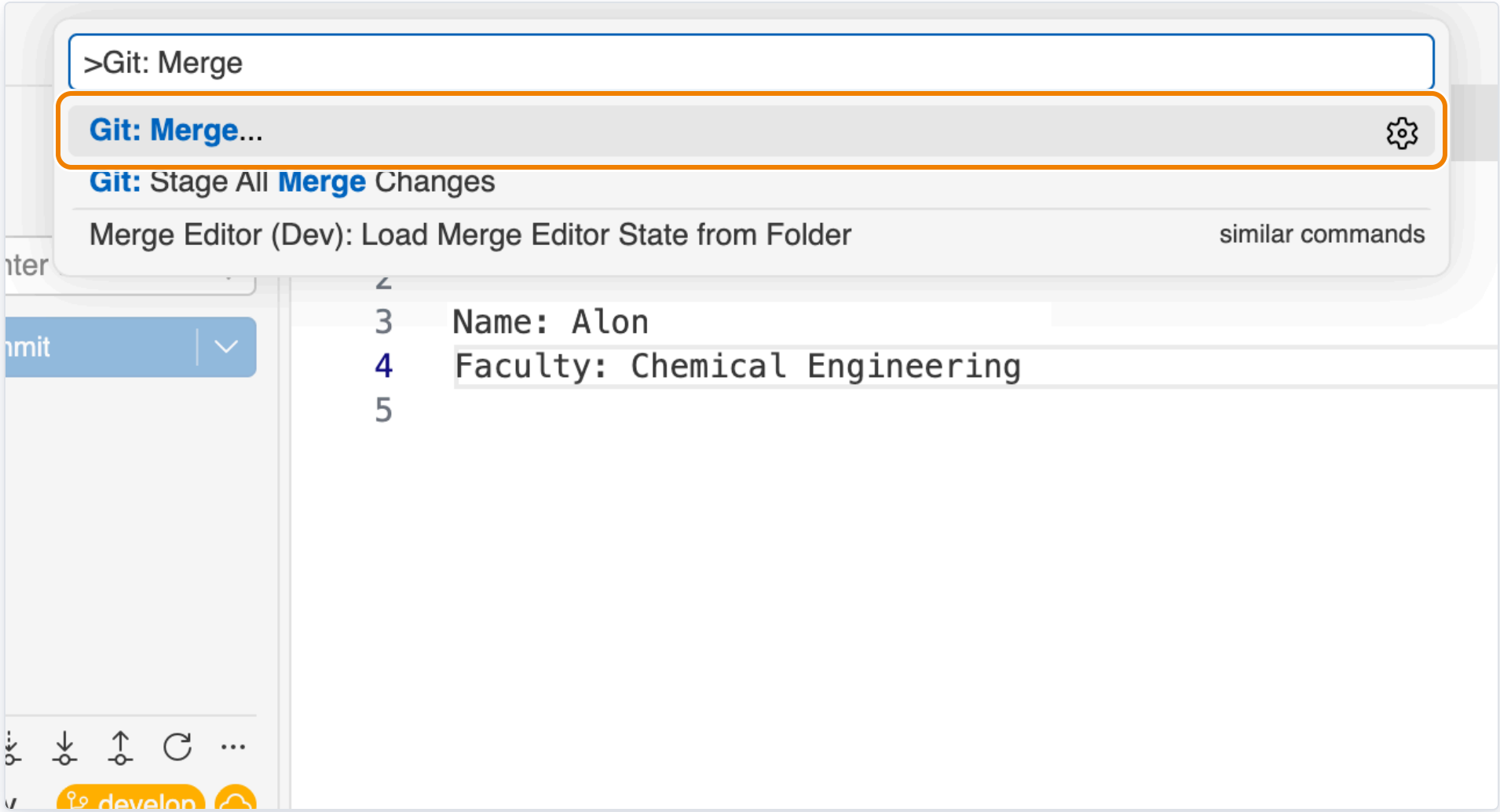
ON BRANCH
main

On main: Alon's details again

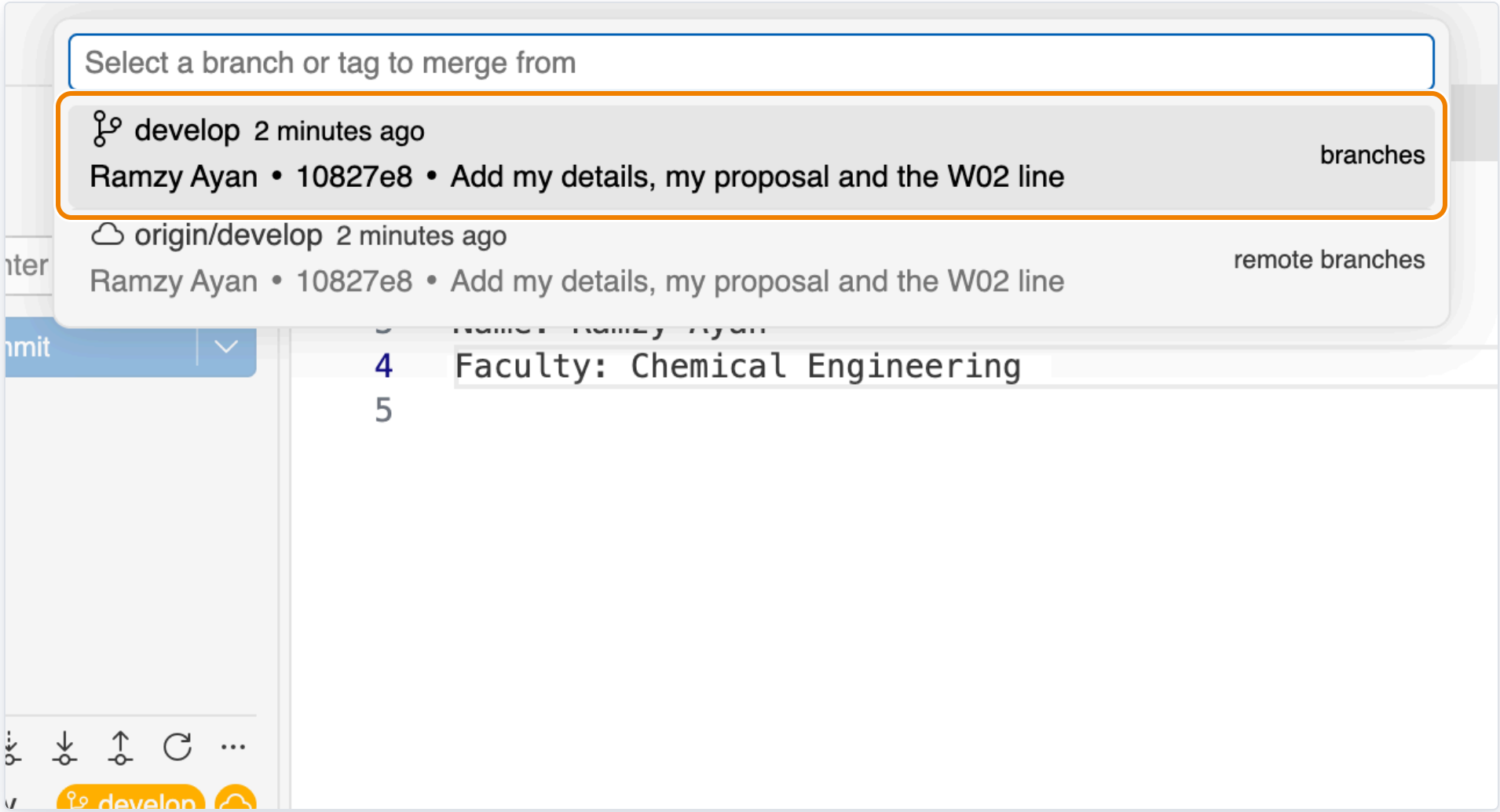
Your work is safe on develop



Ctrl+Shift+P → Git: Merge...

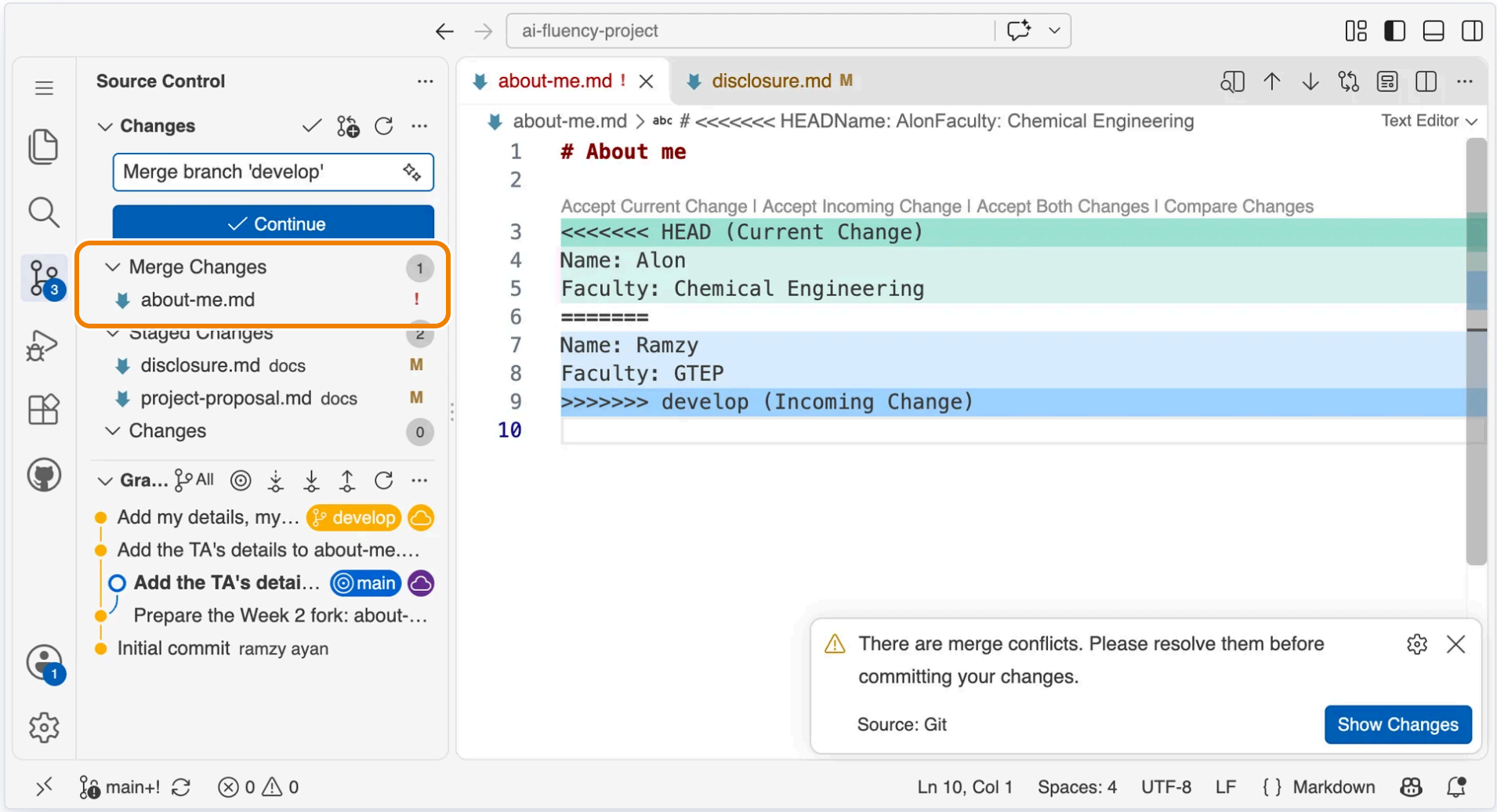


Pick develop



A conflict

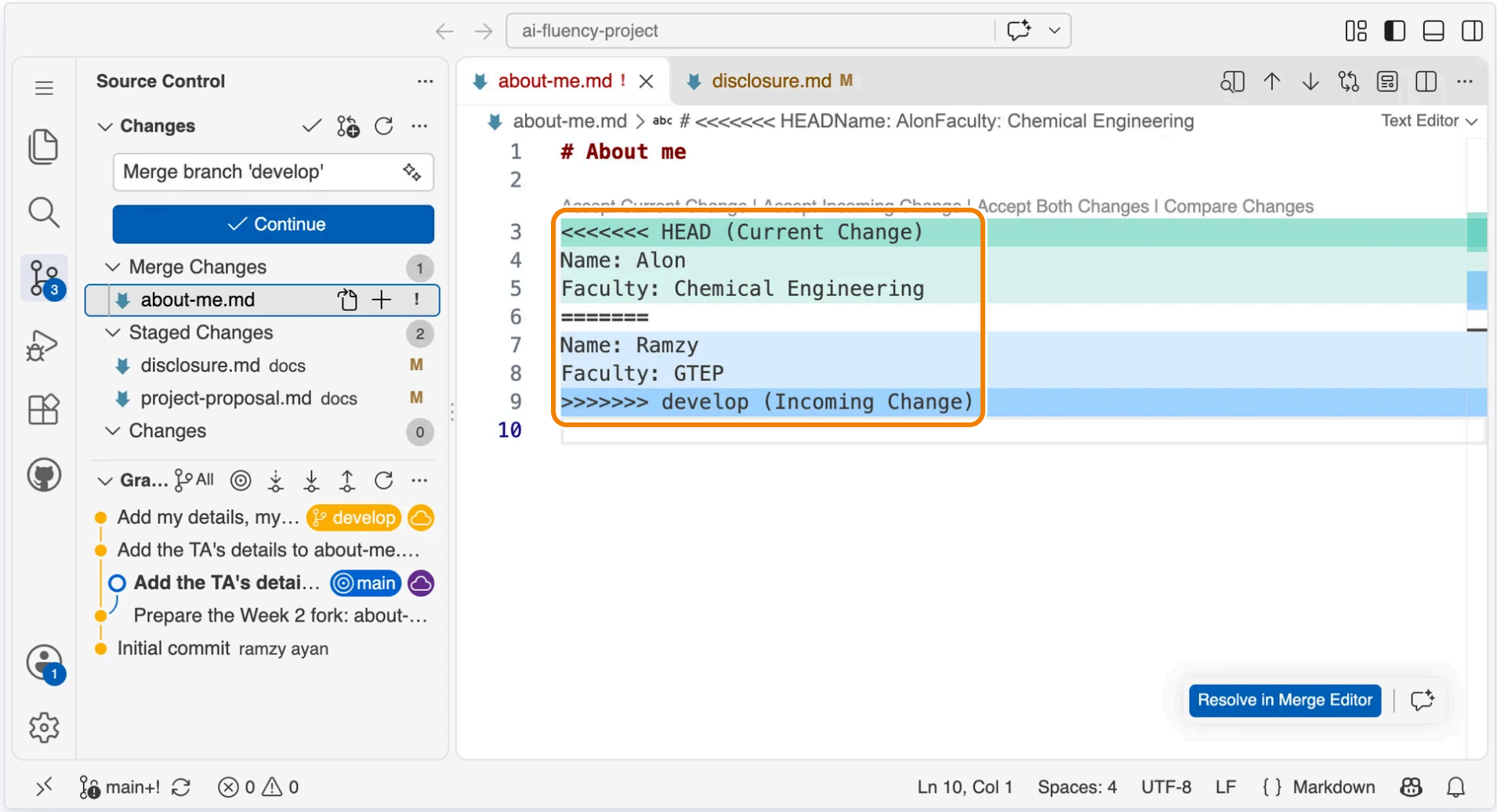
Expected · Merge Changes: about-me.md · the rest came in



WATCH MERGE: A CONFLICT · VS CODE → GITHUB · 6 OF 14

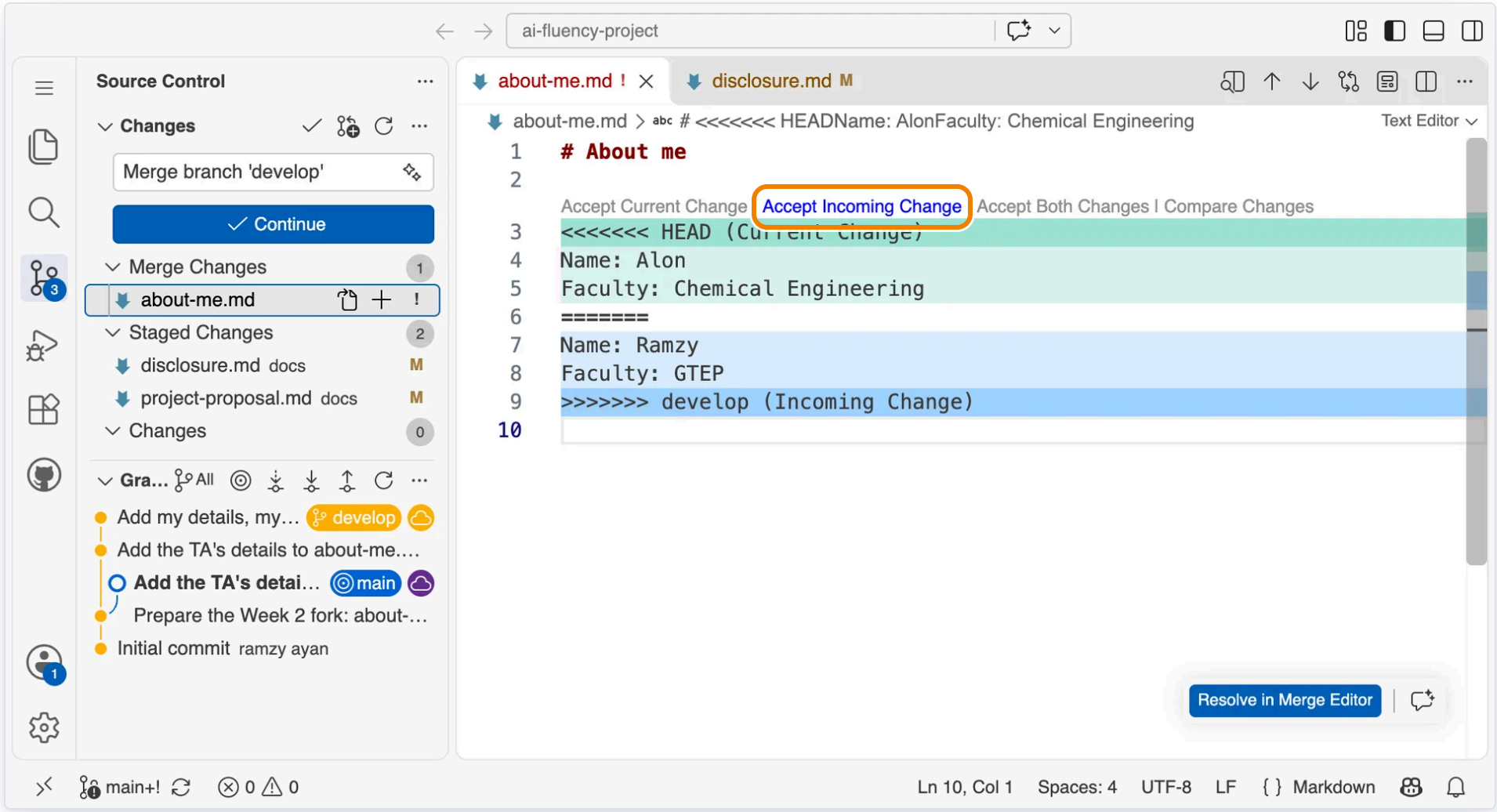
The markers

Click about-me.md · Current = main (Alon's) · Incoming = develop (yours)



Accept Incoming Change

Keep your lines (not Resolve in Merge Editor)



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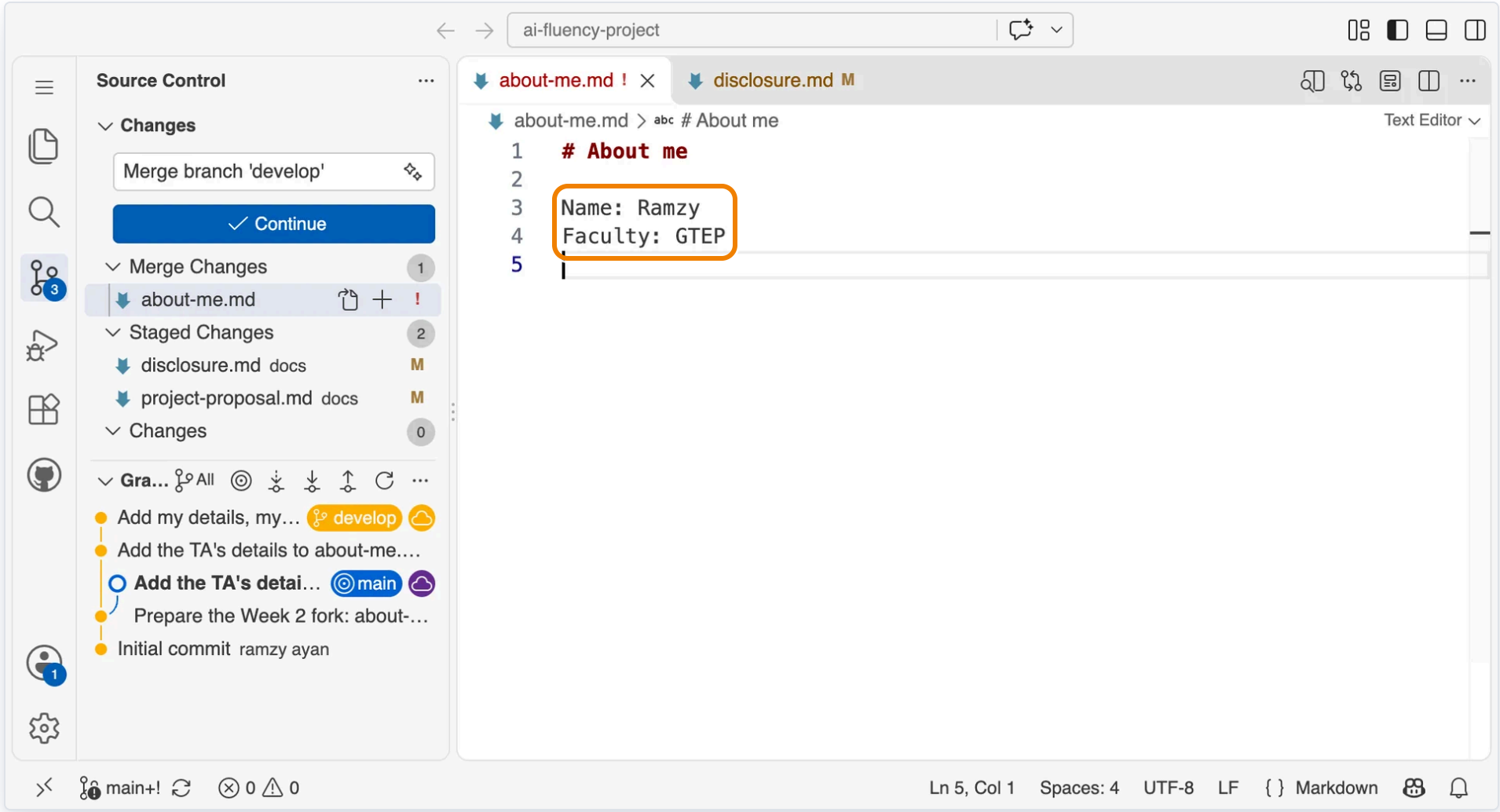
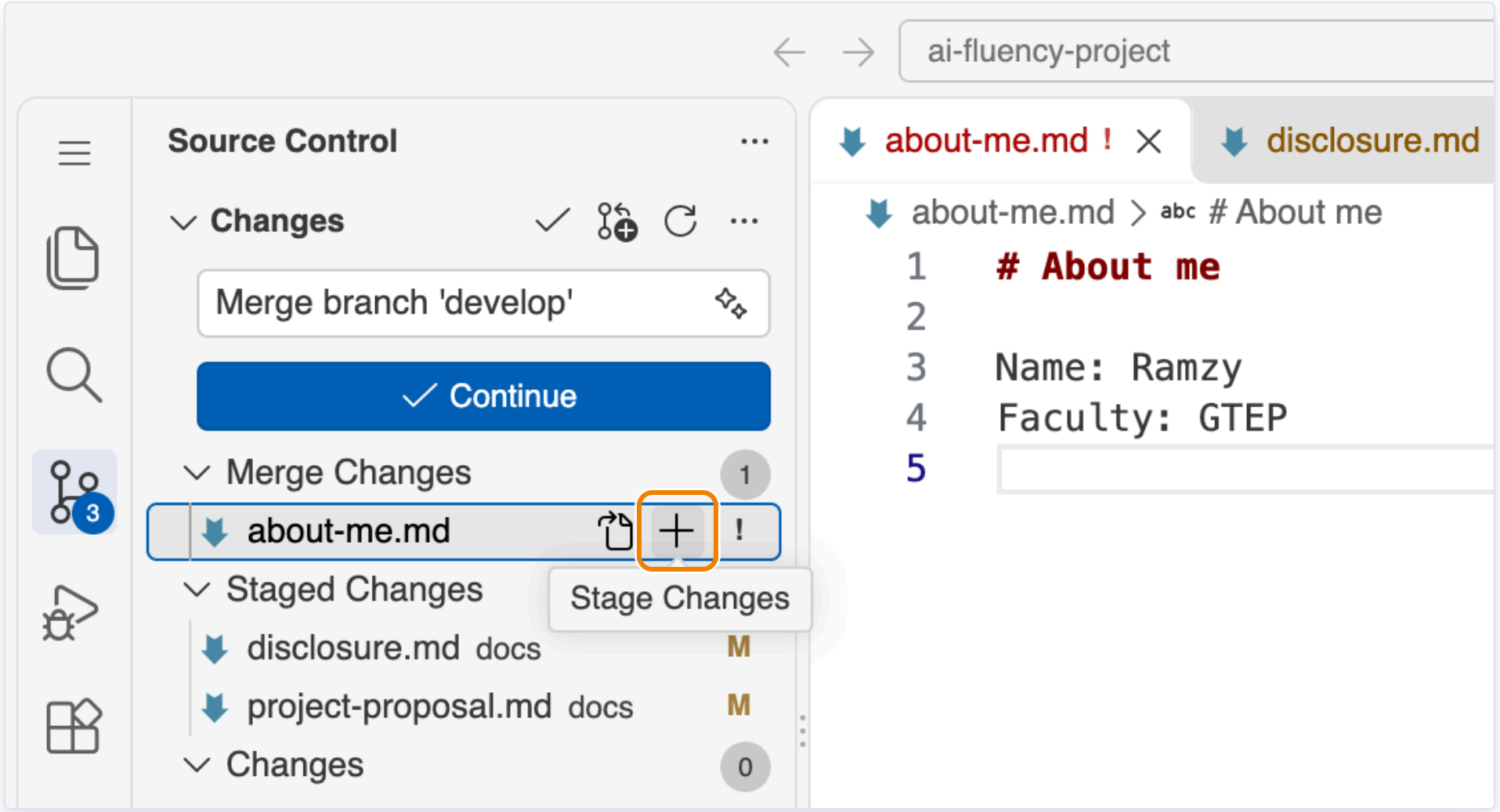
ON BRANCH
main

Your two lines · Ctrl+S



+ Stage



VS CODE → GITHUB

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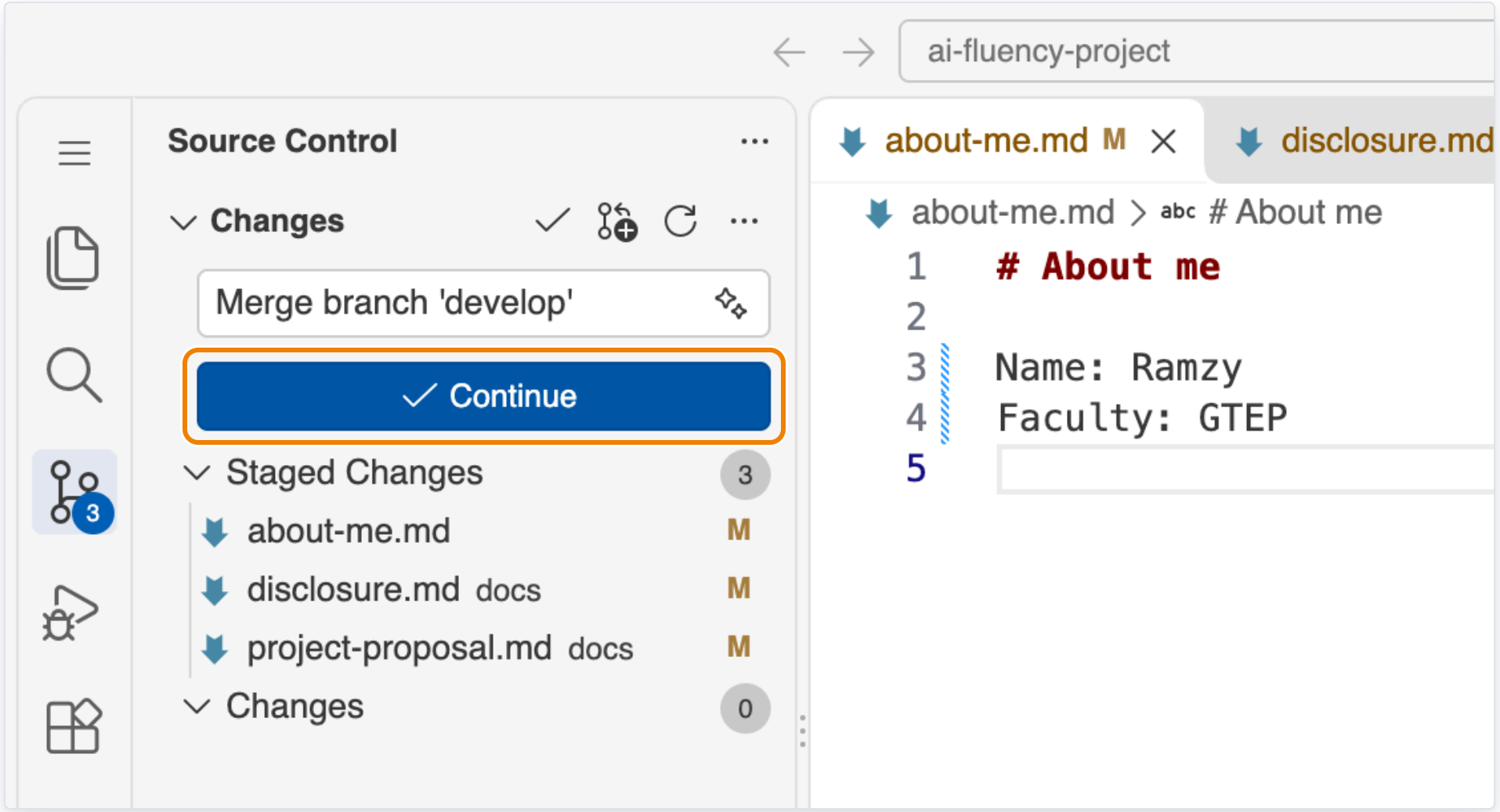
ON BRANCH
main

Continue



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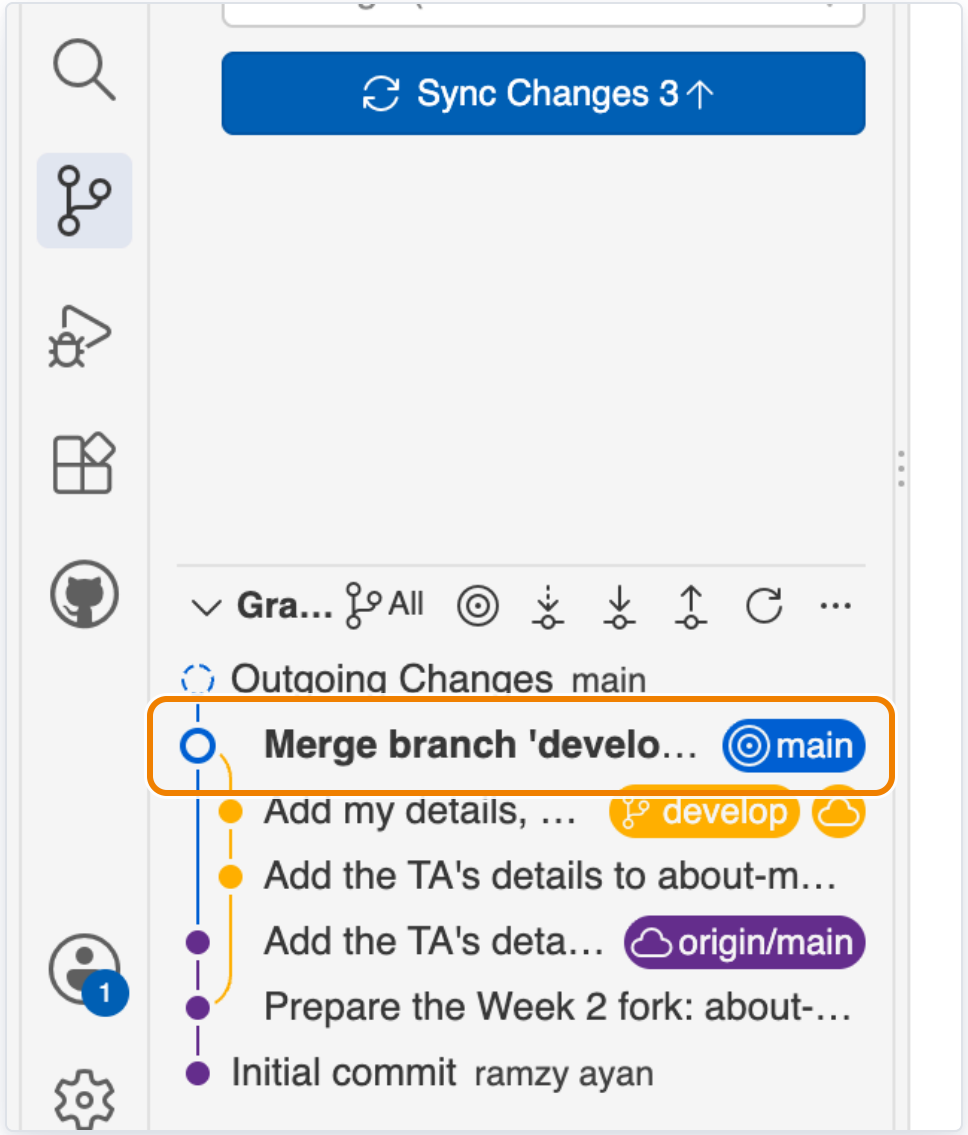
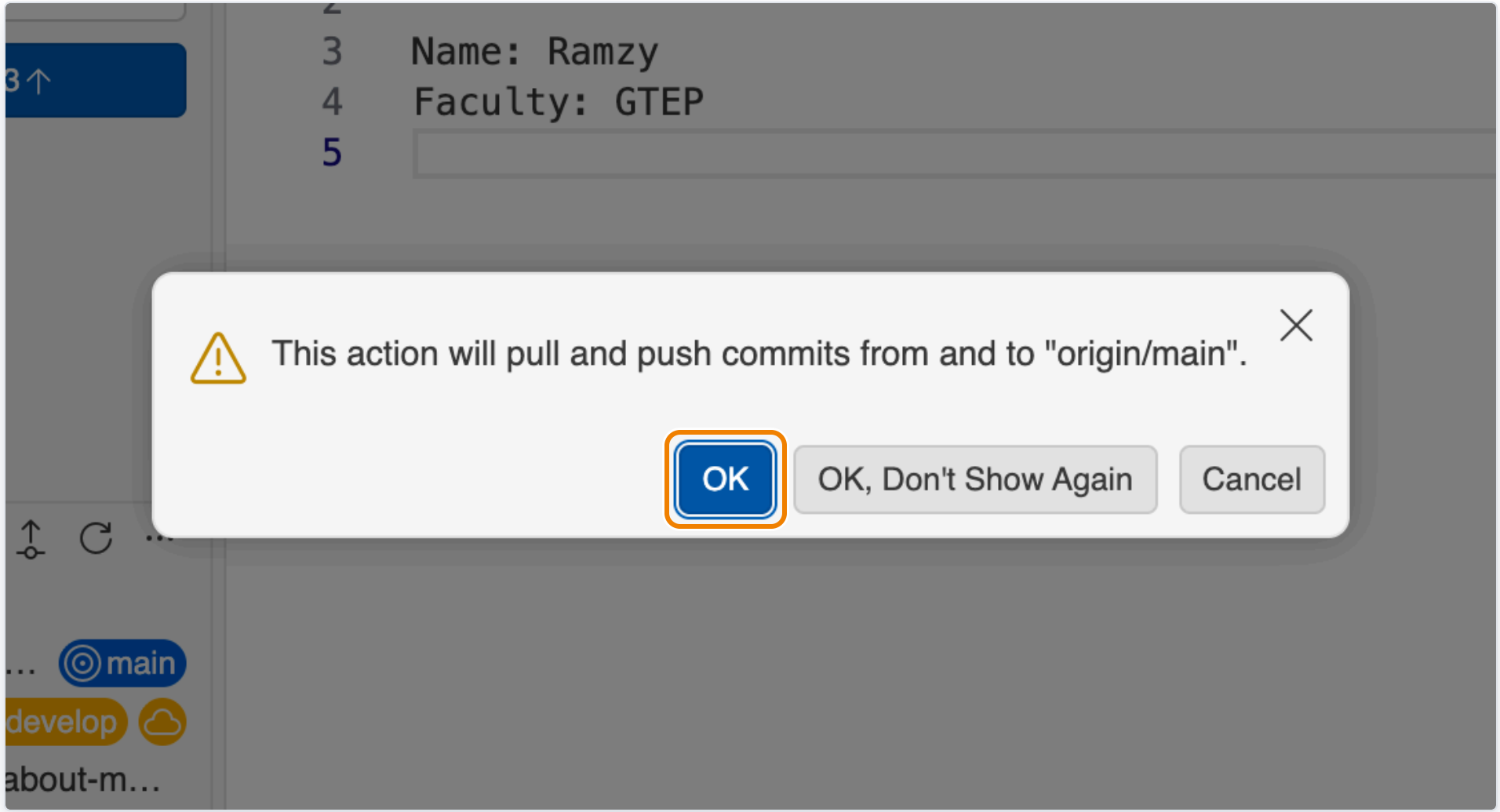
ON BRANCH
main

Graph: the lines join

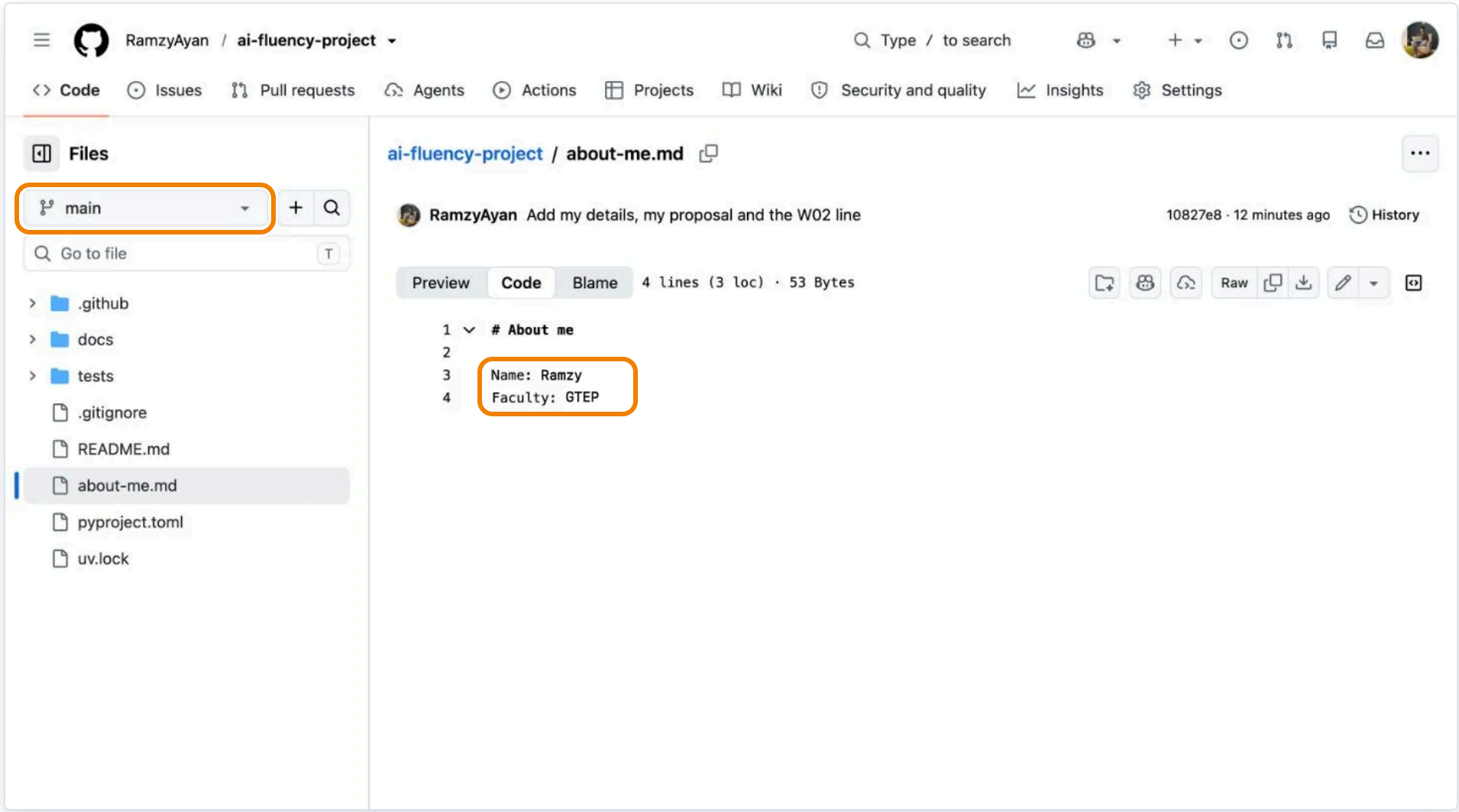


Sync Changes 3↑ → OK



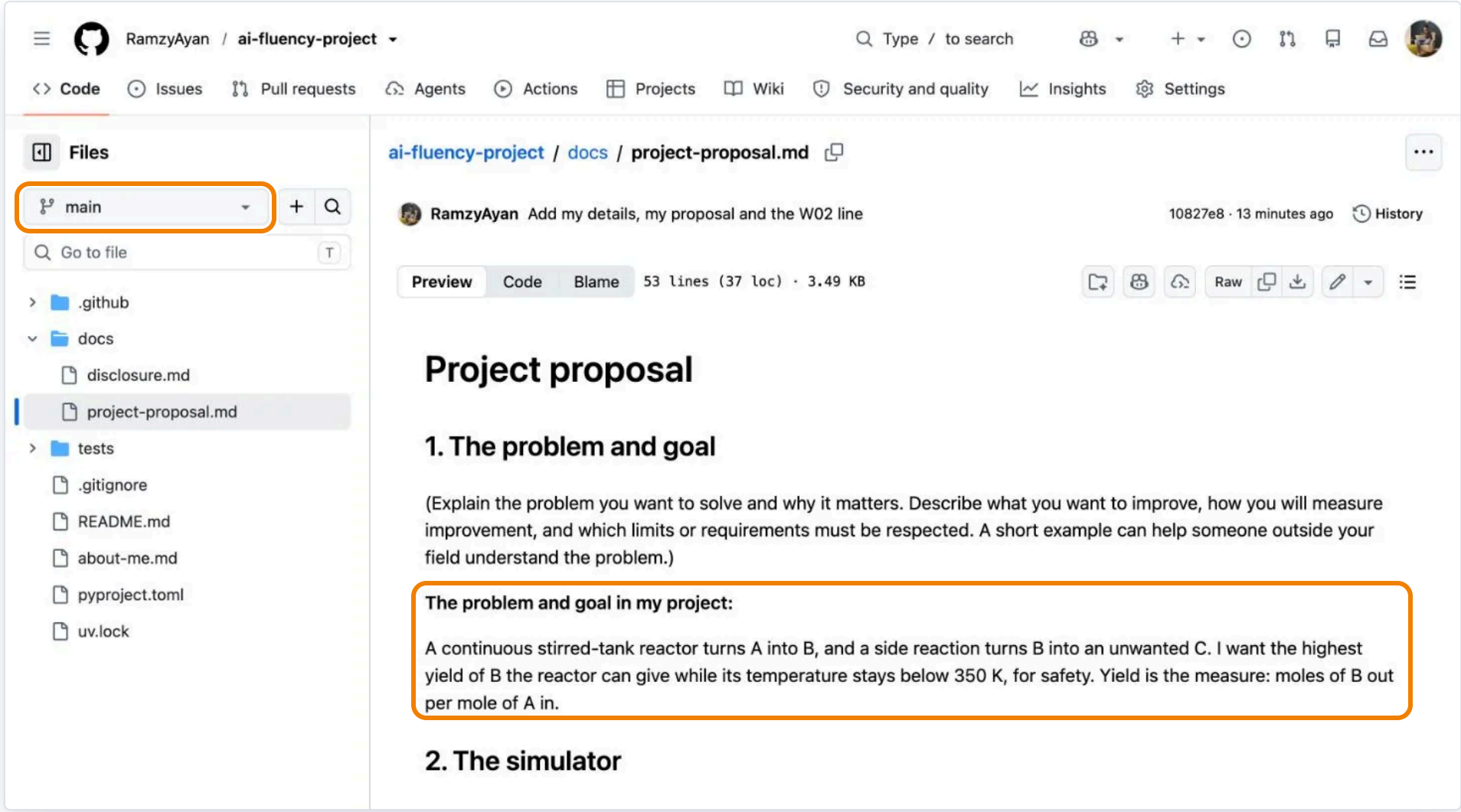
On GitHub: your details

Refresh · main → about-me.md → Code



And your proposal

main → docs → project-proposal.md



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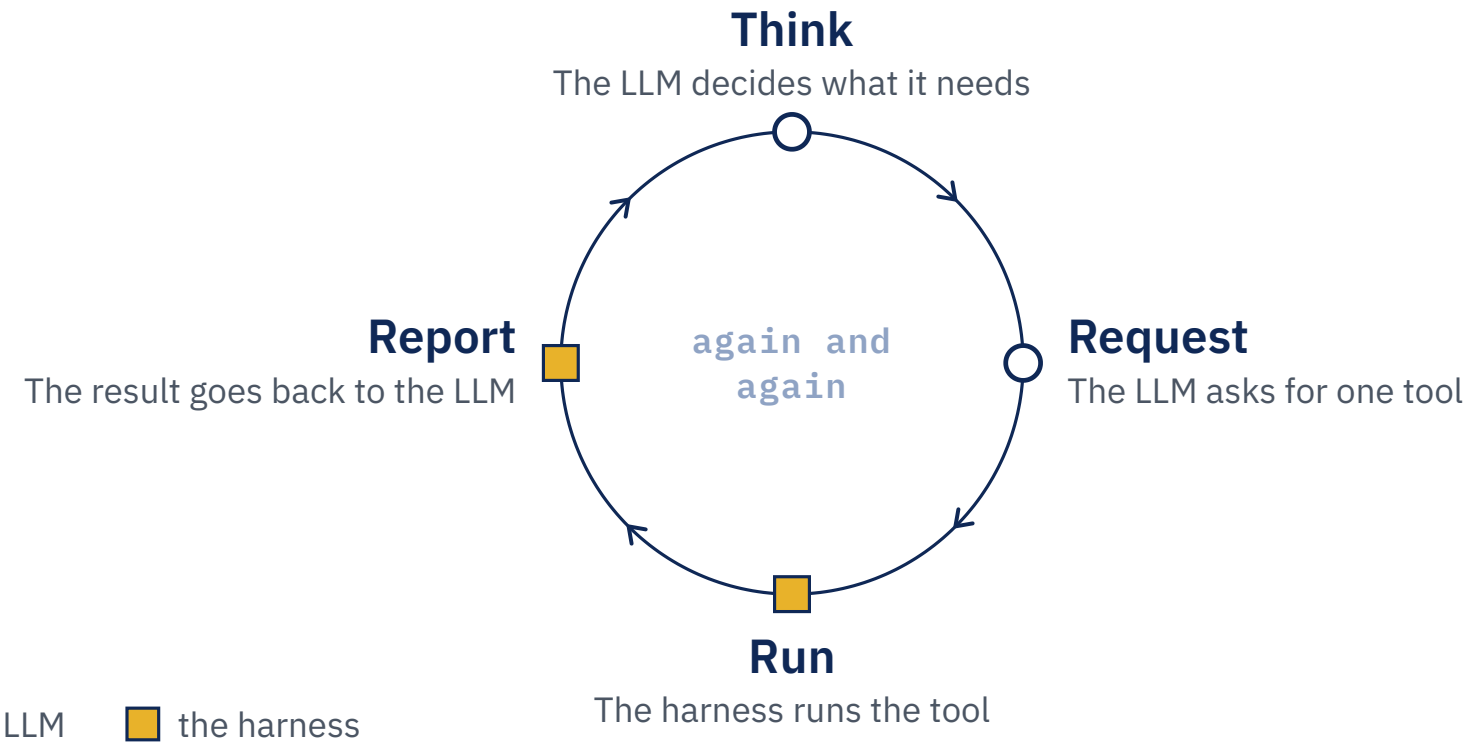
ON BRANCH
main

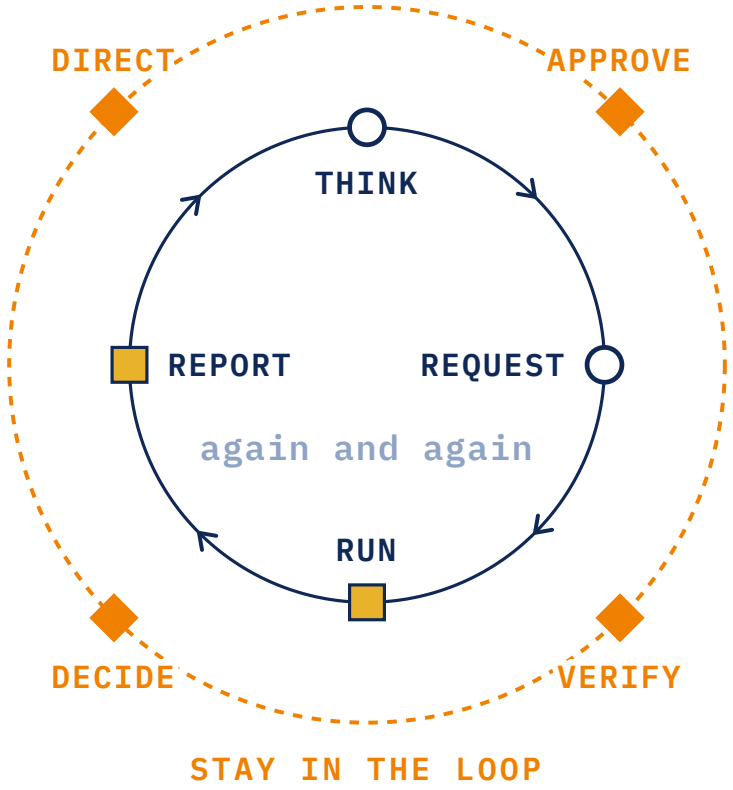
Week 1: the agent loop

Agents may repeat this many times, by themselves.



Week 1: the human in the loop

You are responsible for what the agent does.



YOU, THE HUMAN

- **Direct**
  Give the agent a clear task
- **Approve**
  Check requests before approving
- **Verify**
  Check the result against your goal
- **Decide**
  Keep it, or undo it (Git!)

Week 1: the for-loop test

Take the LLM out: can a simple search do just as well?

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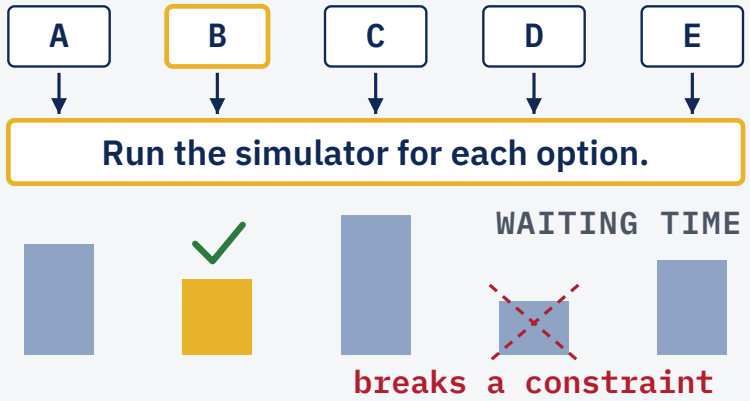
GITHUB

A fixed search may be enough

Try a fixed set of traffic-light timings.



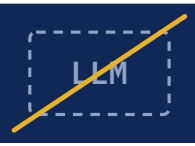
Choose the option with the shortest waiting time that meets the constraints.

An LLM might also...

1. Turn a written request, such as “give buses priority”, into goals and constraints.
2. Use the results to suggest and explain the next experiment.
3. Help adapt the plan when the requirements change.

THE FOR-LOOP TEST

Remove the LLM. Can a simple search solve the task just as well?



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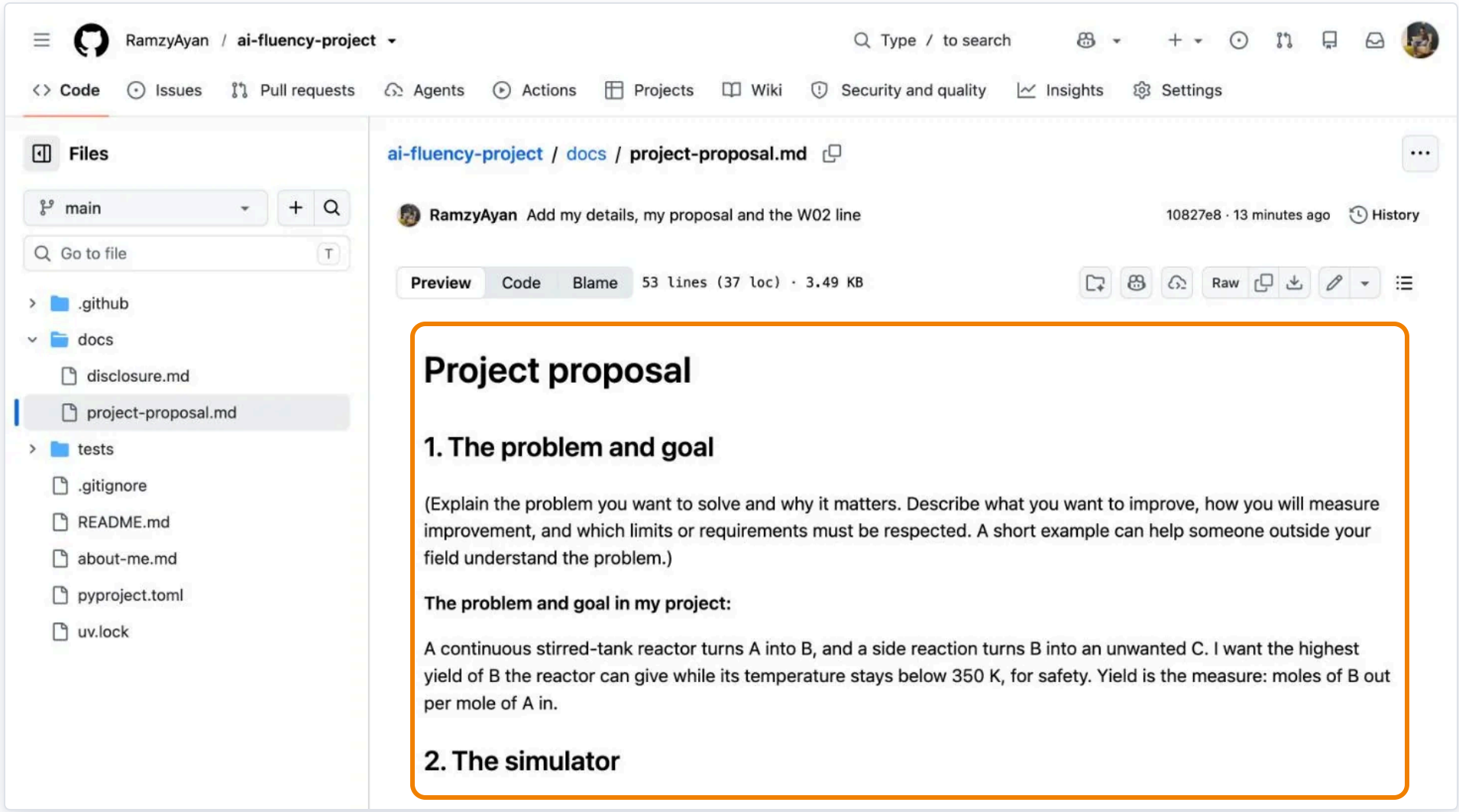
●●●◐○○

YOUR TURN PEER REVIEW · GITHUB · 4 OF 6

Pairs: open your partner's proposal

github.com/<partner>/ai-fluency-project → docs → project-proposal.md



Read it with four questions

One for each section of the Week 1 template

1 THE PROBLEM AND GOAL
What do they want to improve, and how will they measure it?

2 THE SIMULATOR
What goes in, what comes out? Could it run without the AI?

3 THE AGENT'S ROLE
What does the model decide that a for loop could not?

4 OUTSIDE YOUR FIELD
Did the plain-language part make sense to you?

One clear thing, one question

4 minutes each · swap at 12:13 · note the question you got

THE ROUTE · EVERY STAGE DONE

What you did

Fork · clone · switch · commit · push · merge · a conflict · peer review

STARTS	STAGE	TOOL	BRANCH	STEPS
10:38	**Student package**	GITHUB		●●●●●●●●●●●●
10:46	**Meet GitHub**	GITHUB		●●●●●●
10:51	**Meet VS Code**	VS CODE		●●●●●●
10:55	**wsl → code .**	WSL		●●●●●●●●●●
11:02	**Sign in to GitHub**	VS CODE		●●●●●●
11:07	**Fork**	GITHUB		●●●●◌
11:12	**Clone**	VS CODE	main	●●●●●●
11:18	**Switch to develop**	VS CODE	main → develop	●●●●●●
11:24	**Your details**	VS CODE	develop	●●●●●●●●
11:34	BREAK, BACK AT 11:44			
11:44	**Push develop**	VS CODE → GITHUB	develop	●●◌●●●
11:50	**Merge: a conflict**	VS CODE → GITHUB	develop → main	●●●●●●●●●●●●●
12:07	**Peer review**	GITHUB		●●●●●●

WRAP-UP

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GITHUB

Before Thursday

M1 · due Thursday 12 November, before midnight

See you next week

Questions: ask your TA